

# The Economic Consequences of Foreclosure Suspensions in the Great Depression\*

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## Abstract

We study the effects of foreclosure moratoria on local economic activity, individuals, and credit markets. During the Great Depression, twenty-five U.S. states enacted laws that temporarily restricted creditors' ability to foreclose on delinquent farms. We examine the effects of these policies using county-level agricultural, manufacturing, and banking data along with linked full-count census records. States most exposed to the moratoria experienced a persistent 15% increase in the number of farms. Protected farmers and their children were more likely to remain in agriculture and less likely to transition into manufacturing almost two decades after enactment. The policy pushed agricultural production toward smaller, more labor-intensive farms operating on more marginal land with lower capital intensity. Foregone consolidation thus depressed farm revenues and values. At the same time, the moratoria increased local interest rates by 30bps (4%). We also find suggestive evidence that highly exposed areas developed weaker manufacturing sectors in the long run, consistent with slower labor reallocation out of agriculture. The results show that temporary foreclosure suspensions can stabilize incumbent land users but impose persistent costs by distorting local land markets, slowing labor reallocation, and raising the cost of local credit.

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# I Introduction

Debt crises force governments to choose between letting creditors seize collateral and protecting distressed borrowers. When asset prices collapse and credit markets freeze, foreclosure can destroy going concern value, push assets into fire sales, and transmit distress to other sectors (Shleifer and Vishny, 1992; Kiyotaki and Moore, 1997). Foreclosures also impose large costs on borrowers, neighborhoods, and local economies (Campbell et al., 2011; Mian et al., 2015; Diamond et al., 2020). Therefore, suspending foreclosures can keep viable borrowers in place until prices recover, refinancing becomes available, or income rebounds.

The same policies can also create misallocation, especially when the collateral is a productive asset. Foreclosures facilitate the reallocation of productive assets from distressed owners to new operators who are better able to finance, consolidate, or redeploy them. A moratorium, therefore, may slow reallocation, prevent consolidation, and keep labor in low productivity sectors (Caballero and Hammour, 1996; Acharya et al., 2022). Moreover, if lenders expect weaker enforcement in future downturns, they may cut back on lending (Pence, 2006).

Governments have repeatedly resorted to moratoria when faced with mass distress. Most recently, during the COVID-19 crisis, the federal government enacted an eviction moratorium that banned foreclosure on rental properties with mortgages backed by federal programs. The share of households covered by this law has been estimated at 28% (Goodman et al., 2020). Separately, the Department of Agriculture suspended foreclosures all federally backed farm loans (USDA, 2021).

This paper studies the consequences of foreclosure suspensions during the Great Depression. Between 1933 to 1935, twenty-five states passed laws that prevented lenders from foreclosing on delinquent farmers (Alston, 1983). At the time, farm mortgage debt equaled one quarter of total farm value and nearly 200,000 farms were foreclosed upon (Alston, 1983; Hausman et al., 2019). The Supreme Court upheld Minnesota’s statute in *Home Building & Loan Ass’n v. Blaisdell* (1934), emphasizing the emergency conditions caused by the Great Depression. The moratoria targeted the main legal channel through which distressed farms changed hands. Bankruptcy was rare and costly while foreclosures could be initiated by a single creditor and ended, if redemption failed, in forced sales (Wickens, 1936; Messer and Rieder, 2025). The laws could therefore change which farms survived, who operated them, and how quickly land and labor left agriculture.

We exploit the fact that the moratoria differentially impacted regions where more farms were exposed to foreclosure. We compare post-1930 changes across counties with different baseline mortgage exposure, in states that did and did not adopt moratoria using a triple differences design. We include county, year, and state-year fixed effects to absorb permanent county differences, aggregate shocks, and time varying state-level confounders. We also control directly

for county-level New Deal spending (Fishback et al., 2005a) and Dust Bowl erosion (Hornbeck, 2012).

This empirical strategy does not by itself address the fact that moratoria were not adopted randomly. States that intervened were under greater economic and political pressure (Alston, 1984), and treated counties were more populous, had larger farms, and had deeper financial markets before the policy. Therefore, our main empirical design restricts the analysis to a set of 476 “border counties” – neighboring counties on opposite sides of state borders where moratorium status differed. In this sample, we verify that treatment appears as-good-as-randomly assigned with respect to a range of baseline observables – including mortgage exposure. The trade-off to making tighter geographic comparisons is the increased risk of cross-border factor mobility generating SUTVA violations. All our main results hold without restricting to this sample and remain comparable in magnitude to the main design.

The paper is organized around four key results. First, the moratoria substantially reduced farm exit by preventing consolidation. Counties in the top quartile of exposure experienced a 15% increase in the number of farms, or roughly 300 additional farms per county, and this effect persists at least through 1960. The increase is entirely due to foregone consolidation, as total farm acreage remains unchanged. Together, average farm size falls, and the average farm in more exposed counties is worth about 20% less. These findings suggest that the policy durably altered the composition of operating farms, rather than merely delaying a temporary dislocation.

Second, the policy slowed the structural reallocation of labor out of agriculture. Using linked census data from 1930 to 1940, we show that in more exposed counties, a larger share of individuals remained in agriculture and a smaller share moved into manufacturing. The effect persisted to 1950 and was observable in the next generation. Moving from the median to the 90th percentile of the county mortgage-share distribution increases the probability that the child of a protected farm household remains tied to the family farm by about 2.5 percentage points. The moratoria thus did not simply preserve farms in place; they also altered the occupational trajectories of the families attached to those farms.

Third, we study how the moratoria affected farm productivity. Protected farms devoted a larger share of land to crops and operated with lower capital intensity. Farmers brought more marginal acres into cultivation, harvested a smaller share of cropland, and experienced more crop failure. Crop revenues per cropland acre also fell. First, the moratoria changed selection by keeping marginal operators in agriculture who would otherwise have exited through foreclosure. Second, surviving farmers appear to have re-optimized under tighter credit conditions and smaller farm sizes by utilizing more labor and less capital.

Fourth, the suspensions altered local credit conditions. In the counties most exposed to the moratoria, agricultural mortgage interest rates rise by about 4%, or roughly 30 basis points. Be-

tween 1930 and 1940, county-level farm mortgage rates had declined and converged significantly across the country. Our estimates imply that foreclosure suspensions account for roughly 30 percent of the remaining gap in county-level interest rates in 1940. This increase cannot be explained by local banking conditions as the rate of bank failures in moratoria and non-moratoria states were identical due to Federal protections. Instead, we argue that the rate rises are driven by a localized and persistent risk premium. Suspending foreclosure rights weakened creditor protection, and the prevention of consolidation left lenders facing a riskier pool of smaller, less capitalized, and less productive farms.

Taken together, these findings show that crisis-era foreclosure protection came with persistent costs. We do not provide a full welfare analysis and our results do not imply that moratoria were undesirable at every margin. In a period of extreme distress, preventing fire sale liquidation may have generated important short-run benefits. But the evidence suggests that the legal suspension of foreclosure altered the adjustment path by impeding the reallocation process that often accompanies recovery and subsequent growth.

We caution against interpreting these results too broadly. First, the dramatic rise of the US manufacturing sector following WWII, and hence its desirability relative to agriculture, was not evident to policymakers during the depths of the Depression. Our finding on the delayed transition to manufacturing is thus not an indictment of the policy per se. Second, more recent debt interventions tend to be accompanied by explicit government funding guarantees ([Agarwal et al., 2017](#)). By contrast, states did not compensate lenders for moratoria in our setting. This distinction is clearly relevant to the effect of moratorium policies on the financial system. Regardless, our central results on farmer selection and misallocation remain relevant even in settings with lender compensation as long as the underlying collateral has productive use.

**Related Literature.** This paper is related to three strands of literature. The first studies foreclosure, bankruptcy and debt forbearance. Forced liquidation depresses prices, imposes spillovers on neighbors, and amplifies downturns ([Shleifer and Vishny, 1992](#); [Campbell et al., 2011](#); [Ellen et al., 2013](#); [Gupta, 2019](#)). Weak enforcement can create the opposite problem by keeping inefficient borrowers alive and distorts later allocation ([Caballero and Hammour, 1996](#); [Acharya et al., 2022](#)). The Depression-era moratoria suspended liquidation at scale, allowing us to study both forces within a single policy. Our findings speak directly to this tradeoff since foreclosure moratoria relieved short-run distress but increased capital costs and preserved inefficient producers. [Gabriel et al. \(2021\)](#) study foreclosure prevention laws passed following the Great Financial Crisis in California and find limited effects on new mortgage credit. We do find a credit response, likely driven by the difference between owner-occupied housing and productive assets such as farms.

The second literature studies land allocation, agricultural productivity, and structural change. Frictions in land markets shape farm size, productivity, and occupational choice (Adamopoulos and Restuccia, 2014; Chen et al., 2023; Aragón et al., 2024; Haws et al., 2025). Classic development accounts treat the movement of labor out of agriculture as a core driver of growth (Johnston and Mellor, 1961; Matsuyama, 1992). We identify a specific legal mechanism that slowed this process. Foreclosure moratoria preserved marginal farms and kept the subsequent generation attached to agriculture.

The third literature is the economic history of the Great Depression. A central theme in this literature is how shocks and New Deal programs reshaped farm structure and rural credit markets. Alston (1984) documents the scale of farm mortgage distress and the political economy of foreclosure interventions. Other work emphasizes how credit availability and banking institutions mediated the Depression and influenced structural change (Kislev and Peterson, 1982; Depew et al., 2013; Rajan and Ramcharan, 2016; Hausman et al., 2019; Hornbeck, 2023), while related studies show that agricultural policies had enduring impacts on labor structure and land use (Depew et al., 2013; Hausman et al., 2019; Kislev and Peterson, 1982). Environmental shocks such as the Dust Bowl also altered farm structure, migration, and long-run adjustment ((Hornbeck, 2012, 2023)). We study one of the few large interventions outside national New Deal programs and provide new evidence on how crisis-era legal relief shaped both agricultural productivity and the transition into manufacturing.

**Roadmap.** The paper is organized as follows. **Section II** describes the institutional setting, the farm crisis and the legal design of the moratoria. **Section III** describes the data and empirical strategy. **Section IV** presents our results on farm counts, farmers, farmers’ children and farmer migration. **Section V** reports our results on agriculture and manufacturing productivity. **Section VI** studies local credit markets. **Section VII** concludes.

## II Historical and Legal Context

### II.A Agricultural Distress 1920—1933

The First World War shifted relative prices in favor of American agriculture. European disruption boosted demand for U.S. grain and cotton as average crop prices doubled between 1915 and 1919. At the same time, land values rose by more than 50 percent between 1910 and 1920 (Rajan and Ramcharan, 2015). Farmers responded by expanding acreage and buying more machinery. This expansion was financed by mortgages that required interest-only payments during the term and a large balloon payment at maturity. Outstanding farm-mortgage loans peaked in 1924, when

the ratio of mortgage debt to value per acre approached 300 percent ([Horton et al., 1942](#)).

However, once European supply recovered in 1920, commodity prices collapsed. This transformed the nominally fixed mortgage obligations into a heavy burden ([Messer and Rieder, 2025](#)). As farmers faced lower cash flows, repayment for loans issued near the 1919 peak was very difficult. Involuntary transfers of farm real estate rose sharply after 1922, with foreclosure serving as the primary means of involuntary transfer rather than bankruptcy, which was much more expensive ([Messer and Rieder, 2025](#); [Wickens, 1936](#)). We plot these trends in [Figure 2b](#).

The agricultural sector was already weak by the time of the Depression, which then multiplied the damage ([Bernanke, 1983](#)). The Dust Bowl, starting as early as 1930, imposed a highly uneven environmental shock that lowered yields, values, and population in the worst-affected areas ([Hornbeck, 2012](#)). Farm foreclosures reached 96,000 per year between 1921 and 1940, peaking near 200,000 in 1933 ([Alston, 1983](#)). States passed foreclosure moratoria between 1933 and 1935.

## II.B Design and Operation of the Moratoria

The U.S. legal system, until that point, was firmly against debtor relief. The Contracts Clause (Art. I, §10) was interpreted to prohibit moratorium-style legislation. Chief Justice John Marshall warned that state interference with contracts “destroy[ed] all confidence between man and man” ([Poteat, 1938](#); [Prosser, 1934](#)). From the Civil War through the Panic of 1893, state attempts to delay or suspend foreclosures were repeatedly invalidated ([Anonymous, 1939](#)). Other tools for debtor relief such as postponing foreclosures, limiting deficiency judgments and extending redemption periods had met the same fate.

The Depression changed the calculus. Between 1929 and 1933, personal income fell by nearly half and unemployment reached 25 percent ([Wheelock, 2008](#)). Farm mortgage debt stood at roughly \$7.7 billion, equaling a quarter of farm real estate value in 1933 ([Hausman et al., 2019](#)). In Minnesota, half the land was mortgaged and prices fell below production costs ([Prosser, 1934](#)). Farmers disrupted foreclosure sales and marched on state capitols. From 1933 to 1935, 25 states enacted foreclosure moratoria ([Wheelock, 2008](#)). Adopting states were more likely to have higher farm foreclosure rates, a higher percentage of mortgaged farms, and greater reliance on private credit markets ([Alston, 1984](#); [Wheelock, 2008](#)). Political control does not appear to be a predictor of adoption. [Figure A1](#) shows that moratoria were passed by state legislators under Democratic, Republican and divided control at roughly equal propensity. Federally held loans, such as those administered through federal land banks, were less likely to result in foreclosure. After 1933, the FCA became a dominant force while life-insurance companies reduced their exposure to farm mortgages by half ([Wall and Cone, 1937](#)).

This distress paved the way for states to adopt the moratoria. Foreclosure procedures varied widely across states. We give a stylized overview in [Figure 5](#). The moratoria added to this baseline by pausing sales or extending redemption and, commonly, by conditioning any stay on payments so the creditor received some compensation during the hiatus. The laws did not permanently re-write debts but instead delegated to courts the power to grant temporary, conditional relief case by case. The most common tools were judicial stays of foreclosure and extensions of statutory redemption periods. In New York, for example, foreclosures on pre-1932 mortgages were suspended for principal-only defaults so long as the borrower kept current on interest and taxes ([Wheelock, 2008](#)). Minnesota’s 1933 statute authorized courts to extend redemption as late as May 1, 1935, and required use-and-occupancy payments during the extension.

The Supreme Court’s decision in *Home Building & Loan Ass’n v. Blaisdell* (1934) upheld Minnesota’s law and emphasized that emergency conditions combined with the temporary and limited nature of relief, fair protection for creditors, and ongoing judicial supervision made the policies legal under the Contracts Clause. In *Blaisdell*, the borrowers were allowed to remain upon paying \$40 per month, applied first to taxes and upkeep and then to the debt holder ([Prosser, 1934](#)). Crucially, not all such policies survived scrutiny. In *W.B. Worthen Co. v. Kavanaugh*,<sup>1</sup> the Supreme Court invalidated Arkansas’s 1933 statutes because the law gutted a mortgagee’s security for many years which allowed debtors to retain property without paying taxes, interest, or rent and without showing hardship. Finally, the moratoria remained constitutionally valid only as long as the emergency lasted. For instance, in 1938, the state supreme courts in both Nebraska and Mississippi ruled that the emergency had ended and stopped the moratoria ([Poteat, 1938](#)).

Consequently, the moratoria prevented foreclosures only at the margin. Courts granted relief only to debtors that had a plausible path to recovery and weren’t “hopelessly insolvent” owners who did not have a “reasonable probability” of saving the land ([Poteat, 1938](#)). Most statutes came with a requirement of partial repayments which meant that farmers with no hope of survival received no protection.

Although the moratoria preserved farmer occupancy, they did not guarantee retention of legal ownership. The original statutes and judicial opinions clearly distinguished between possession and the final transfer of title. Minnesota’s statute left the foreclosure sale valid, meaning the lender became the purchaser and was positioned to obtain full title if redemption failed. The statute simply extended the redemption period which allowed the debtor to retain physical possession during the interval as long as payments were made. The Supreme Court explicitly noted that the statute “does not affect the validity of the sale or the right of a mortgagee-purchaser to title in fee” if redemption ultimately failed.<sup>2</sup> In essence, legal title returned to borrowers only if debts were fully redeemed or refinanced within the statutory window.

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<sup>1</sup>295 U.S. 56 (1935)

<sup>2</sup>*Home Building & Loan Assn. v. Blaisdell*, 290 U.S. 398, 425 (1934)

Therefore, we can confidently hypothesize that the policy ought to increase the number of farms at the county level but the implications for ownership structure are ambiguous. Since the laws preserved possession without guaranteeing title, the same farm could appear as owner-operated, lender-owned with the former owner in place, or effectively tenant-operated in the Census. Lender behavior also muddies these categories, as creditors often bought at sale, deferred possession, leased back, or negotiated informal workouts. We expect a clear effect on the count of farms, but the composition among owner-operators, tenants, and manager-operated farms can plausibly move in either direction.

## **II.C Bridge to the New Deal**

Beginning in mid-1933, the federal government intervened directly in farm markets. The Agricultural Adjustment Act (AAA) paid farmers to plant fewer acres, which increased farm income by more than 50 percent by 1935. AAA rental and benefit payments accounted for about a quarter of that improvement. The Roosevelt administration also devalued the dollar, which made American crops cheaper for foreign buyers and pushed commodity prices up almost immediately in spring 1933. Counties where crop prices rose the most saw faster recovery, and the effect was largest where farmers carried the heaviest debt burden ([Hausman et al., 2019](#)).

At the same time, the Farm Credit Administration started to refinance debt. From 1933 to 1939, the Federal Land Banks alone held 48 percent of the total farm mortgage debt, up from a peak of just 9 percent in 1922 ([Rose, 2013](#)). The Land Bank Commissioner made roughly 680,000 loans. By March 31, 1940, the FLBs and LBC together had nearly \$2.7 billion of farm mortgages outstanding against total farm-mortgage debt of about \$12 billion. Although a third of all FCA loans became delinquent, only 7 percent were ultimately foreclosed before 1944 ([Rose, 2013](#)).

These efforts by the federal government made the effects of the moratoria persistent. The moratoria acted as a bridge, helping farmers preserve title until prices had rebounded from the 1932 lows and the federal government had refinanced debt.

Finally, we want to emphasize that foreclosure remained the primary channel for distressed land transfers. Between 1930 and 1934, farmer bankruptcies ran at 0.75 per 1,000 farms, while foreclosure rates ranged from 20 to over 35 per 100,000 ([Messer and Rieder, 2025](#)). In 1934, only 4,464 farm bankruptcies were recorded nationwide. Foreclosure could be initiated by a single creditor, did not stay collection on other debts, and culminated in a sale if redemption failed. Despite the institutional changes in bankruptcy law, foreclosures remained the primary involuntary method for land reallocation during the Depression ([Wickens, 1936](#)).

The Frazier–Lemke Farm Bankruptcy Act, passed in 1934, allowed indebted farmers, after bankruptcy had been adjudged, to retain possession for up to five years while paying a court-set

“reasonable rental,” with the right to repurchase at appraised value over six years at one percent interest. However, this law was unanimously struck down by the Supreme Court.<sup>3</sup> Congress passed a revised version in 1935 which reduced the moratorium to three years, included a formal declaration of emergency, and allowed public sale under certain conditions with farmer redemption rights. The revised law was upheld.<sup>4</sup> These reforms offered farmers new legal tools but never displaced foreclosure as the main channel for distressed land transfers, especially during the 1930s.

### III Data and Empirical Strategy

This section describes the data and research design. We code state foreclosure moratoria, proxy pre-policy exposure using the 1930 county mortgage share, and construct both county-level and individual-level outcomes. We identify causal effects through a triple-difference design—exploiting variation in policy adoption, pre-existing exposure, and time—supplemented by border-county comparisons that improve covariate balance. [Table 1](#) displays summary statistics for our primary analysis variables at the county level in 1930.

#### III.A Variable Construction

##### Treatment: Foreclosure Policies

Our main treatment is a state-level indicator for whether a state enacted a foreclosure moratorium during the Depression-era rollout of 1933–1934. We follow [Alston \(1984\)](#) and build on a contemporaneous USDA report titled *State Measures for the Relief of Agricultural Indebtedness in the United States, 1933 and 1934* ([Olcott and Bercaw, 1934](#)) and the 1936 version of the same report.

The report covers measures aimed at preventing farmers from losing real estate through mortgage foreclosure or tax sale – including sale moratoria, fair-value deficiency rules, extended redemption periods, and tax-sale relief – while explicitly excluding programs that extend new credit. We code each state’s law according to these criteria. Using these excerpts, we produce [Table 2](#), which summarizes the key features for each state: year of passage, length of the moratoria, and whether the law included a stay of foreclosures, extended the redemption period, or limited a lender’s right to sue farmers for remaining debt after foreclosures. In total, 25 states adopted a moratorium. Our results are robust to alternative codings of three borderline cases, which we discuss in [Appendix B](#), and the full legal texts appear in ??.

<sup>3</sup>*Louisville Joint Stock Land Bank v. Radford*, 295 U.S. 555 (1935)

<sup>4</sup>*Wright v. Vinton Branch of Mountain Trust Bank of Roanoke*, 300 U.S. 440 (1937).

## Exposure: Farm Mortgage Adoption

We use county-level data on farming activity from [Haines et al. \(2018\)](#), who compile the U.S. Agricultural Census over many years. The data are based on farm-level enumeration, but only the county aggregations are systematically available. We collect data at decennial intervals beginning in 1900, with additional data available every five years from 1925 until we end our sample in 1960. This gives a broadly symmetric 30-year window on either side of the policy change, allowing us to examine pre-trends and longer-term outcomes.

Our exposure measure is the county share of farms with mortgage debt in 1930, which proxies for the extensive margin of farms at risk of foreclosure. The share of farms under mortgage is assessed as the proportion of returns indicating non-zero mortgage debt on farm land or buildings. Strictly speaking, respondents could report mortgage debt on a farm located in a different county, though wording revisions introduced before 1925 made the question more specific to the enumerated property. We therefore treat responses as referring to the farm being counted. Farmers were asked to report both mortgage repayments and debt (in dollars) in 1920, 1930, and 1940, which we use to compute the mortgage interest rate.

We prefer the mortgage share over debt-to-value ratios as the measure of exposure because the denominator of the ratio—land values—was typically assessed at boom-era prices that collapsed sharply during the 1920s. As distressed farms sought refinancing, these ratio measures became volatile over time, making them a noisier proxy for foreclosure exposure than the simpler extensive-margin indicator.

In [Figure A2](#) we show the predictors of the mortgage share in the 1930 cross-section of counties. High-mortgage counties have larger farms, more tenant farmers, and are generally more valuable in our measures of machinery, crop value and farm value per acre. Lower interest rates are also a bi-variate predictor of mortgage adoption, but not a multivariate predictor, suggesting that lenders engage in risk pricing beyond the observables we control for.

## Outcomes

**Farms, Farm Values and Farmland** Our most direct outcome is the number of farms in a county, also from the U.S. Agricultural Census. Crucial to our analysis is the definition of a farm. From 1925, this definition was standardized as follows: A tract or tracts of land directly farmed by one person (a “farmer”), either alone or with household members and/or hired labor. Each separate tract operated by “tenants, renters, croppers, or managers,” even under the same land owner, is counted as a distinct farm. Thus, by definition, the number of farms equals the number of farmers. This definition matters for our measurement: foreclosure transfers legal title but need not interrupt the farm’s operation if the new owner retains the same tenant or continues farming

directly. Farm counts, therefore, capture the number of distinct operators, but may undercount ownership changes.

A farm must additionally be of at least three acres or produce at least \$250 worth of produce. These criteria differed in enumerations prior to 1925, when farms smaller than three acres were counted as long as they required the full-time labor of one farmer. The exclusion of these small, subsistence, or lifestyle farms from the 1925 definition onward resulted in a net reduction in the number of farms. Year-fixed effects capture the national-level decline in farm counts arising from this definitional adjustment. To the extent that this definition change had differential regional impacts, the decrease in farm counts was largest in the Northeast, the region least important for the agricultural crisis and foreclosure policy.

Farmland is defined as all land under the farmer’s control, whether or not it is under cultivation, but excluding land under the control of a tenant or cropper. Farm values are collected to include the farmer-reported value of all land and buildings under normal (as opposed to stressed) conditions; importantly, these are not market transaction or foreclosure-sale prices.

**Linked Census Outcomes for Adults** We use individual-level census microdata from the U.S. Decennial Census, as provided by [Ruggles et al. \(2024\)](#). In particular, we collect data on age, family structure, sector and occupational choices, schooling, and residence (including living on a farm). The linking procedure of [Ruggles et al. \(2025\)](#) connects individuals across censuses to build the Multigenerational Longitudinal Panel, which we study for the censuses from 1910 to 1950. Their method incorporates ancillary genealogical data from the Social Security Administration, Ancestry.com, and FamilySearch to enable the linking. Although [Ruggles et al. \(2025\)](#) have substantially improved linkage quality for women, match rates remain lower than for men in this era. We therefore focus on men to minimize attrition bias in our linked sample. We regard this as a conservative choice and recognize that women’s labor market and mobility responses may differ.

With this sample, we construct occupational transitions and migration among workers. We define occupational transitions (migration) at the county level as the share of individuals from a given occupation (county) in census  $t - 1$  who are still in the same occupation (county) in census  $t$ . Both occupational-retention and county-retention rates are computed only over individuals successfully linked across census waves.<sup>5</sup>

The census provides several ways to identify farmers: occupational code, industry code, and farm residence. The latter provides the broadest definition, as it captures the adult household of a farmer who may provide farm labor but is not formally counted as employed. Under

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<sup>5</sup>Given that our main treatment is at the county level, we lose nothing by collapsing this data to the county level for the regression. We can achieve similar results at the individual level with appropriate weighting and clustering.

occupational codes, the census distinguishes between *farm owner*, who may be full or partial owners of the land or tenant farmers, and *farm workers*, who are hired labor on farms but not themselves farmers in the sense of the agricultural census definition discussed above. We use all three definitions, as they capture subtly distinct economic content.

**Linked Census Outcomes for Children** We focus in particular on the children of farmers to understand the effect of foreclosure policy on farm retention. We take the 1930 census and identify children who live on farms. We link these children to their census returns as adults in 1950.

We restrict this link to children who do not migrate from their county of birth, to clarify that observed differences in outcomes arise from the 1930 county rather than a potentially different 1950 county. Moreover, this restriction helps alleviate concerns about the nature of selective attrition from the linked sample under migration, which is hard to observe. This measurement choice restricts the estimand, such that our child-outcome estimates pertain only to individuals whose adult outcomes remain tied to their childhood county environment.

To assess robustness, we also use an alternative measure of migration. In 1940, the census asked respondents to recall their migration history: in which county did they live five years ago? The variable is commonly used in the migration literature ([Derenoncourt, 2022](#); [Hornbeck, 2023](#)). This provides an alternative approach to understanding migration without relying on establishing successful links between censuses, which may be compromised by differential attrition under migration.<sup>6</sup>

## Other Data

**Manufacturing** We access county-level data on the number of manufacturing establishments, wage earners, total wages, and value added from the Census of Manufacturers. The data are compiled by [Hornbeck \(2012\)](#) and supplemented with additional years from the ICPSR historical county data series ([Haines et al., 2010](#)). Where definitions have changed over time, we harmonize the data in accordance with ([Hornbeck and Keskin, 2015](#)).

**New Deal and Dustbowl** To understand how our policy interacts with other important forces in 1930s agriculture, we use data from [Hornbeck \(2012\)](#) on the extent of dust-bowl erosion (high, medium, and low) in each county and the total expenditure of various New Deal programs, originally collated from the Office of Government Reports in [Fishback et al. \(2005a\)](#).

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<sup>6</sup>The MLP procedure of [Ruggles et al. \(2025\)](#) under-represents lifetime migrants in the linked sample by approximately 15% (4 percentage points) relative to the full census.

**Geographic Data** [Schroeder et al. \(2025\)](#) provides GIS files for US counties in each decade of our sample. We use the method of [Eckert et al. \(2020\)](#) to generate an area-weighted crosswalk from each county-year to a baseline set of 1930 counties, which we use as the consistent unit of analysis. This entails that whenever a county boundary differs from the 1930 definition, we compute the area of intersection with the 1930 county and re-aggregate the data from the altered county back into the 1930 unit in proportion to the overlap.<sup>7</sup>

The majority of county border changes in our sample are concentrated in the West, Southwest, and Texas. We exclude the independent cities of Virginia, Hawaii, and Alaska, as well as U.S. territories, from our analysis.

### III.B Empirical Strategy

We aim to identify the effects of foreclosure laws on farmers and county economies over time. Our empirical strategy is based on the idea that moratoria should matter most where more farms were ex ante exposed to foreclosure risk. Since county-level data on foreclosure rates are unavailable, we proxy for that exposure using the 1930 mortgage share. [Alston \(1983\)](#) reports a state-level correlation of 0.65 between foreclosure incidence and the farm mortgage share, supporting our use of the mortgage share as a proxy for pre-policy foreclosure risk. [Figure 1](#) compares the spatial distribution of foreclosure rates in 1932 (at the state level) to mortgage shares (at the county level) in 1930. The overall correlation reported above is striking, while sufficient county-level variation in mortgage rates creates the opportunity to make even tighter within-state comparisons. However, because high-exposure counties in treated states are not randomly located, we cannot rely on simple treated-versus-untreated comparisons.

**Treatment Balance** The central challenge in this setting is the non-random assignment of the foreclosure policy across states and, hence, counties, as shown in column 1 of [Table 3](#), which examines balance in the 1930 cross-section. Counties treated by a moratorium have a larger share of farms under mortgage, are larger in population, and are more agricultural. Treated counties have roughly 20% more farms, and those farms are roughly 40% larger in area. These factors align with the political drivers of policy adoption documented by [Alston \(1984\)](#).

In column 2, we include indicator variables for each decile of the mortgage share distribution to assess the validity of comparisons between counties with similar mortgage shares. This approach mechanically captures nearly all differences in the mortgage share exposure, but generally fails to mitigate imbalance in other observables within the 1930 cross-section. Moratoria adoption is an endogenous political choice that reflects the agricultural characteristics of the

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<sup>7</sup>This assumes a uniform spatial density of the data within the county.

marginally viable farmer, not only the size of this margin.

**Triple Difference Design** This empirical problem lends itself to a triple difference design (Gruber, 1994). It nests three sources of variation: (1) variation over time, (2) variation across states induced by the policy changes, and (3) variation across counties in their baseline mortgage shares. Our analysis also uses a rich set of fixed effects, including state–year fixed effects. We expect the largest effects in counties that received the foreclosure moratorium and had the highest share of mortgaged farms in 1930. The standard triple difference estimator is  $\beta_0$  in the regression:

$$\begin{aligned}\ln(Y_{ct}) = & \beta_0(D_s \times \ln(M_{c,1930}) \times \mathbb{I}_{t>1930}) \\ & + \beta_1(\ln(M_{c,1930}) \times \mathbb{I}_{t>1930}) \\ & + \eta_c + \sigma_t + \chi_{st} + \epsilon_{ct}\end{aligned}$$

where  $s$ ,  $c$ , and  $t$  index state, county, and time, respectively.  $D_s$  indicates that the state adopts a moratorium law.  $\ln(M_{c,1930})$  is the log of the share of farms with mortgages in 1930 in the county.  $\mathbb{I}_{t>1930}$  indicates the post-treatment period. In words,  $\beta_0$  measures whether, after 1930, the within-state gradient of outcomes with respect to baseline mortgage exposure is steeper in moratorium states than in non-moratorium states.

The post-1930 indicator denotes a reduced-form exposure period that begins after 1930. Although the underlying moratoria were temporary, this specification is intended to capture both their immediate effects and any persistent consequences operating through subsequent re-financing, debt restructuring, or farm-support programs. The single post-period indicator is also practical: census observation years are sparse, and the laws were passed within a tight two-year window (1933–1935), making a more granular rollout design difficult to implement.

We include county, year, and state-year fixed effects. As a result,  $\beta_0$  is identified from within-state, within-year differences across counties with different baseline mortgage exposure, comparing how that exposure gradient changes after 1930 in moratorium states relative to non-moratorium states. The inclusion of these fixed effects makes the inclusion of  $D_s$ ,  $\ln(M_{c,1930})$ , and  $\mathbb{I}_{t>1930}$  and other lower-order interactions redundant.

Standard errors are clustered at the state level to account for potential observed shocks at the level of treatment assignment.<sup>8</sup> Regressions are weighted by 1930 farmland area, such that the estimand has the interpretation as the effect on the average unit of agricultural land.

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<sup>8</sup>Olden and Møen (2022) argue that treatment unit-clustered standard errors are appropriate for triple-difference designs in most applications, while Ortiz-Villavicencio and Sant’Anna (2025) show that efficiency gains are possible using doubly-robust estimators. Our results remain significant at standard levels when we adopt county-level clustering.

The identifying assumption requires that the relative outcome of high and low-mortgage share counties in the treated states trend in the same way as the relative outcome of high and low-mortgage share counties in the control states in the (counterfactual) absence of treatment (Olden and Møen, 2022). Note that this does not require two sets of parallel trend assumptions, despite the triple difference being interpretable as the difference of two double difference (DD) estimates. The DD parallel trends assumption can be violated, provided that the violation is stable across groups (Ortiz-Villavicencio and Sant’Anna, 2025). We provide evidence consistent with the required stability of these parallel-trend violations below.

*Testing Assumptions.* We complement the cross-sectional balance analysis with pre-trend tests for the identifying restriction underlying the DDD design, following Lin and Peri (2025). In our setting, this restriction implies two diagnostic implications.

**Assumption I:** Before foreclosure moratoria are adopted, the treated-minus-control difference must evolve similarly across high- and low-mortgage-share counties.

**Assumption II:** Before foreclosure moratoria are adopted, the high-minus-low mortgage-share difference must evolve similarly across adopting and non-adopting states.

We analyze the pre-trends corresponding to these restrictions in Figure 6 for two outcomes: log value per farm (Figure 6a) and log number of farms (Figure 6b). Our specification follows the standard TWFE event study specification and is estimated separately for each subgroup (indicated by color). For the dynamic specification, we discretize exposure using a top-decile indicator ( $\geq 63\%$  mortgage share) to improve visual interpretability and retain power.

The upper figure in each panel corresponds to Assumption I. We plot the time-varying effect of the moratoria treatment for each group. The lower panel corresponds to Assumption II, which tests for the effects of the 1930 mortgage share. Although we observe some pre-trends, the differences between groups are broadly parallel until after the treatment, which reassures us that the identifying restriction is valid.

*Other specifications.* In presenting our results, we use a standard dynamic triple difference to complement our static specification:

$$\begin{aligned} \ln(Y_{ct}) = & \sum_{\tau \neq 1930} \beta_{\tau} (D_s \times \mathbb{I}_{M_c > p90} \times \mathbb{I}_{t=\tau}) \\ & + \sum_{\tau \neq 1930} \gamma_{\tau} (D_s \times \mathbb{I}_{t=\tau}) \\ & + \sum_{\tau \neq 1930} \theta_{\tau} (\mathbb{I}_{M_c > p90} \times \mathbb{I}_{t=\tau}) \\ & + \eta_c + \sigma_t + \chi_{st} + \epsilon_{ct} \end{aligned}$$

As an additional specification, we compare our triple-difference results with a standard double-difference specification estimated in a sample restricted to counties in the top quartile of the 1930 mortgage-share distribution. This has two benefits: allowing for a nonlinear effect of mortgage share in the most exposed counties and making effect sizes easier to interpret.

**Improving Comparison Validity** The DDD design compares how post-1930 outcomes vary with baseline mortgage exposure in moratorium states relative to the corresponding exposure gradient in non-moratorium states. It is a continuous-exposure policy design: identification comes from differences in post-treatment outcome slopes with respect to pre-treatment exposure, not simply from treated-versus-untreated level differences. Recent work on exposure-based designs emphasizes that such comparisons remain most credible when treated and untreated units have good overlap in baseline characteristics at a given level of exposure, since differencing alone does not ensure support or rule out selection on exposure (Callaway et al., 2024; Borusyak et al., 2025).

Table 3 shows that this concern is empirically substantial in our setting: in the full sample, counties in moratorium and non-moratorium states remain observably different even after flexibly conditioning on mortgage exposure. We therefore complement the baseline DDD with a border-county specification that restricts comparisons to geographically contiguous counties on either side of a treatment border. This restriction targets the same underlying exposure contrast on a more local support.

To demonstrate the utility of these tighter geographic comparisons, column 3 of Table 3 restricts the sample to the 476 counties adjacent to a state border where the treatment status differs. The location of these counties is shown in Figure 3b. Geographic contiguity helps address nearly all observable imbalances in treatment assignment. Moreover, this contiguity is particularly useful for addressing (potentially unobservable) differences in soil and climatic characteristics that are important for agricultural outcomes. While one could control for some of these characteristics directly, Ortiz-Villavicencio and Sant’Anna (2025) argue that covariate adjustment in DDD designs can alter the identifying comparison in nontrivial ways. Thus, we treat the border restriction as a design-based way to improve comparability rather than relying solely on regression adjustment.

There are two potential drawbacks to the border county design. First, border counties might not be representative of the full set of US counties over which we want to make inferences. In Figure A3, we examine observable representativeness of the border county sample. We find that unconditionally, within-state and with mortgage exposure controls, that these counties are broadly representative across our key co-variates. Border counties have approximately 10% larger farms and are 10% less machine intensive, but are balanced on population, the credit environment, and

number of farms with the full sample.

The second potential drawback is that shorter geographic distances increase the plausibility of factor mobility between units, which yield SUTVA violations. We acknowledge this possibility is acute for labor. In all our key results, we compare the triple difference design across the full sample and in the border sample, and generally find the compression of the effect size (which would be the effect of factor mobility) to be minor.

Column 4 of [Table 3](#) adds a set of indicators to the border sample specification for each specific border segment. For example, all counties along the Arizona/New Mexico border are included in a single indicator, so our treatment-control comparison is made within this fine-grained set of counties. There are 46 distinct state-border pairs among the 25 states that have adopted moratoria. This approach further controls the likely unobservable differences, following the minimum wage literature, which extensively uses state-level legislative changes ([Allegretto et al., 2017](#)).

The outstanding imbalance is in the agricultural mortgage interest rate. The specification in column 4 indicates that, even in neighboring counties, farmers faced interest rates approximately 3% lower in states that adopted foreclosure policies. This is evidence that, even while other features are held constant, banking conditions were highly local.<sup>9</sup>

We summarize the treatment balance visually in [Figure 4](#). In summary, we find that treated counties are observably different, and conditioning on mortgage-share exposure does not solve the imbalance. We therefore present border-county estimates alongside full-sample DDD estimates throughout, treating geographic contiguity as a design-based robustness check rather than an alternative identification strategy.

## IV Effects on Farms, Farmers, and their Children

This section discusses how foreclosure moratoria reshaped the allocation of households to farming. Moratoria insulated incumbent farmers in counties experiencing severe mortgage distress. This protection had two immediate consequences. First, it increased the number of operational farms and slowed occupational reallocation away from agriculture. Second, these initial dislocations had significant intergenerational consequences for the children of farmers. This generation was much more likely to stay in agriculture instead of transitioning to manufacturing. Finally, we find small and unstable migration effects.

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<sup>9</sup>Interstate banking was highly restricted until at least the 1970s ([Jayaratne and Strahan, 1996](#)).

## IV.A Number of Farms

We begin by examining the most direct consequence of foreclosure moratoria—the number of farms in a county. Event-study estimates indicate stable pre-trends and a large and sustained increase in farm counts beginning in 1940 (Figure 7a). Given that most moratoria were enacted in 1933, it is plausible that their effects took time to materialize. Additionally, as discussed in Section II.C, New Deal agricultural policies introduced after 1935 likely reinforced and extended these effects, explaining the absence of an immediate impact in 1935.

The event study plots rely on the DDD specification, which estimates whether the rise in farm counts in treated states and in counties that were more exposed ex ante (higher mortgage shares), beyond (i) generic post-1930 changes and (ii) any differential post-1930 changes in high-mortgage places that occurred everywhere. The triple interaction is the key parameter of interest, and including lower-order interactions eliminates policy-only and exposure-only post-treatment trends.

The double-difference (DD) estimator restricts the analysis to counties in the highest quartile of mortgage exposure and directly compare the most severely treated counties with both untreated and less-treated counties. The estimate is 0.133 and is significant at the 1 percent level in the full sample. In the border county specification (DD), it ranges from 0.182 to 0.287 (Table 4). These coefficients imply roughly 15%–30% higher farm counts in the top quartile of mortgaged counties in moratoria states after the policy. For the median county in this top quartile, this would translate to roughly 300 more farms.

The DDD specification utilizes continuous variation in exposure. We estimate a coefficient of 0.231 that is statistically significant at the 1% level (Table 4). Since both the dependent and independent variables are in log form, this coefficient can be interpreted as an elasticity. Counties with a 10% higher baseline mortgage share experienced approximately a 2.3% greater increase in farm numbers. The border-county specification yields even larger coefficients (ranging from 0.369 to 0.397).

From 1930 to 1940, the total number of farms in the U.S. declined from 6.3 million to 6.1 million; the trend was one of farmers leaving agriculture. The effects of this policy mitigate the overall decline in the number of farms during this period by approximately 125,000–262,000 in the top quartile of counties.

These results are not driven by any single region. In Table A3, we estimate the triple differences specification while individually excluding Census Bureau Regions (Northeast, Midwest, South Atlantic, South, and West). The coefficient on the number of farms is largely stable across these specifications. The largest deviation from the baseline coefficient occurs when we exclude the South which is characterized by a greater emphasis on cotton and sharecropping and also a

lower reliance on mortgages.

A natural concern with this analysis is that concurrent New Deal programs could drive the observed impact of the moratoria. We control for county level exposure to major New Deal programs—Agricultural Adjustment Administration (AAA) benefit payments and Farm Security Administration (FSA) credit to farms, as well as Works Progress Administration (WPA), Civilian Conservation Corps (CCC), and Public Works Administration (PWA) relief and public-works spending—using the county-level series assembled by [Fishback et al. \(2005b\)](#). We fully interact each of these variables with our treatments. This leaves the DDD estimate essentially unchanged ([Table 4](#), Column 5).

We also test the sensitivity of our results to the Dust Bowl using erosion severity data from [Hornbeck \(2012\)](#). By interacting county-level Dust Bowl severity indicators with our treatments, we confirm that our key coefficient remains robust and statistically significant ([Table 4](#), Column 6). Counties experiencing the most severe erosion show a substantial reduction in the number of farms. Counties with medium erosion exhibit no meaningful change. Together, these tests indicate that neither New Deal spending patterns nor Dust Bowl dynamics are driving the farm-count results.

## IV.B Individual Farmers

The increase in farm counts corresponds directly to an increase in the number of farmers. Using individual-level data from the U.S. Census linked between 1910 and 1940, we find that moratoria significantly slowed the reallocation of labor out of agriculture.

The coefficient of interest for remaining in agriculture by 1940 is 0.04, while the coefficient for transitioning into manufacturing is  $-0.007$  ([Table 7](#)). Moving from the median mortgage share (approximately 43%) to the 90th percentile (approximately 63%) corresponds to roughly 0.8 percentage points greater agricultural retention and 0.14 percentage points fewer exits into manufacturing in moratorium states. Border-pair estimates exhibit the same directional effect even though it is less precisely estimated.

These occupational effects are persistent. After linking farmers from 1930 to 1950, we find stronger agricultural retention in Panel A of [Table A1](#). The coefficient for remaining in agriculture increases to 0.058 and remains statistically significant at the 1% level. Moving from the median to the 90th percentile of mortgage share implies approximately 1.1 percentage points greater agricultural retention and 0.36 percentage points fewer exits into manufacturing by 1950.

These results are driven by keeping people in the agricultural sector broadly rather than preserving their specific status as farm owners. The coefficient on the probability of remaining a farm owner is small and not statistically significant (Column 3). The same is true for the county-

level results ([Table A2](#)). This is consistent with the institutional design of moratoria. The policy allowed judicial stays and extended redemption periods which lowered the foreclosure threshold and kept leveraged farms in operation even if it didn't preserve ownership. The near-zero effect on owner retention indicates that the policy mainly preserved agricultural employment rather than legal title.

We conduct a placebo test by replicating our empirical design in a period entirely unaffected by the policy. We estimate occupational transitions from 1920 to 1930 and present the results in Panel B of [Table A1](#). The coefficients for remaining in agriculture, continuing farm ownership, and transitioning from agriculture to manufacturing are all statistically indistinguishable from zero. These null findings are consistent across the border-pair specification as well.

#### **IV.C Intergenerational Effects of the Moratoria**

This section discusses how foreclosure moratoria shaped the economic trajectories of farmers' children. Using individual-level records from the 1950 Census linked to parental occupations in 1930, we ask whether children of farmers in counties with high pre-moratorium mortgage exposure, within moratorium states, followed different adult occupations than otherwise similar children.

We identify children living on farms in 1930, link them to their 1950 census returns, and restrict the sample to non-movers who remained in their childhood county between 1930 and 1950. Our analysis is at the individual level and controls for age, race, and county economic characteristics. The average age of these children is 6 years in 1930 and 26 by 1950.

Children of farmers in high mortgage counties in moratorium states are significantly more likely to be in agriculture in 1950 ([Table 8](#)). The triple-interaction coefficient is 0.071 in the parsimonious specification and 0.079 with controls. Both are significant at the 1 percent level. Moving from the median to the 90th percentile of mortgage share raises the probability that a farmer's child works in agriculture by 5%.

This effect is additive to the baseline occupational inheritance among children of farmers. Children of farmers are 39 percentage points more likely to be in agriculture in 1950 and 13 percentage points less likely to be in manufacturing than other children. The triple interaction captures an additional policy-induced tilt toward agriculture beyond baseline intergenerational persistence.

To determine whether the additional farms we observe under moratoria are 1930 farms that remain in family ownership and operation, we follow the approach in [Haws et al. \(2025\)](#) to proxy for farm succession. Our outcome is an indicator variable indicating that a child remains in the same county, is employed in agriculture, and shares common neighbors (defined as shared

census returns within 50 observations) with their linked child return. The results in columns 3 and 4 show that the triple interaction coefficient is around 0.03. Moving from the median to the 90th percentile of the mortgage share distribution in a treated state increases by around 2.5% the probability that a child retains the family farm. These specifications also reveal no effect of farm succession on those who are not the children of farmers, which helps clarify that our measurement captures an inheritance channel.

The moratoria also reduced the entry of farmers' children into manufacturing. The triple interaction is negative and statistically significant in the parsimonious specification ( $-0.051$ ), but it becomes smaller and statistically insignificant with the inclusion of demographic and economic controls ( $-0.014$ ). This interpretation is supported by the large, significant coefficient on the interaction between the mortgage share and moratoria in both specifications. By delaying or preventing foreclosures, the policy directly retained marginal farm households and increased the likelihood that farmer children remained in agriculture, including on their family farms.

#### IV.D Migration

Foreclosure relief may change migration incentives in opposing ways. Suspending foreclosure can reduce distress migration by preventing displacement. But once the farm is saved, relief can loosen constraints and increase the option to leave agriculture by preserving housing wealth and reducing the immediate threat of eviction.

Contrary to popular imagination of this era, there was little net migration during the Great Depression. Net movement away from farms averaged about 630,000 per year in the 1920s but fell to roughly 195,000 per year between 1930 and 1935 and even reversed in 1932, when about 325,000 more people moved *to* farms than left them (Bernert, 1944; Wilse-Samson and Boone, 2021).

Table 15 reports DDD estimates using two migration measures. *Linked Migration* (1930–1940) measures whether an individual linked between the 1930 and 1940 censuses lives in a different county in 1940. *Migration* (1935–1940) uses the 1940 census “county of residence in 1935” question. A key advantage of the 1935–1940 measure is that it does not mechanically depend on the linkage procedure for migration status (Witse-Samson and Boone, 2021).

The results are modest. Moving from the median to the 90th percentile of mortgage exposure in moratoria states reduces the linked 1930–1940 migration rate by about 0.28 percentage points and the 1935–1940 rate by about 0.82 percentage points. These results are not stable across specifications. In the border-county comparisons (cols. (3) and (6)), the interaction flips sign and becomes imprecise. The evidence suggests that foreclosure moratoria do not generate large or robust changes in county-to-county migration, consistent with the broader Depression-era

pattern documented by [Wilse-Samson and Boone \(2021\)](#).

## V Effects on the Farm Economy

This section examines how foreclosure moratoria affected farm values, farm size, and cropland productivity. The moratoria prevented consolidation which kept farms smaller on average. Per-farm values fell sharply and persistently, driven entirely by this foregone consolidation. Farmers responded to distorted factor prices by expanding cropland, at the expense of pasture, and spreading a diminished capital stock across more acres. This led to a high-labor, low-capital farming model with lower yields per cropland acre. At the same time, the number of manufacturing establishments did not change, but we find suggestive evidence that output per plant fell modestly, consistent with reduced local demand from poorer, smaller farms.

### V.A Farm Values and Size

The moratoria kept farms smaller. The previous results confirm that the moratoria increased the number of farms significantly while county-level total farmland acres remained fixed. We estimate a treatment elasticity of  $-0.244$  for average farm size ([Table 6](#)). Total acres in farmland stays flat throughout our sample which implies that the increase in farm counts translates directly into a decrease in acres per farm. This foregone consolidation is the central mechanism behind the decline in per-farm values.

The triple-difference specification yields a treatment elasticity of  $0.107$  for the average value of farm land and buildings per farm ([Table 5](#)). Moving from the median to the 90th percentile of mortgage exposure corresponds to a decline of roughly 6–7 percent. The border-pair specification produces an even larger elasticity of  $-0.236$ . Double difference estimates that focus solely on the most exposed counties imply a roughly 20 percent decline in average farm values following the moratoria. Event study figures confirm no discernible pre-trends and a sharp, persistent fall in value after 1940, remaining persistent through 1960 ([Figure 8](#)).

Per-farm land value represents the discounted present value of expected net income from the farm as an operating unit. If markets had fully adjusted after the moratoria through consolidation, refinancing, or reallocation of land to more productive users, then per-farm values should have rebounded. Instead, the persistent decline suggests that the policy durably lowered expected returns on farm operations.

The decline in value per farm is not driven by a reduction in per-acre values. We find that the moratoria led to a slight but persistent *increase* in the average value of farm land and buildings per acre. The triple difference specification yields a positive treatment elasticity of  $0.156$  for the

average value of land and buildings per farmland acre (Table A4, Column 1). This pattern can be captured by the following accounting identity:

$$\log\left(\frac{\text{value}}{\text{farm}}\right) = \log\left(\frac{\text{value}}{\text{acre}}\right) + \log\left(\frac{\text{acres}}{\text{farm}}\right). \quad (1)$$

Because total farmland acres and total value of land and buildings both stay flat, the entire drop in *value per farm* reflects the loss of scale economies due to foregone consolidation.

We posit two explanations for the increase in value per farmland acre. First, the moratoria artificially inflated observed land prices by altering the type of transaction reported in the market. By legally preventing lenders from forcing distress sales, the moratoria effectively removed the left tail of the market’s price distribution (Campbell et al., 2011). The remaining observable sales used for appraisal were skewed toward voluntary and non-distressed transactions, mechanically driving up the average observed value per acre.

Second, the land economics literature finds that per-acre prices rise as parcel size falls. One study found that average price per acre of 40-160 acre farms is almost twice as high as per acre prices of farms greater than 640 acres for transactions recorded in 1971 (Brorsen et al., 2015). As farms get smaller, many improvements on the land (house, barn) are lumpy and partly fixed, so dividing by fewer acres raises per-acre value mechanically. This puzzle can also be partially explained by liquidity effects, higher road frontage per unit of area (Tsoodle et al., 2006), and increased potential for non-agricultural or “hobby” farming uses (Sengupta and Osgood, 2003).

## V.B Farm Productivity

We now discuss how the moratoria affected cropland productivity. Faced with higher interest rates and relatively abundant labor, farmers substituted away from capital-intensive production and toward labor-intensive production. The primary margin for this substitution was land allocation. We find that the share of farmland dedicated to cropland increases substantially in response to the moratoria. Our triple-difference specification yields a treatment elasticity of 0.252 (Table A7). For a county moving from the median (43%) to the 90th percentile (63%) of mortgage exposure, this implies a 10 percent increase in the share of farmland being used for crops.

This expansion of cropland came at the expense of capital-intensive uses such as pasture. Hornbeck (2012) finds that more-eroded counties with better access to banking immediately shifted land from cropland to pasture after the Dust Bowl. In contrast, there was no difference in shifting land from wheat to hay. The reduction in capital per acre is also consistent with the large reduction in the capital-labor ratio (Table A5).

Given this endogenous expansion along the cropland margin, examining outcomes *per farm-*

*land acre* would be misleading, as that would conflate the intensive margin (productivity on existing land) with the extensive margin (the addition of new, likely lower-quality, land). The correct metric is, therefore, per cropland acre.

We find clear evidence of lower capital intensity per cropland acre. The amount of machinery available per cropland acre decreases significantly (Table A6). Our primary state-year fixed effect specification yields an elasticity of  $-0.339$ , significant at the 1 percent level. We also see a reduction in physical capital. The share of farms with tractors decreases with an elasticity of  $-0.065$  (Table A10).

This combination of more land being worked with less capital had predictable negative consequences for output. We find suggestive evidence that the total value of crop sold per cropland acre fell with an elasticity of  $-0.235$  (Table A8). We find stronger evidence that the share of cropland acres actually harvested declined after the policy (Table A9). This decline can be directly attributed to an increase in crop failure or cropland being left idle.<sup>10</sup>

The decline in productivity per cropland acre reflects two complementary channels. Farmers spread their diminished capital stock too thinly over expanded cropland acres, leading to lower yields. And the newly activated cropland was of marginal quality, inherently less productive and more prone to failure.

The fall in productivity per cropland acre, combined with flat total farmland acres and flat total crop output, implies that farmers were cultivating a much larger but less productive cropland base just to produce the same total output. The moratoria locked in a high-labor, low-capital, low-yield farming model, cementing the inefficiency that ultimately drove the permanent decline in per-farm asset values.

## V.C Manufacturing

We measure manufacturing outcomes using the historical Censuses of Manufacturing, which report the number of establishments, wage earners, total wages, and value added. We present the manufacturing results using two standardized indices that capture the extensive and intensive margins since the underlying variables are noisy and strongly correlated. Each index is the first principal component of the underlying variables, scaled to have a mean of 0 and unit standard deviation for the pre-moratorium period. The underlying regressions for each component are similar to the index results (reported in the appendix).

The median county in 1930 had 17 manufacturing establishments employing 247 workers,

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<sup>10</sup>In the Agricultural Census, total cropland is the sum of acres harvested, failed, and idle. Beginning in 1950, the reported columns for failed and idle cropland appear to be data errors, reporting all zeros. Harvested cropland, however, remains complete. We therefore use “cropland failure” for 1925-1945 and “harvested share” (which moves inversely to failure and idleness) as a consistent measure throughout the sample (1925-1960).

paying \$206,000 in wages, and generating \$519,000 in value added. The bottom quartile had 6 or fewer establishments; the top decile had 90 or more. Seventeen percent of counties had no manufacturing. Among counties with at least one establishment, the median plant employed about 18 workers and paid roughly \$1,020 per worker.

We find a clear null on the extensive margin. In the preferred triple difference specification, the coefficient is indistinguishable from zero (Table A11). Event study estimates indicate no systematic divergence between high-mortgage counties in moratorium states and their comparison counties, and no evidence of pre-trends (Figure 10). The moratoria do not measurably change the number of manufacturing establishments, employment, payroll, or total value added in treated counties.

For the intensive margin index, we find suggestive evidence that the moratoria had a negative impact. In the full county sample, the triple difference coefficient is negative and statistically significant from 1920 to 1947. Moving from the median-exposure county to the top decile of pre-moratorium farm mortgage exposure is associated with an intensive margin index 0.05 standard deviations lower than that of otherwise similar counties, almost a decade after the moratoria expired (Table A13). The results are less strong when we extend the sample to 1957 (Table A12). The individual regressions indicate 5% fewer workers per plant, 6% lower payroll per plant, and 5% lower value added per plant, while wages per worker remain flat.

The event study for the intensive margin index shows the same pattern (Figure 10). Before the moratoria, treated and comparison counties moved in parallel. After implementation, the treated coefficients fall below zero and remain moderately negative through 1947. When we restrict the sample to border counties, the estimated coefficients shrink and become imprecise. By 1958, the coefficients are statistically indistinguishable from zero. Therefore, we treat the intensive margin evidence as suggestive. The moratoria may have temporarily weakened manufacturing activity within existing plants in counties with high mortgage exposure, but there is no sign of a large or permanent manufacturing collapse.

These patterns are consistent with the moratoria primarily distorting the agricultural sector and affecting manufacturing indirectly through reduced local demand for manufactured goods. Earlier results show that high-exposure counties remain characterized by smaller, less capitalized farms and lower land values, implying a persistent decline in demand for farm implements and locally purchased consumer durables. When per-farm wealth falls and credit is tight, rural households reduce purchases of locally produced durables and inputs, just as foreclosure-driven wealth losses depressed consumption during the Great Recession (Mian et al., 2015). Models of irreversible and lumpy investment show that when demand falls, firms optimally delay or forgo large, discrete capital upgrades even if they remain in operation, precisely the kind of intensive margin response that our data capture (?).

Classic development economics treated agricultural progress as a precondition for industrialization (Johnston and Mellor, 1961). Matsuyama (1992) challenges this view, arguing that comparative advantage and market integration can pull resources toward agriculture and away from manufacturing. Asher et al. (2022) find that large irrigation-driven productivity gains in India generate persistent increases in agricultural outcomes but do not translate into commensurate long-run increases in nonfarm employment, with structural transformation occurring largely through the growth of regional towns. In a setting very similar to ours, Hornbeck and Keskin (2015) find that technology enabling access to the Ogallala aquifer raised agricultural revenue but produced no long-run expansion in manufacturing, because higher farm income bids up local factor prices and crowds out non-agricultural activity.

Our results line up with that logic but in the opposite direction. The moratoria operate as a negative agricultural shock that reduces farm wealth, so the demand channel points toward weaker local manufacturing, while the cost channel points toward lower wages and land costs that could attract tradable manufacturing. Consistent with Hornbeck and Keskin (2015), we find essentially no response in manufacturing establishment counts, which suggests that these forces may roughly offset on the extensive margin.

## VI Effects on Credit Markets

The moratoria had a significant effect on local credit markets. Rural credit markets during this period were highly local due to both physical barriers and regulatory constraints. Most farmers did not own cars, and telephones were rare, so repeated in-person interaction with a lender was effectively a prerequisite for obtaining and maintaining credit (Rajan and Ramcharan, 2016). The Federal Reserve Act of 1913 explicitly limited national banks' ability to originate real-estate loans across district borders, and although the McFadden Act of 1927 modestly relaxed these constraints, interstate branching remained largely prohibited (Jayaratne and Strahan, 1996). State banks operated under similar de facto geographic limits. Following Rajan and Ramcharan (2016), we take "the banks within a county and in physically proximate counties" to be the relevant local market.

Within this locally segmented market, the foreclosure moratoria raised the cost of credit. Our baseline estimates in Table 11 indicate that by 1940, farm mortgage interest rates (as reported in the Agricultural Census) in the most-exposed counties were about 30 basis points higher than in unexposed counties. This is roughly a 4 percent increase relative to the sample mean interest rate of around 7.5 percent, and it is robust to our standard suite of specifications. The estimate is economically meaningful when set against the dramatic convergence in farm interest rates between 1930 and 1940. County-level rates were highly dispersed in 1930 but had mostly

converged by 1940 (Figure A4). Using the 1930 cross-sectional dispersion as a baseline, our estimates imply that foreclosure suspensions account for roughly 30 percent of the remaining gap in county-level interest rates in 1940.

The remainder of this section asks *why* a temporary suspension of foreclosures translated into a persistent spread. Using annual data on national banks, we first rule out a bank-distress channel. We then show that loan quantities do not respond to the moratorium. We then use a simple loan-pricing decomposition to show that neither the mechanical expected loss from future moratoria nor the observable deterioration of the borrower pool can account for the spread when calibrated at reasonable levels. What remains is a large residual, which are consistent with either a creditor-rights (political) risk premium and/or to unobserved adverse selection.

## VI.A Credit Mechanisms

A first possibility is that moratoria weakened local banks' balance sheets, either by preventing them from foreclosing on delinquent borrowers or by triggering depositor runs (Mitchener and Richardson, 2024) and producing a classic credit crunch (Correia et al., 2026). This mechanism has little support in the data. Rates of bank failure in the late 1930s are strikingly similar across moratoria and non-moratoria states (Figure A5), and the creation of the FDIC in 1933 and subsequent recapitalization programs had largely stabilized the banking system by mid-decade.

The spread therefore reflects the *pricing* of surviving credit rather than a disruption to its supply.

Consider a risk-neutral, competitive lender with funding cost  $\rho$ . The spread on a farm mortgage must compensate for the expected loss the lender bears if a future crisis triggers a moratorium. Writing the annualized zero-profit condition as  $(1 + r) - \lambda dL = 1 + \rho$  yields the pricing identity

$$\Delta r \approx \lambda dL, \quad (2)$$

where  $\lambda$  is the annual probability of a future crisis in which a moratorium is imposed,  $d$  is the crisis-conditional default rate, and  $L$  is the *incremental* recovery loss a lender suffers from being forced to wait out a stay rather than foreclosing at default. Interpreting the cross-sectional adopter-minus-non-adopter spread requires the difference  $\Delta(\lambda dL)$ ; under the disciplined assumption that adoption signals a state will re-suspend with probability one while non-adopters will not,  $\Delta\lambda = \lambda$  and the object collapses to (2), which we read as an *upper bound* on the expected-

loss channel.<sup>11</sup> We decompose the observed spread into three economically distinct pieces:

$$\underbrace{\Delta r}_{\text{observed spread}} = \underbrace{\lambda d L}_{\text{expected moratorium loss}} + \underbrace{\beta' \Delta X}_{\text{priced observable selection}} + \underbrace{\text{residual}}_{\text{creditor-rights risk premium} + \text{unobserved selection}}. \quad (3)$$

We apply this decomposition separately by statutory regime, distinguishing states whose statutes only stayed sales from those that also regulated or abolished deficiency judgments, and report the results in [Table 14](#). The same top-quartile double-difference design that delivers the headline rate result implies observed spreads of 27.9 basis points in stay-only states and 35.4 basis points in deficiency-regulation states, bracketing the pooled headline of roughly 30 basis points.

We calibrate the components of (2) directly. The crisis-conditional default rate  $d$  is taken from state-level 1932 foreclosure rates (the series underlying [Figure 1](#)), scaled by the share of distressed farmers who would plausibly have qualified for a stay, since courts denied relief to “hopelessly insolvent” owners without a “reasonable probability” of saving the land ([Poteat, 1938](#)). The stay horizon  $T$  comes from our statute coding, and  $\rho$  from the 1930 funding cost in non-moratorium counties. The incremental loss  $L$  is the gap between immediate-foreclosure recovery and post-stay recovery, which turns on the ratio  $\theta_M/\theta_0$  of collateral recovery after a stay to recovery at default. We calibrate  $\theta_M/\theta_0 = 1.15$  from Depression-era estimates of farm-collateral recovery in bank failures ([Rajan and Ramcharan, 2015, 2016](#)) and treat it as a horizon-matched lower bound<sup>12</sup>. Because collateral values recovered over 1933–1940 through fire-sale avoidance, dollar devaluation, and price supports,  $L$  is (weakly) negative: a lender forced to wait typically recovered *more*, not less. The mechanical expected-loss benchmark is therefore essentially zero, and if anything slightly negative, across a wide sensitivity grid for  $\lambda$  ([Table 14](#)). Since a larger  $\theta_M/\theta_0$  only makes  $L$  more negative, our lower-bound calibration renders the residual conservative.

As shown in [Section IV](#) and [Section V](#), moratoria left a pool of smaller, less valuable, and less capitalized farms. If lenders reprice this compositional shift using the historical relationship between rates and borrower characteristics, it justifies a spread  $\beta' \Delta X$ . We estimate the pricing gradient  $\beta$  by regressing pre-treatment (1930) interest rates on log borrower-risk characteristics—value per farm, machinery per cropland acre, and the tractor share—absorbing state fixed effects, and measure the moratorium-induced shift  $\Delta X$  with the same top-quartile double-difference design used for the rate itself.<sup>13</sup> The surviving pool is riskier on these observables, so the term

<sup>11</sup>Loan maturity does not enter (2): a longer mortgage spans more crisis-years but earns the spread over correspondingly more years, and the two offset in the annualized flow condition.

<sup>12</sup>This magnitude is also conservative relative to the modern forced-sale literature, in which foreclosure discounts of 10–30 percent ([Campbell et al., 2011](#)) imply a recovery ratio of  $1/(1 - \text{discount}) \approx 1.11\text{--}1.43$  for collateral that avoids a distress sale; we anchor to the lower end because those are point-in-time discounts on residential housing rather than the three-year farm-collateral horizon we require.

<sup>13</sup>We deliberately exclude value per acre, which is linked to value per farm by an accounting identity and rises mechanically post-moratoria as the distress tail is removed from the comparison set ([Section V](#)); we also exclude farm size, whose cross-sectional gradient is not sign-stable, and crop revenue per cropland acre, which proxies for regional crop-mix risk. Standard errors on  $\beta' \Delta X$

is positive, but it is small—1.7 basis points in stay-only states and 3.8 basis points in deficiency-regulation states—and imprecisely estimated.

Netting the expected-loss and priced-selection components from the observed spread leaves a residual that accounts for 92–104 percent of the spread in both regimes and across the full  $\lambda$  grid (Table 14). The decomposition is remarkably insensitive to the calibration precisely because both quantifiable channels are near zero: a rational, risk-neutral lender pricing only the direct expected loss and the observable deterioration of the pool would charge almost nothing. Two forces the exercise cannot separately identify remain in the residual. The first is a creditor-rights risk premium. A large literature shows that stronger legal protection for creditors lowers borrowing costs and expands credit supply (La Porta et al., 1997). The moratoria were an extreme, highly salient instance of the state overriding private contracts and suspending enforcement of secured claims, and even though they were temporary, a lender may rationally revise beliefs about such interventions recurring in future downturns and charge more—particularly on long-maturity farm mortgages exposed to multiple business cycles. Matching a positive spread when the mechanical loss is negative requires exactly this kind of ambiguity or risk aversion beyond the risk-neutral benchmark. The second is unobserved negative selection: riskiness of the surviving pool not captured by observable farm characteristics, in the spirit of classic adverse-selection models of credit markets (Stiglitz and Weiss, 1981). That elevated rates persist in moratorium counties after netting out observable composition, and are concentrated where foreclosures were actually suspended rather than in neighboring areas facing similar macroeconomic shocks, is consistent with both.

In summary, foreclosure moratoria left a clear imprint on local credit markets, raising farm borrowing costs by an amount that explains a sizable share of the residual cross-county dispersion in interest rates by 1940. A decomposition suggests that this spread is not the mechanical price of an expected moratorium loss—which our calibration puts near zero—nor a repricing of the observable deterioration in the borrower pool, but a large residual consistent with a creditor-rights risk premium and adverse selection on unobservables.

## VII Conclusion

Foreclosure moratoria are a recurring policy response to economic crises. The Great Depression offers a unique opportunity to study what happens when governments actually deploy these tools at scale. We exploit cross-state variation in whether moratoria were adopted and within-state variation in the extent of exposure of counties to mortgage distress, using a triple-difference design supplemented by a state-border empirical strategy.

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combine two independently estimated objects via the delta method, assuming zero cross-sample covariance.

We show that the moratoria worked, at least in the narrow sense that they kept farms in operation. Counties where the policy had the most bite saw a persistent 15-30% rise in farm counts and a corresponding fall in average farm size. The linked census evidence corroborates this account. We find that farmers in high-exposure treated counties were more likely to stay in agriculture and less likely to move into manufacturing. Their children followed suit. A generation later, the sons of protected farmers were still disproportionately working the family plot.

We also show that preserving farms came at a cost. Moratoria pushed up local borrowing costs, likely reflecting both weakened creditor protections and a riskier pool of surviving borrowers. The remaining farms were smaller, more labor-intensive, and less productive. Cropland expanded onto marginal acres, capital per acre declined, crop failures increased, and per-acre revenues declined. Per-farm values dropped and stayed down. We find no evidence that manufacturing collapsed on the extensive margin, although there is suggestive evidence that existing plants operated at lower intensity, consistent with weaker local demand and slower labor reallocation.

Two key limitations of our paper point to directions for future work. First, while our data provide evidence consistent with selection on surviving operators, they cannot fully disentangle selection from post-crisis re-optimization within farms. Second, a complete welfare analysis would require quantifying short-run insurance benefits and potential spillovers (e.g., avoided fire-sale losses and community stabilization) against the long-run distortions we document.

Our results speak to a fundamental tension in crisis-era policymaking. Temporary suspensions of creditor rights can provide valuable breathing room during periods of acute distress, but such interventions entail substantial costs. We cannot speak to overall welfare, but our evidence demonstrates that the long-run costs of foreclosure moratoria—in terms of misallocation, reduced productivity, and slower structural change—were substantial and persistent. Policymakers considering similar interventions should weigh these trade-offs carefully, recognizing that short-run protection may generate long-run distortions that outlast the crisis itself.

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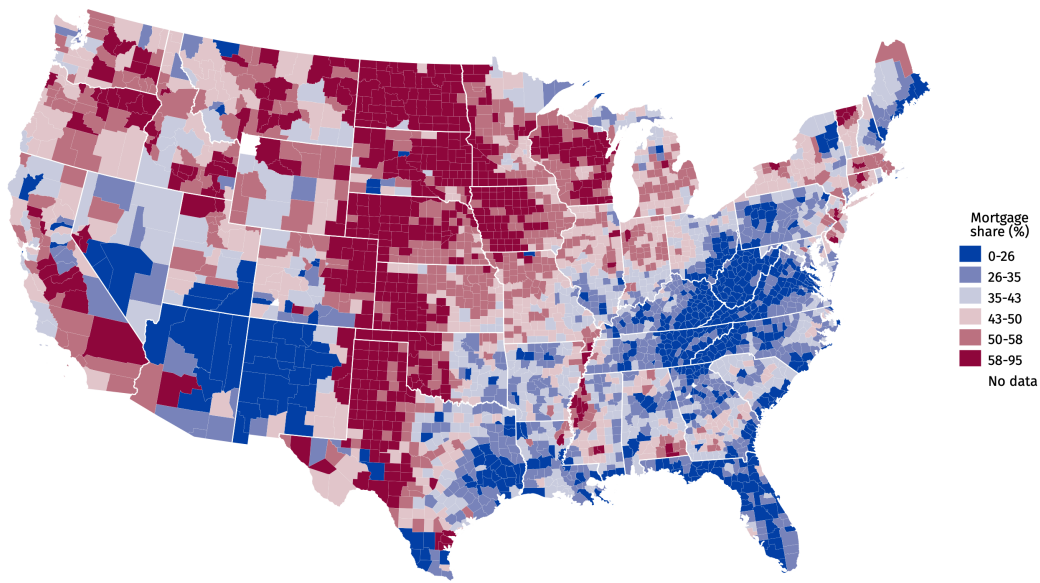
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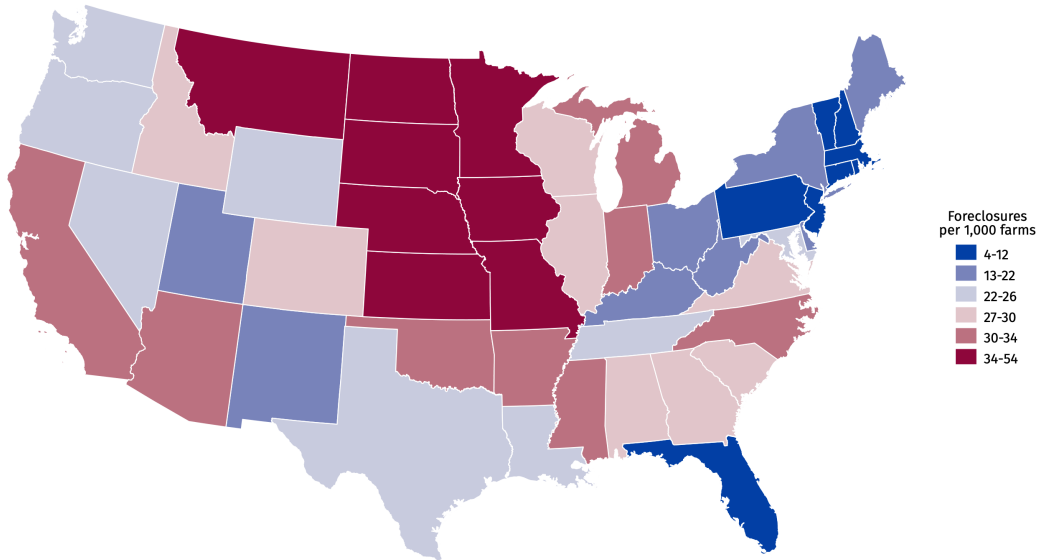
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Figure 1: Geography of the Foreclosure Crisis



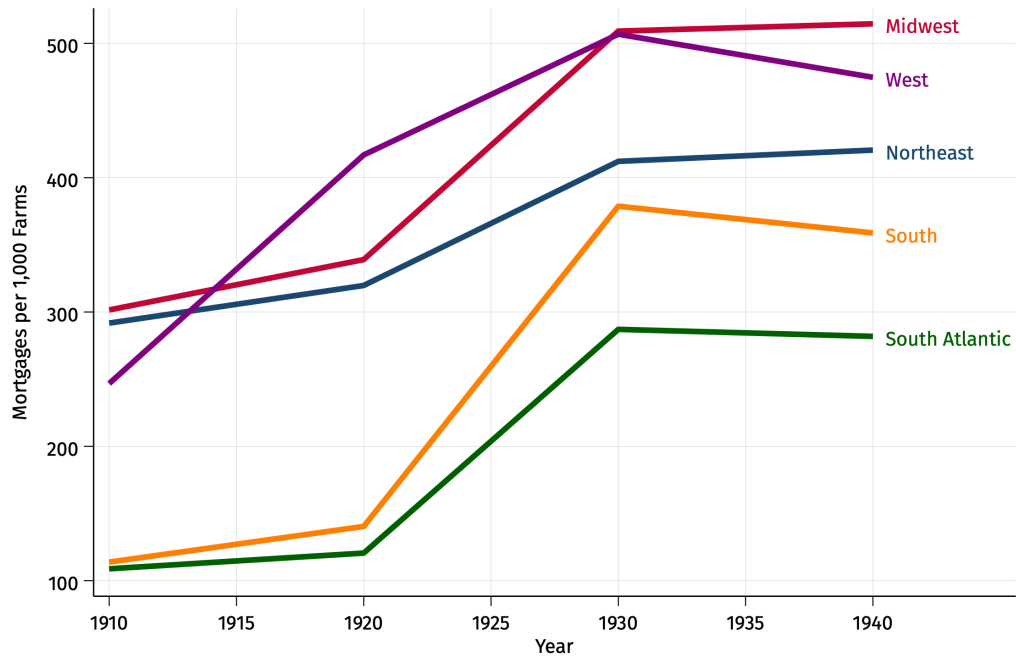
(a) Share of farms with a mortgage in 1930



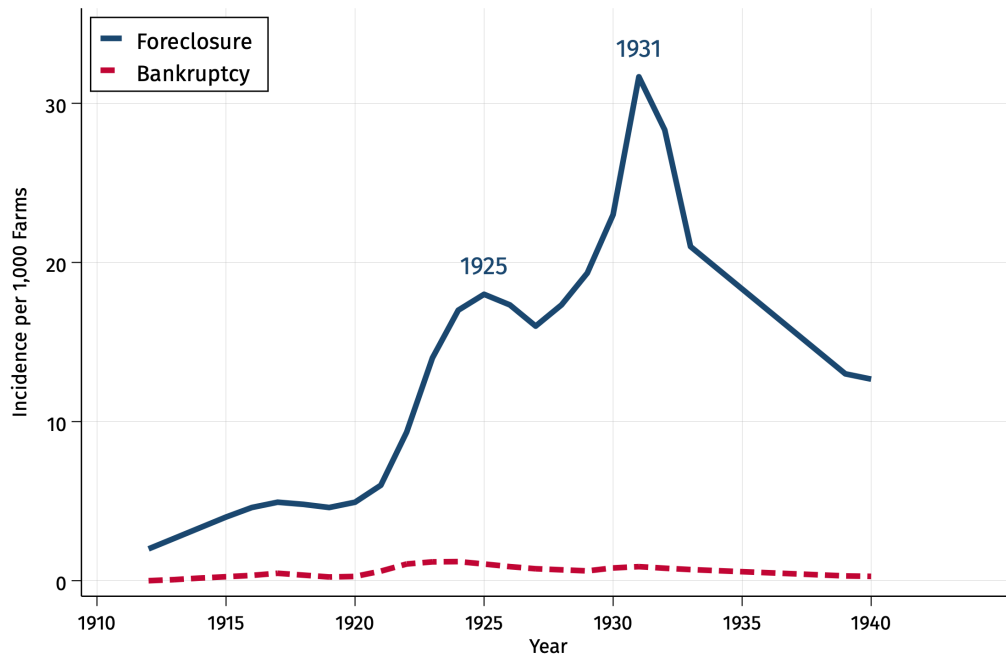
(b) Farm foreclosure rates in 1932

*Notes:* The upper panel maps the county-level share of farms under mortgage in 1930, as reported in the United States Census of Agriculture ([Haines et al., 2018](#)). Missing counties reflect unavailable county-level mortgage data. The lower panel maps state-level farm foreclosure rates in 1932 from [Stauber \(1933\)](#).

Figure 2: Timeline of the Foreclosure Crisis



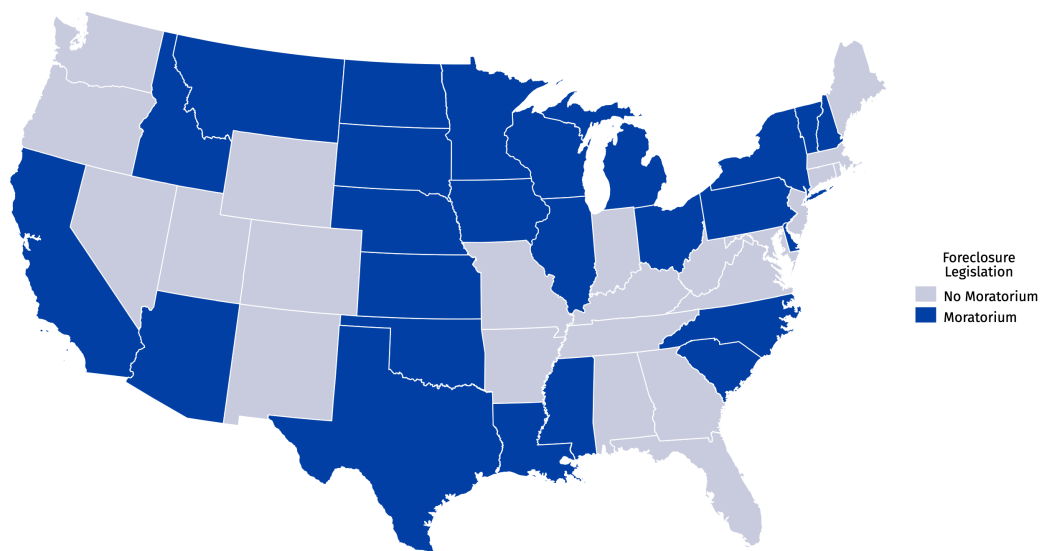
(a) Share of farms under mortgage by region



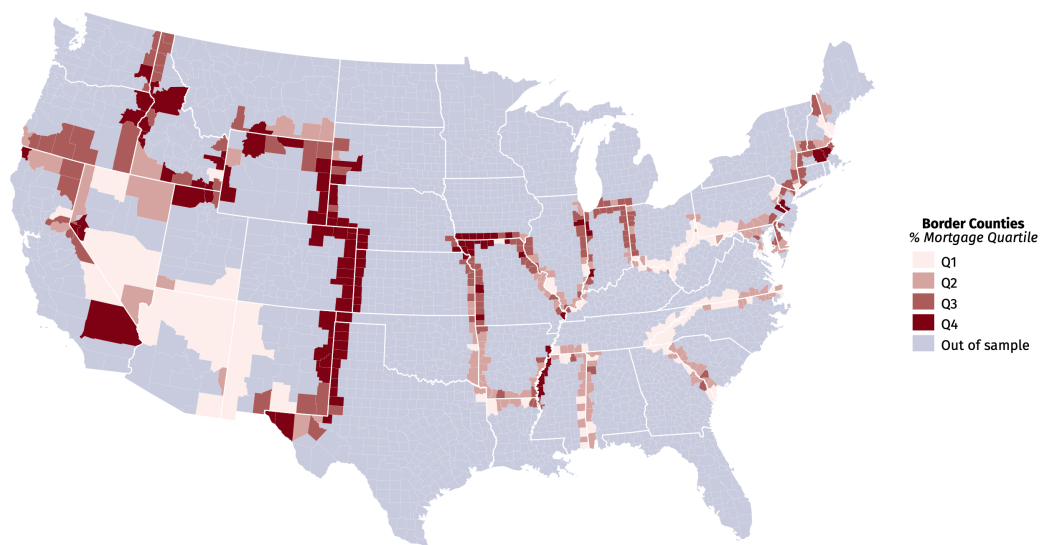
(b) Share of farms under involuntary transfer

Notes: The upper panel plots the regional share of farms under mortgage using Agricultural Census data (Haines et al., 2018). The lower panel plots the share of farms under involuntary transfer using the Works Progress Administration sample constructed by Messer and Rieder (2025).

Figure 3: Geography of the Foreclosure Moratoria



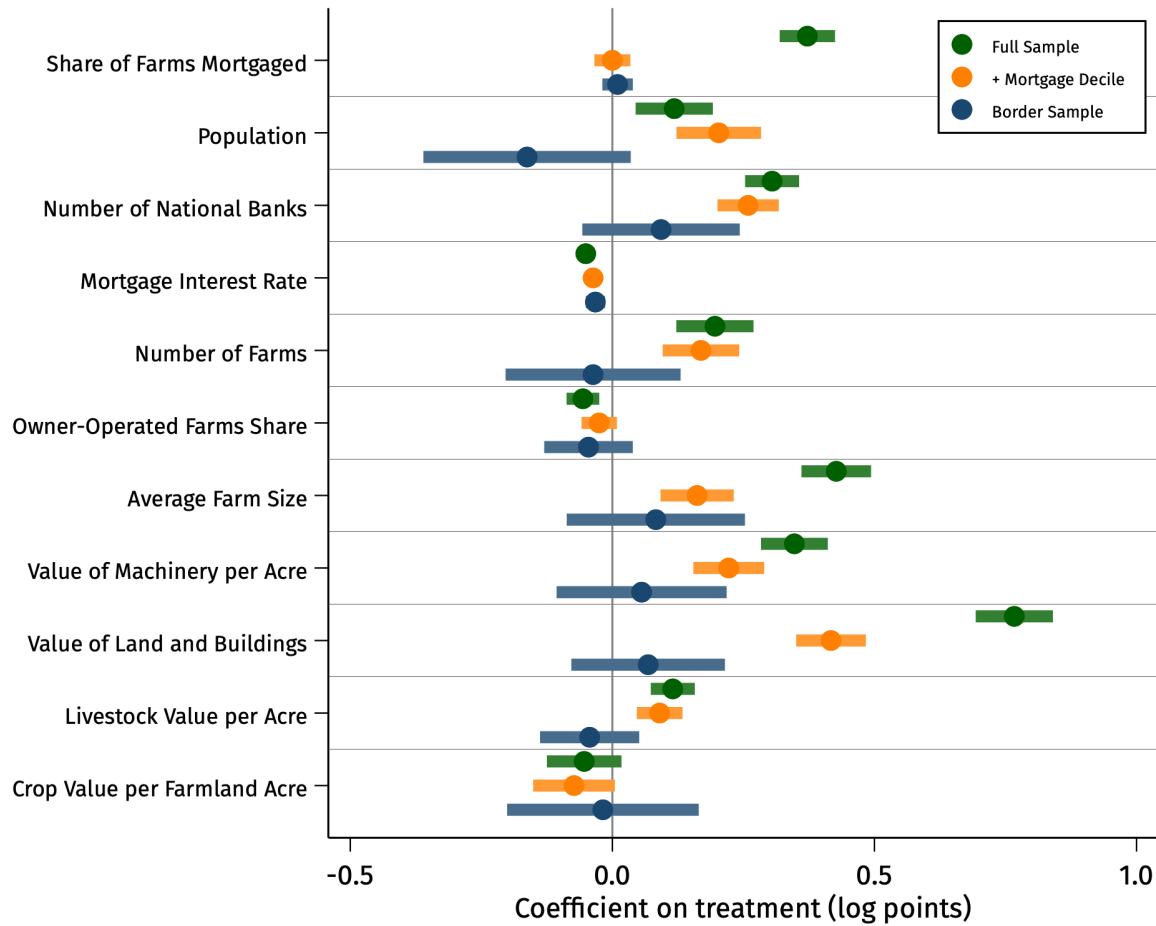
(a) States adopting foreclosure moratorium laws



(b) Counties in the border sample and their mortgage share ( $N = 476$ )

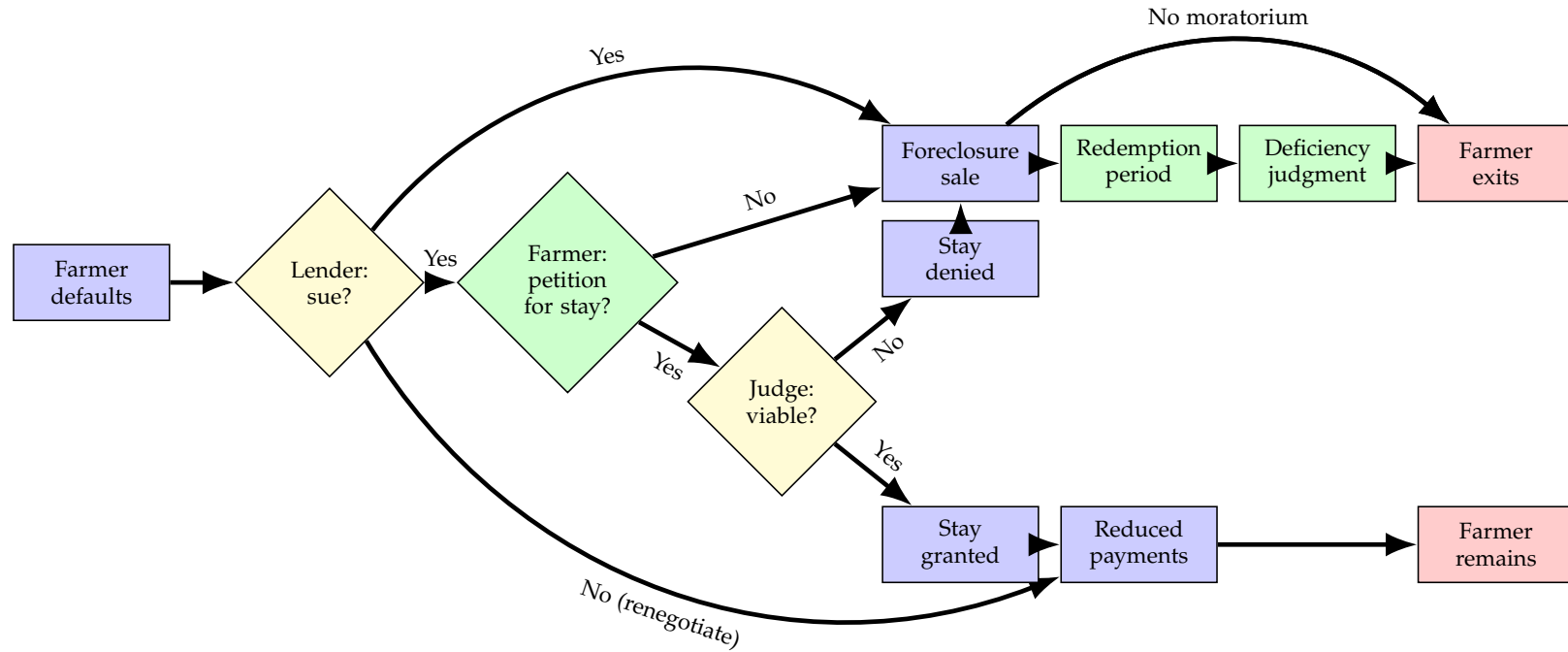
Notes: The upper panel indicates states that adopted foreclosure moratoria, based on [Alston \(1984\)](#). The lower panel shows the border sample, defined as counties adjacent to a state border across which moratorium status differs. The shading denotes quartiles of the county share of farms under mortgage in 1930.

Figure 4: Covariate Balance under Moratorium Treatment



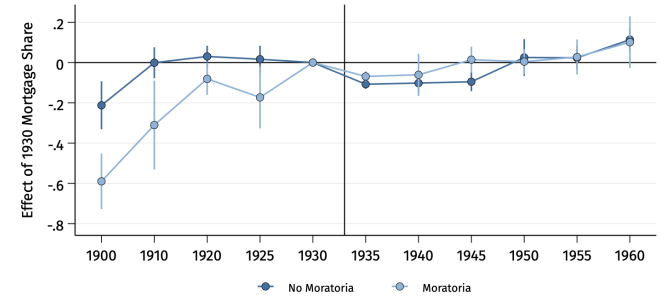
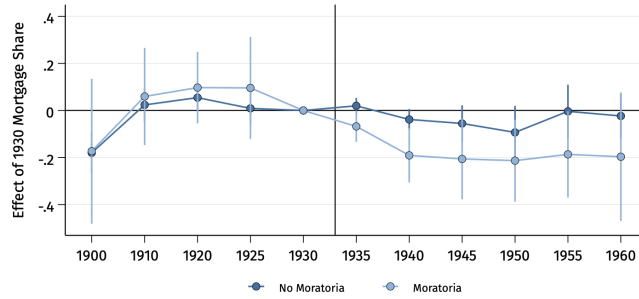
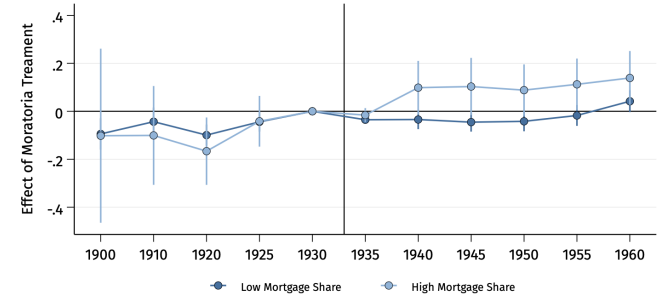
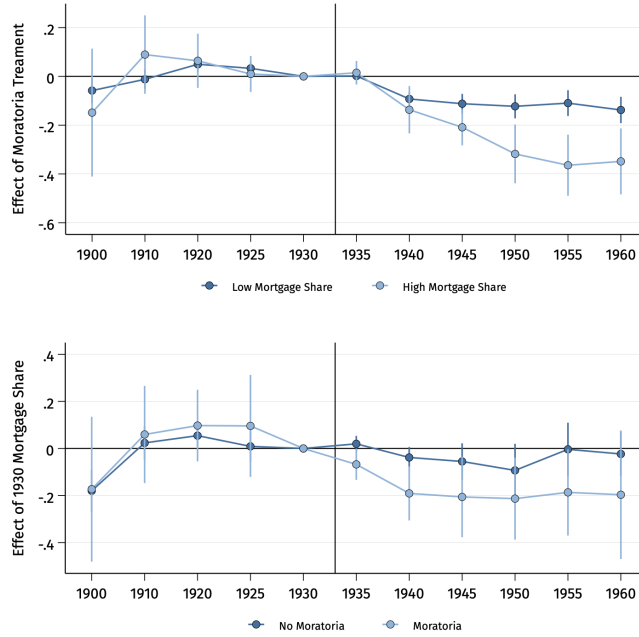
*Notes:* This figure compares baseline covariate differences between counties in moratorium and non-moratorium states across three samples. *Full Sample* reports unconditional differences in 1930 characteristics. *Mortgage Decile* adds fixed effects for deciles of the 1930 county mortgage share. *Border Sample* restricts the sample to counties adjacent to a state border across which treatment status differs.

Figure 5: The Process of Mortgage Foreclosure and Moratorium



Notes: The diagram illustrates the mortgage-foreclosure process and the points at which moratorium laws intervene. Yellow nodes denote decision points. Green nodes denote pathways created by moratorium laws. The three types of green nodes correspond to the legal categories summarized in Table 2. The diagram summarizes the discussion in Section II.B.

Figure 6: Pre-trends in Triple-Difference Estimation

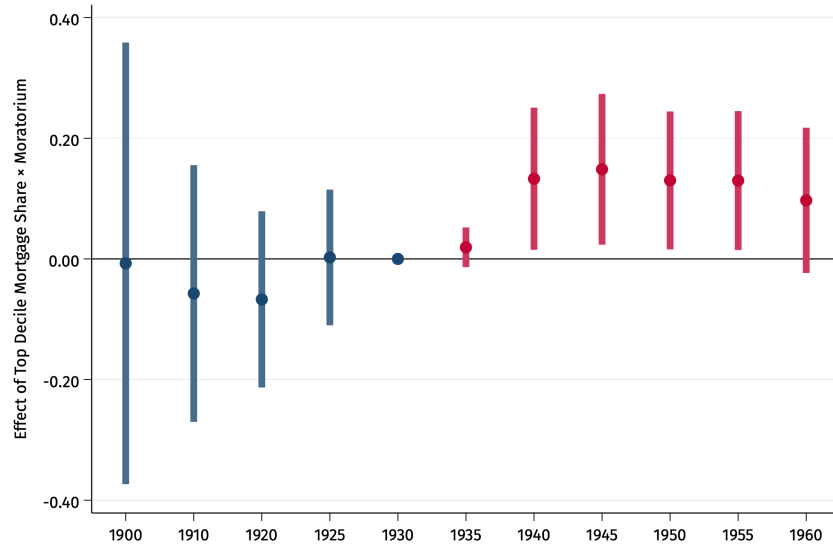


(a) Log average farm value

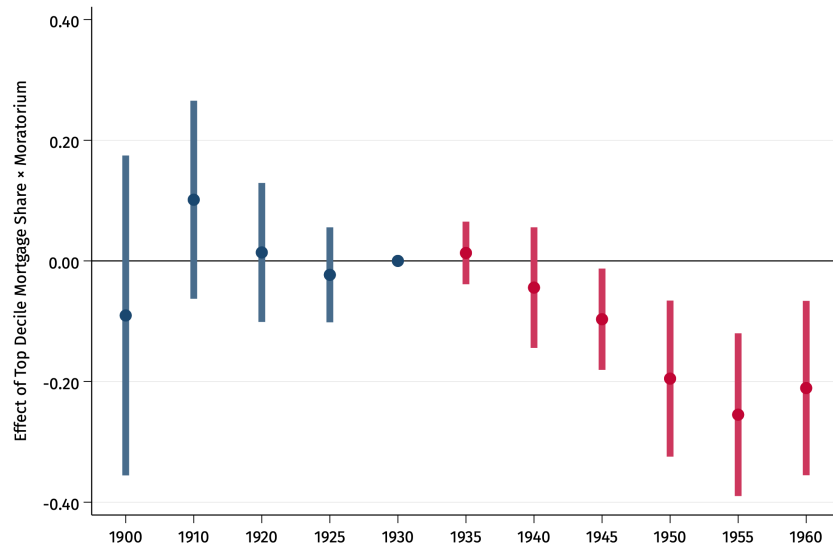
(b) Log number of farms

Notes: This figure reports event-study estimates used as pre-trend diagnostics for the triple-difference design, as described in Section III.B. The left panel uses log average farm value as the outcome; the right panel uses log number of farms. Each subplot reports separate event-study estimates by subgroup: high- versus low-mortgage-share counties (top) and moratorium versus non-moratorium states (bottom). Data are from [Haines et al. \(2018\)](#).

Figure 7: Triple-Difference Event Studies for Farm Outcomes



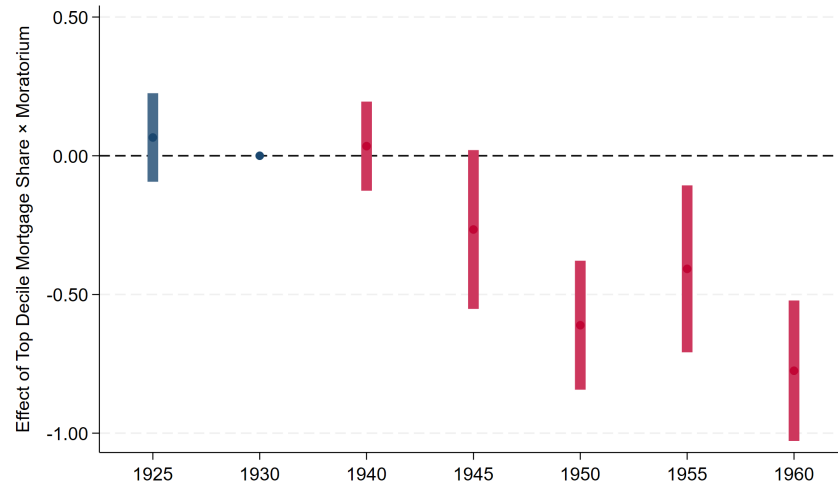
(a) Log number of farms



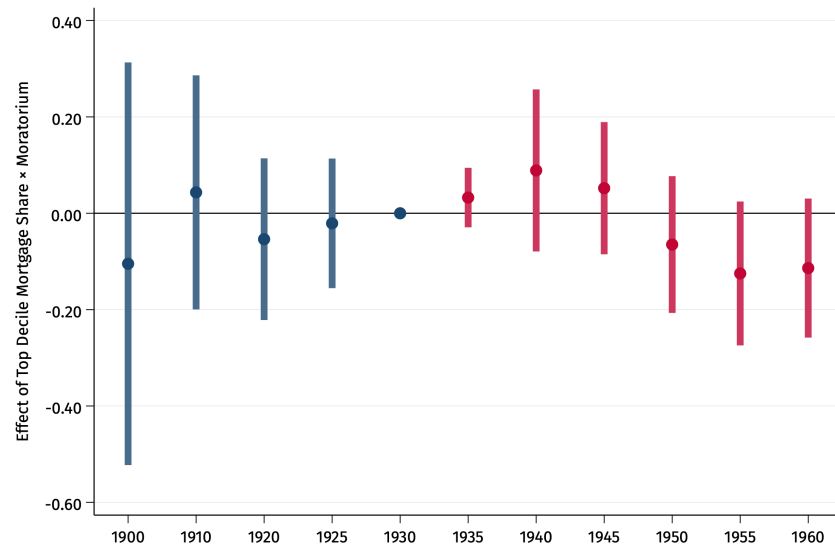
(b) Log average farm value

*Notes:* Observations are county-years. The upper panel plots event-study estimates for the log number of farms in a county. The lower panel plots event-study estimates for the log average value of farm land and buildings per farm, in current dollars. Data are from the Agricultural Censuses, 1900–1960. Each coefficient corresponds to the interaction between moratorium status and an indicator for counties in the top decile of the 1930 mortgage-share distribution. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Figure 8: Triple-Difference Event Studies for Farm Outcomes II



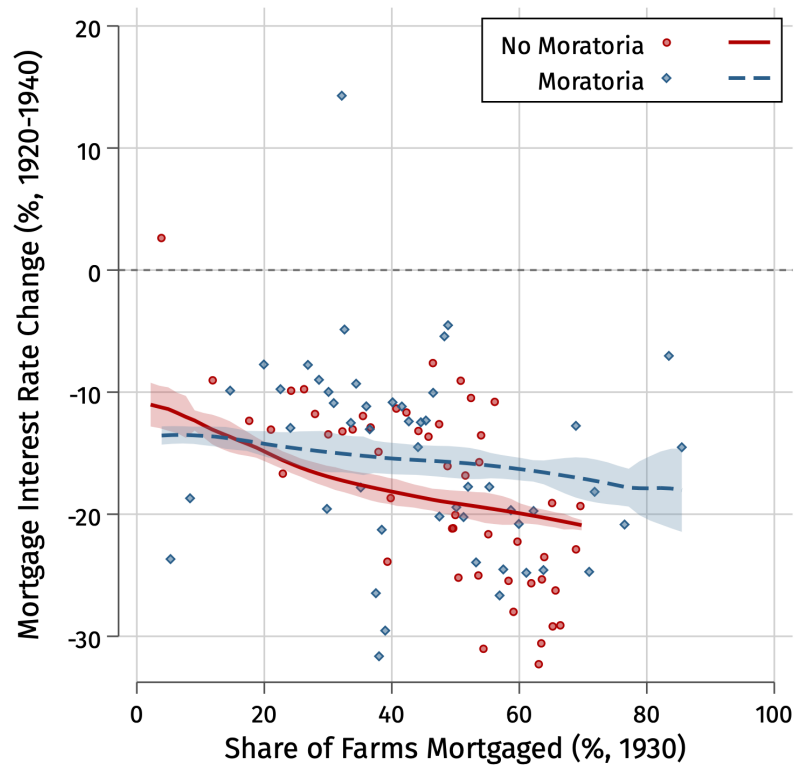
(a) Log crop revenue per acre of cropland



(b) Log total farm value

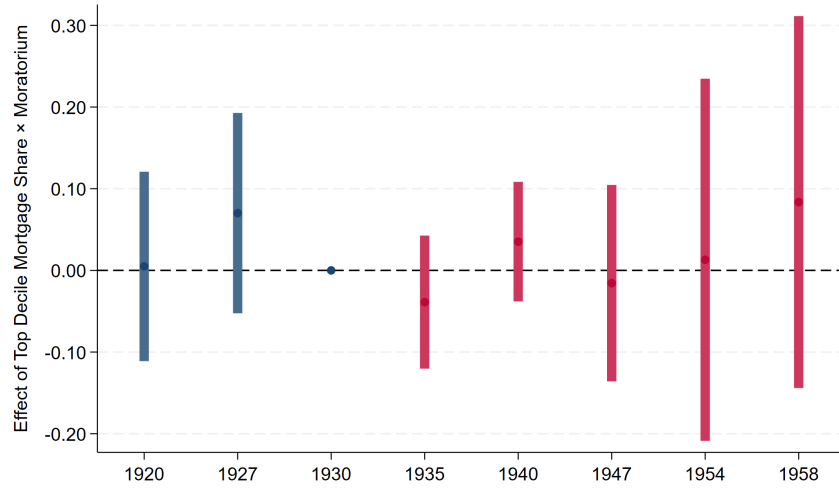
*Notes:* Observations are county-years. The upper panel plots event-study estimates for log crop revenue per acre of cropland. The lower panel plots event-study estimates for the log total value of farm land and buildings in a county. Data are from the Agricultural Censuses, 1900–1960. Each coefficient corresponds to the interaction between moratorium status and an indicator for counties in the top decile of the 1930 mortgage-share distribution. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Figure 9: Moratoria and the Interest Rate



*Notes:* Observations are counties in the border-pair sample. The horizontal axis is the county share of farms under mortgage in 1930. The vertical axis is the percentage change in the farm mortgage interest rate between 1920 and 1940, as reported in the Agricultural Census. Data are grouped into 50 equally sized bins. Local polynomial estimates are weighted by 1930 farmland area.

Figure 10: Triple-Difference Event Studies for Manufacturing Outcomes



(a) Extensive-margin index



(b) Intensive-margin index

*Notes:* Observations are county-years. The upper panel plots event-study estimates for the manufacturing extensive-margin index. The lower panel plots event-study estimates for the manufacturing intensive-margin index. The indices are constructed from county-level manufacturing outcomes described in the text. Each coefficient corresponds to the interaction between moratorium status and an indicator for counties in the top decile of the 1930 mortgage-share distribution. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table 1: County Summary Statistics in 1930

|                                          | N     | mean  | s.d  | min   | p50   | p90   | max   |
|------------------------------------------|-------|-------|------|-------|-------|-------|-------|
| Foreclosure Moratoria                    | 3,085 | 0.58  | 0.49 | 0     | 1     | 1     | 1     |
| Number of Farms (log)                    | 3,085 | 7.30  | 0.98 | 0     | 7.50  | 8.24  | 9.68  |
| Owner-Operated Farms Share               | 3,082 | 0.61  | 0.20 | 0.03  | 0.64  | 0.87  | 1     |
| Share of Farms Mortgaged                 | 3,083 | 0.42  | 0.16 | 0     | 0.43  | 0.63  | 0.95  |
| Average Farm Size (log)                  | 3,083 | 5.01  | 0.97 | 0.81  | 4.78  | 6.44  | 11.03 |
| Value of Machinery per Acre (log)        | 3,083 | 1.03  | 0.92 | -6.34 | 1.02  | 2.09  | 6.55  |
| Value of Land and Buildings (log)        | 3,083 | 16.07 | 1.08 | 9.39  | 16.15 | 17.36 | 19.99 |
| Number of National Banks                 | 3,085 | 2.36  | 3.83 | 0     | 1     | 6.00  | 61    |
| Mortgage Interest Rate (%)               | 3,066 | 6.48  | 0.69 | 2.88  | 6.38  | 7.39  | 10    |
| Population (log)                         | 3,085 | 9.82  | 1.04 | 4.38  | 9.78  | 11.04 | 15.20 |
| Farm Population Share                    | 3,045 | 0.46  | 0.25 | 0     | 0.51  | 0.77  | 0.99  |
| Farmer Age                               | 3,085 | 46.49 | 1.92 | 41    | 46.26 | 49.12 | 58.50 |
| Farmer Share White                       | 3,085 | 0.90  | 0.18 | 0.14  | 0.99  | 1     | 1     |
| Ratio Farm Workers to Owners             | 3,085 | 0.50  | 0.54 | 0.04  | 0.43  | 0.74  | 24    |
| Pr[Stay Farm Owner] (1930-1940)          | 3,085 | 0.47  | 0.12 | 0     | 0.49  | 0.59  | 0.77  |
| Pr[Stay in Agriculture] (1930-1940)      | 3,085 | 0.69  | 0.13 | 0     | 0.72  | 0.81  | 0.90  |
| Pr[Transition Ag. to Manuf.] (1930-1940) | 3,085 | 0.06  | 0.04 | 0     | 0.05  | 0.12  | 0.50  |

*Notes:* Observations are counties in 1930, unless otherwise indicated. Summary statistics are unweighted. Data are from [Alston \(1984\)](#), [Haines et al. \(2010\)](#), [Haines et al. \(2018\)](#), [Pawel Janas \(2024\)](#), and [Ruggles et al. \(2024, 2025\)](#).

Table 2: State Foreclosure Moratorium Legislation (1933–1934)

| State          | Year | Length<br>(Months) | Foreclosure<br>Stay | Redemption<br>Extension | Deficiency<br>Regulation |
|----------------|------|--------------------|---------------------|-------------------------|--------------------------|
| Arizona        | 1933 | 24                 | Yes                 | No                      | Yes                      |
| California     | 1933 | 45                 | Yes                 | Yes                     | Yes                      |
| Delaware       | 1933 | 14                 | Yes                 | No                      | No                       |
| Idaho          | 1933 | 48                 | Yes                 | Yes                     | Yes                      |
| Iowa           | 1933 | 49                 | Yes                 | Yes                     | Yes                      |
| Illinois       | 1933 | 24                 | Yes                 | No                      | No                       |
| Kansas         | 1933 | 46                 | No                  | Yes                     | Yes                      |
| Louisiana      | 1934 | 21                 | Yes                 | No                      | Yes                      |
| Michigan       | 1933 | 45                 | Yes                 | Yes                     | Yes                      |
| Minnesota      | 1933 | 46                 | Yes                 | Yes                     | Yes                      |
| Mississippi    | 1934 | 13                 | Yes                 | No                      | Yes                      |
| Montana        | 1933 | 24                 | Yes                 | Yes                     | Yes                      |
| Nebraska       | 1933 | 48                 | Yes                 | No                      | Yes                      |
| New Hampshire  | 1933 | 48                 | Yes                 | Yes                     | No                       |
| New York       | 1933 | 46                 | Yes                 | No                      | Yes                      |
| North Carolina | 1933 | 24                 | Yes                 | No                      | Yes                      |
| North Dakota   | 1933 | 52                 | Yes                 | Yes                     | Yes                      |
| Ohio           | 1933 | 47                 | Yes                 | No                      | No                       |
| Oklahoma       | 1933 | 24                 | Yes                 | No                      | No                       |
| Pennsylvania   | 1933 | 46                 | Yes                 | No                      | Yes                      |
| South Carolina | 1933 | 53                 | Yes                 | No                      | Yes                      |
| South Dakota   | 1933 | 48                 | Yes                 | Yes                     | Yes                      |
| Texas          | 1933 | 18                 | Yes                 | No                      | Yes                      |
| Vermont        | 1933 | 48                 | Yes                 | Yes                     | No                       |
| Wisconsin      | 1933 | 49                 | Yes                 | Yes                     | Yes                      |

*Notes:* This table summarizes state foreclosure-relief laws discussed in [Section II](#). *Year* is the year the law took effect. *Length* is the approximate duration of relief, including extensions. *Foreclosure Stay* indicates laws allowing postponement of foreclosure proceedings. *Redemption Extension* indicates laws extending the period during which property could be redeemed after sale. *Deficiency Regulation* indicates laws limiting lenders' ability to pursue deficiency judgments after foreclosure.

Table 3: Balance of Treated Counties in 1930

|                             | (1)<br>Full Sample      | (2)<br>+ Mortgage Decile | (3)<br>+ Border Sample  | (4)<br>+ Border Pair FE |
|-----------------------------|-------------------------|--------------------------|-------------------------|-------------------------|
| Share of Farms Mortgaged    | 0.372***<br>(0.0269)    | 0.00107<br>(0.0175)      | 0.00996<br>(0.0148)     | -0.0000371<br>(0.0201)  |
| Total Population            | 0.117***<br>(0.0376)    | 0.202***<br>(0.0412)     | -0.163<br>(0.101)       | -0.125<br>(0.0967)      |
| Number of National Banks    | 0.304***<br>(0.0263)    | 0.259***<br>(0.0298)     | 0.0929<br>(0.0765)      | 0.0839<br>(0.0860)      |
| Mortgage Interest Rate      | -0.0507***<br>(0.00366) | -0.0368***<br>(0.00377)  | -0.0323***<br>(0.00937) | -0.0326***<br>(0.0104)  |
| Number of Farms             | 0.195***<br>(0.0376)    | 0.169***<br>(0.0373)     | -0.0368<br>(0.0850)     | 0.0282<br>(0.109)       |
| Owner-Operated Farms Share  | -0.0561***<br>(0.0159)  | -0.0252<br>(0.0172)      | -0.0455<br>(0.0430)     | -0.0572*<br>(0.0300)    |
| Average Farm Size           | 0.428***<br>(0.0339)    | 0.162***<br>(0.0356)     | 0.0829<br>(0.0865)      | -0.00769<br>(0.0675)    |
| Value of Machinery per Acre | 0.347***<br>(0.0325)    | 0.222***<br>(0.0345)     | 0.0558<br>(0.0825)      | 0.0903<br>(0.0804)      |
| Value of Land and Buildings | 0.766***<br>(0.0376)    | 0.418***<br>(0.0339)     | 0.0683<br>(0.0745)      | 0.0815<br>(0.0954)      |
| Livestock Value per Acre    | 0.115***<br>(0.0214)    | 0.0904***<br>(0.0223)    | -0.0434<br>(0.0481)     | -0.0134<br>(0.0393)     |
| Crop Value per Acre         | 0.110**<br>(0.0546)     | 0.0346<br>(0.0521)       | -0.0942<br>(0.107)      | 0.0133<br>(0.0984)      |
| Proportion Treated          | 0.58                    | 0.58                     | 0.52                    | 0.52                    |
| Obs                         | 3,089                   | 3,089                    | 476                     | 476                     |
| Pairs                       |                         |                          |                         | 46                      |

*Notes:* Observations are counties in 1930. Each coefficient is from a regression of the indicated baseline characteristic on an indicator for whether the county is in a state that adopted a foreclosure moratorium ([Alston, 1984](#)). Unless otherwise noted, outcomes are in logs. *Mortgage Decile* indicates specifications that include fixed effects for deciles of the 1930 farm mortgage share. *Border Sample* restricts the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment. Standard errors are heteroskedasticity-robust in columns 1–3 and clustered at the border-pair level in column 4.

Table 4: Effects of Moratoria on Number of Farms

|                                 | Outcome: Log of Number of Farms |                     |                     |                     |                     |                     |                     |                     |                   |
|---------------------------------|---------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|-------------------|
|                                 | (1)                             | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 | (7)                 | (8)                 | (9)               |
| Post=1 × Moratoria              | 0.251***<br>(0.049)             |                     | 0.311***<br>(0.112) | 0.417***<br>(0.117) | 0.242***<br>(0.050) | 0.251***<br>(0.077) | 0.133***<br>(0.031) | 0.295***<br>(0.102) | 0.182*<br>(0.098) |
| Post=1 × Moratoria × % Mortgage | 0.230***<br>(0.060)             | 0.355***<br>(0.069) | 0.258**<br>(0.100)  | 0.392***<br>(0.119) | 0.231***<br>(0.061) | 0.264***<br>(0.060) |                     |                     |                   |
| County FE                       | ✓                               | ✓                   | ✓                   | ✓                   | ✓                   | ✓                   | ✓                   | ✓                   | ✓                 |
| Year FE                         | ✓                               | ✓                   | ✓                   | ✓                   | ✓                   | ✓                   | ✓                   | ✓                   | ✓                 |
| State × Year FE                 |                                 | ✓                   |                     |                     |                     |                     |                     |                     |                   |
| Border segment × Post FE        |                                 |                     |                     | ✓                   |                     |                     |                     |                     | ✓                 |
| New Deal controls               |                                 |                     |                     |                     | ✓                   |                     |                     |                     |                   |
| Dust Bowl controls              |                                 |                     |                     |                     |                     | ✓                   |                     |                     |                   |
| Border counties                 |                                 |                     | ✓                   | ✓                   |                     |                     |                     | ✓                   | ✓                 |
| High % mortgage counties        |                                 |                     |                     |                     |                     |                     | ✓                   | ✓                   | ✓                 |
| Period                          | 1900-1959                       | 1900-1959           | 1900-1959           | 1900-1959           | 1900-1959           | 1900-1959           | 1900-1959           | 1900-1959           | 1900-1959         |
| Adjusted $R^2$                  | 0.88                            | 0.90                | 0.87                | 0.89                | 0.88                | 0.88                | 0.85                | 0.81                | 0.84              |
| Obs                             | 33,689                          | 33,689              | 5,677               | 5,189               | 33,689              | 33,689              | 8,425               | 1,147               | 1,039             |
| Counties                        | 3,089                           | 3,089               | 523                 | 478                 | 3,089               | 3,089               | 768                 | 105                 | 95                |
| Outcome mean                    | 2,148                           | 2,148               | 2,058               | 2,108               | 2,148               | 2,148               | 1,868               | 1,708               | 1,755             |
| Outcome SD                      | 1,561                           | 1,561               | 1,737               | 1,799               | 1,561               | 1,561               | 1,589               | 1,905               | 2,053             |

Notes: Observations are county-years. The outcome is the log number of farms in a county. Data are from the Agricultural Censuses, 1900–1960. In the triple-difference specifications, the treatment effect is interacted with the county share of mortgaged farms in 1930. In the double-difference specifications, the sample is restricted to counties in the top quartile of the 1930 mortgage-share distribution (Q4). Columns marked *Border* restrict the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table 5: Effects of Moratoria on Farm Values

|                                 | Outcome: Log of Value per Farm (\$/farm) |                     |                    |                      |                      |                      |                      |                   |                   |
|---------------------------------|------------------------------------------|---------------------|--------------------|----------------------|----------------------|----------------------|----------------------|-------------------|-------------------|
|                                 | (1)                                      | (2)                 | (3)                | (4)                  | (5)                  | (6)                  | (7)                  | (8)               | (9)               |
| Post=1 × Moratoria              | -0.232***<br>(0.046)                     |                     | -0.223*<br>(0.115) | -0.289***<br>(0.095) | -0.218***<br>(0.046) | -0.371***<br>(0.095) | -0.187***<br>(0.042) | -0.114<br>(0.098) | -0.106<br>(0.085) |
| Post=1 × Moratoria × % Mortgage | -0.167***<br>(0.038)                     | -0.107**<br>(0.047) | -0.153<br>(0.103)  | -0.236***<br>(0.086) | -0.148***<br>(0.040) | -0.240***<br>(0.067) |                      |                   |                   |
| County FE                       | ✓                                        | ✓                   | ✓                  | ✓                    | ✓                    | ✓                    | ✓                    | ✓                 | ✓                 |
| Year FE                         | ✓                                        | ✓                   | ✓                  | ✓                    | ✓                    | ✓                    | ✓                    | ✓                 | ✓                 |
| State × Year FE                 |                                          | ✓                   |                    |                      |                      |                      |                      |                   |                   |
| Border segment × Post FE        |                                          |                     |                    | ✓                    |                      |                      |                      |                   | ✓                 |
| New Deal controls               |                                          |                     |                    |                      | ✓                    |                      |                      |                   |                   |
| Dust Bowl controls              |                                          |                     |                    |                      |                      | ✓                    |                      |                   |                   |
| Border counties                 |                                          |                     | ✓                  | ✓                    |                      |                      |                      | ✓                 | ✓                 |
| High % mortgage counties        |                                          |                     |                    |                      |                      |                      | ✓                    | ✓                 | ✓                 |
| Period                          | 1900-1959                                | 1900-1959           | 1900-1959          | 1900-1959            | 1900-1959            | 1900-1959            | 1900-1959            | 1900-1959         | 1900-1959         |
| Adjusted $R^2$                  | 0.93                                     | 0.95                | 0.90               | 0.93                 | 0.93                 | 0.93                 | 0.87                 | 0.84              | 0.86              |
| Obs                             | 33,645                                   | 33,645              | 5,677              | 5,189                | 33,645               | 33,645               | 8,425                | 1,147             | 1,039             |
| Counties                        | 3,088                                    | 3,088               | 523                | 478                  | 3,088                | 3,088                | 768                  | 105               | 95                |
| Outcome mean                    | 12,393                                   | 12,393              | 11,875             | 12,093               | 12,393               | 12,393               | 16,778               | 14,949            | 15,333            |
| Outcome SD                      | 14,313                                   | 14,313              | 14,562             | 15,248               | 14,313               | 14,313               | 13,519               | 7,919             | 8,374             |

*Notes:* Observations are county-years. The outcome is the log average value of farm land and buildings per farm. Data are from the Agricultural Censuses, 1900–1960. In the triple-difference specifications, the treatment effect is interacted with the county share of mortgaged farms in 1930. In the double-difference specifications, the sample is restricted to counties in the top quartile of the 1930 mortgage-share distribution ( $Q4$ ). Columns marked *Border* restrict the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table 6: Effects of Moratoria on Farm Size

|                                 | Outcome: Log of Average Farm Size |                      |                      |                      |                      |                      |                      |                      |                      |
|---------------------------------|-----------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                 | (1)                               | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  | (7)                  | (8)                  | (9)                  |
| Post=1 × Moratoria              | -0.375***<br>(0.073)              |                      | -0.485***<br>(0.159) | -0.385***<br>(0.106) | -0.367***<br>(0.075) | -0.448***<br>(0.132) | -0.245***<br>(0.038) | -0.311***<br>(0.104) | -0.253***<br>(0.087) |
| Post=1 × Moratoria × % Mortgage | -0.323***<br>(0.081)              | -0.244***<br>(0.067) | -0.417**<br>(0.164)  | -0.312***<br>(0.102) | -0.307***<br>(0.085) | -0.481***<br>(0.102) |                      |                      |                      |
| County FE                       | ✓                                 | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    |
| Year FE                         | ✓                                 | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    | ✓                    |
| State × Year FE                 |                                   | ✓                    |                      |                      |                      |                      |                      |                      |                      |
| Border segment × Post FE        |                                   |                      |                      | ✓                    |                      |                      |                      |                      | ✓                    |
| New Deal controls               |                                   |                      |                      |                      | ✓                    |                      |                      |                      |                      |
| Dust Bowl controls              |                                   |                      |                      |                      |                      | ✓                    |                      |                      |                      |
| Border counties                 |                                   |                      | ✓                    | ✓                    |                      |                      |                      | ✓                    | ✓                    |
| High % mortgage counties        |                                   |                      |                      |                      |                      |                      | ✓                    | ✓                    | ✓                    |
| Period                          | 1900-1959                         | 1900-1959            | 1900-1959            | 1900-1959            | 1900-1959            | 1900-1959            | 1900-1959            | 1900-1959            | 1900-1959            |
| Adjusted $R^2$                  | 0.93                              | 0.96                 | 0.93                 | 0.94                 | 0.94                 | 0.94                 | 0.90                 | 0.87                 | 0.88                 |
| Obs                             | 33,647                            | 33,647               | 5,677                | 5,189                | 33,647               | 33,647               | 8,425                | 1,147                | 1,039                |
| Counties                        | 3,088                             | 3,088                | 523                  | 478                  | 3,088                | 3,088                | 768                  | 105                  | 95                   |
| Outcome mean                    | 783                               | 783                  | 1,236                | 1,296                | 783                  | 783                  | 783                  | 934                  | 929                  |
| Outcome SD                      | 2,949                             | 2,949                | 4,645                | 4,895                | 2,949                | 2,949                | 1,487                | 1,314                | 1,415                |

Notes: Observations are county-years. The outcome is log average farm size in a county. Data are from the Agricultural Censuses, 1900–1960. In the triple-difference specifications, the treatment effect is interacted with the county share of mortgaged farms in 1930. In the double-difference specifications, the sample is restricted to counties in the top quartile of the 1930 mortgage-share distribution (Q4). Columns marked *Border* restrict the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment; for example, all counties along the Arizona–New Mexico border are assigned to one border-pair fixed effect. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table 7: Effects of Moratoria on Occupational Transitions, 1930–1940

|                               | Stay in Ag.         |                    | Stay Owner          |                  | Ag. to Manuf.        |                    |
|-------------------------------|---------------------|--------------------|---------------------|------------------|----------------------|--------------------|
|                               | (1)                 | (2)                | (3)                 | (4)              | (5)                  | (6)                |
| Moratoria $\times$ % Mortgage | 0.040***<br>(0.011) | 0.039**<br>(0.017) | 0.005<br>(0.014)    | 0.003<br>(0.019) | -0.007***<br>(0.003) | -0.005<br>(0.008)  |
| Moratoria                     | 0.058***<br>(0.010) | 0.040**<br>(0.017) | 0.028**<br>(0.013)  | 0.017<br>(0.022) | -0.012***<br>(0.003) | -0.008<br>(0.007)  |
| % Mortgage                    | 0.037***<br>(0.008) | 0.004<br>(0.010)   | 0.039***<br>(0.010) | 0.009<br>(0.011) | -0.002<br>(0.002)    | -0.010*<br>(0.005) |
| Demographic Controls          | Yes                 | No                 | Yes                 | No               | Yes                  | No                 |
| Economic Controls             | Yes                 | No                 | Yes                 | No               | Yes                  | No                 |
| Outcome Mean                  | 0.71                | 0.69               | 0.49                | 0.47             | 0.06                 | 0.06               |
| Outcome SD                    | 0.09                | 0.09               | 0.10                | 0.10             | 0.03                 | 0.03               |
| Sample                        | All                 | Border             | All                 | Border           | All                  | Border             |
| Obs                           | 2,333               | 476                | 2,333               | 476              | 2,333                | 476                |
| Adjusted $R^2$                | 0.33                | 0.02               | 0.27                | 0.00             | 0.42                 | 0.04               |
| p50-p90 effect (%)            | 3.08                | 1.70               | 1.72                | 0.46             | -0.37                | -0.58              |

Notes: Observations are counties. Data are from [Ruggles et al. \(2024, 2025\)](#) and [Haines et al. \(2010\)](#). County-level means are constructed from the linked IPUMS full-count sample. The sample is restricted to men active in the labor force in both 1930 and 1940 who worked in agriculture in 1930. Demographic controls are age, age squared, and an indicator for White. Economic controls are the logs of the number of farms and the number of manufacturing establishments in 1930. Standard errors are heteroskedasticity-robust. Regressions are weighted by 1930 farmland area.

Table 8: Effects of Moratoria on the Children of Farmers in 1950

|                                                       | Agriculture         |                     | Operate Family Farm |                     | Manufacturing        |                      |
|-------------------------------------------------------|---------------------|---------------------|---------------------|---------------------|----------------------|----------------------|
|                                                       | (1)                 | (2)                 | (3)                 | (4)                 | (5)                  | (6)                  |
| Moratoria $\times$ Farmer's Child $\times$ % Mortgage | 0.071***<br>(0.021) | 0.079***<br>(0.025) | 0.034***<br>(0.010) | 0.030**<br>(0.013)  | -0.051**<br>(0.026)  | -0.014<br>(0.033)    |
| Moratoria $\times$ % Mortgage                         | 0.027***<br>(0.009) | -0.005<br>(0.016)   | 0.004**<br>(0.002)  | -0.002<br>(0.006)   | -0.050**<br>(0.025)  | -0.076**<br>(0.033)  |
| Moratoria $\times$ Farmer's Child                     | 0.074***<br>(0.023) | 0.061**<br>(0.025)  | 0.022*<br>(0.012)   | 0.008<br>(0.014)    | -0.062**<br>(0.028)  | -0.020<br>(0.037)    |
| Farmer's Child $\times$ % Mortgage                    | 0.054***<br>(0.014) | 0.072***<br>(0.015) | 0.030***<br>(0.007) | 0.043***<br>(0.008) | -0.029**<br>(0.014)  | -0.080***<br>(0.016) |
| Moratoria                                             | 0.023***<br>(0.009) | 0.009<br>(0.014)    | 0.003*<br>(0.002)   | 0.004<br>(0.005)    | -0.035<br>(0.027)    | -0.074*<br>(0.038)   |
| % Mortgage                                            | -0.001<br>(0.003)   | 0.011<br>(0.010)    | -0.001<br>(0.001)   | -0.001<br>(0.003)   | 0.043***<br>(0.012)  | 0.079***<br>(0.015)  |
| Farmer's Child                                        | 0.390***<br>(0.017) | 0.368***<br>(0.019) | 0.158***<br>(0.009) | 0.166***<br>(0.011) | -0.130***<br>(0.018) | -0.136***<br>(0.022) |
| Demographic Controls                                  | No                  | Yes                 | No                  | Yes                 | No                   | Yes                  |
| Economic Controls                                     | No                  | Yes                 | No                  | Yes                 | No                   | Yes                  |
| Outcome Mean                                          | 0.11                | 0.11                | 0.03                | 0.03                | 0.30                 | 0.30                 |
| Outcome SD                                            | 0.31                | 0.31                | 0.18                | 0.18                | 0.46                 | 0.46                 |
| Sample                                                | All                 | All                 | All                 | All                 | All                  | All                  |
| Obs                                                   | 186,857             | 161,506             | 186,857             | 161,506             | 186,857              | 161,506              |
| Adjusted $R^2$                                        | 0.23                | 0.24                | 0.09                | 0.10                | 0.01                 | 0.02                 |
| p50-p90 effect (%)                                    | 5.52                | 5.73                | 2.40                | 2.55                | -3.05                | -3.19                |

Notes: Observations are adult individuals observed in the 1950 Census and linked to their parents' 1930 Census returns. Data are from [Ruggles et al. \(2024, 2025\)](#) and [Haines et al. \(2010\)](#). Treatment is assigned using the county of residence in 1930, when the individual was a child (mean age 6). The sample is restricted to nonmigrants between 1930 and 1950. Demographic controls are age, age squared, and an indicator for White. Economic controls are the logs of the number of farms and the number of manufacturing establishments in 1930. Standard errors are clustered at the childhood-county level.

Table 9: Effects of Moratoria on Land Usage

|                                               | (1)<br>% Farm Cropland | (2)<br>% Idle Cropland | (3)<br>% Failed Cropland | (4)<br>% Harvested Cropland |
|-----------------------------------------------|------------------------|------------------------|--------------------------|-----------------------------|
| Post=1 $\times$ Moratoria $\times$ % Mortgage | 0.303***<br>(0.077)    | 0.397***<br>(0.112)    | 1.072***<br>(0.173)      | -0.138***<br>(0.034)        |
| Post=1 $\times$ % Mortgage                    | -0.033<br>(0.025)      | 0.130***<br>(0.048)    | -0.153**<br>(0.066)      | 0.095***<br>(0.025)         |
| Post=1 $\times$ Moratoria                     | 0.268***<br>(0.060)    | 0.800***<br>(0.119)    | 1.391***<br>(0.162)      | -0.124***<br>(0.032)        |
| County FE                                     | ✓                      | ✓                      | ✓                        | ✓                           |
| Year FE                                       | ✓                      | ✓                      | ✓                        | ✓                           |
| Sample Period                                 | All<br>1925-1959       | All<br>1925-1945       | All<br>1925-1945         | All<br>1925-1959            |
| Adjusted $R^2$                                | 0.94                   | 0.73                   | 0.42                     | 0.34                        |
| Obs                                           | 24,514                 | 15,288                 | 15,265                   | 24,509                      |
| Counties                                      | 3,077                  | 3,071                  | 3,073                    | 3,076                       |
| Outcome mean                                  | 0.42                   | 0.12                   | 0.04                     | 0.84                        |
| Outcome SD                                    | 0.22                   | 0.10                   | 0.06                     | 0.12                        |

*Notes:* Observations are county-years. Data are from the Agricultural Censuses, 1900–1960. Specifications follow the paper’s baseline triple-difference design. Standard errors are clustered at the county level. Regressions are weighted by 1930 cropland area.

Table 10: Effects of Moratoria on Agricultural Outcomes

|                                               | (1)<br>Capital/Acre  | (2)<br>Capital/Labor | (3)<br>Crop Yield   | (4)<br>Animal Yield |
|-----------------------------------------------|----------------------|----------------------|---------------------|---------------------|
| Post=1 $\times$ Moratoria $\times$ % Mortgage | -0.338***<br>(0.064) | -0.209***<br>(0.048) | -0.234**<br>(0.106) | 0.088<br>(0.072)    |
| Post=1 $\times$ % Mortgage                    | -0.001<br>(0.019)    | -0.012<br>(0.034)    | 0.288***<br>(0.064) | 0.067<br>(0.051)    |
| Post=1 $\times$ Moratoria                     | -0.359***<br>(0.048) | -0.243***<br>(0.046) | -0.237**<br>(0.096) | 0.094<br>(0.061)    |
| County FE                                     | ✓                    | ✓                    | ✓                   | ✓                   |
| Year FE                                       | ✓                    | ✓                    | ✓                   | ✓                   |
| Sample Period                                 | All<br>1925-1945     | All<br>1930-1945     | All<br>1925-1959    | All<br>1925-1954    |
| Adjusted $R^2$                                | 0.85                 | 0.93                 | 0.60                | 0.90                |
| Obs                                           | 12,268               | 8,688                | 21,418              | 18,382              |
| Counties                                      | 3,072                | 2,925                | 3,066               | 3,066               |
| Outcome mean                                  | 13                   | 509                  | 20                  | 14                  |
| Outcome SD                                    | 53                   | 18,151               | 18                  | 14                  |

*Notes:* Observations are county-years. Outcomes are expressed per acre of cropland. Data are from the Agricultural Censuses, 1900–1960. Specifications follow the paper’s baseline triple-difference design. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table 11: Effects of Moratoria on Interest Rates

|                                               | Outcome: Mortgage Interest Rate (%) |                     |                     |                     |                  |                    |                    |
|-----------------------------------------------|-------------------------------------|---------------------|---------------------|---------------------|------------------|--------------------|--------------------|
|                                               | (1)                                 | (2)                 | (3)                 | (4)                 | (5)              | (6)                | (7)                |
| Post=1 $\times$ Moratoria                     | -0.023<br>(0.097)                   | 0.606***<br>(0.153) | 0.546***<br>(0.189) | 0.700***<br>(0.179) | 0.345<br>(0.358) | 0.318**<br>(0.137) | 0.319**<br>(0.147) |
| Post=1 $\times$ Moratoria $\times$ % Mortgage | 0.012<br>(0.114)                    | 0.513***<br>(0.141) | 0.475**<br>(0.202)  | 0.610***<br>(0.175) | 0.228<br>(0.383) |                    |                    |
| County FE                                     | ✓                                   | ✓                   | ✓                   | ✓                   | ✓                | ✓                  | ✓                  |
| Year FE                                       | ✓                                   | ✓                   | ✓                   | ✓                   | ✓                | ✓                  | ✓                  |
| Border segment $\times$ Post FE               |                                     |                     | ✓                   |                     |                  |                    | ✓                  |
| New Deal controls                             |                                     |                     |                     | ✓                   |                  |                    |                    |
| Dust Bowl controls                            |                                     |                     |                     |                     | ✓                |                    |                    |
| Border counties                               |                                     | ✓                   | ✓                   | ✓                   | ✓                | ✓                  | ✓                  |
| High % mortgage counties                      |                                     |                     |                     |                     |                  | ✓                  | ✓                  |
| Period                                        | 1920-1940                           | 1920-1940           | 1920-1940           | 1920-1940           | 1920-1940        | 1920-1940          | 1920-1940          |
| Adjusted $R^2$                                | 0.76                                | 0.67                | 0.72                | 0.67                | 0.68             | 0.76               | 0.72               |
| Obs                                           | 9,196                               | 1,558               | 1,421               | 1,558               | 1,558            | 314                | 280                |
| Counties                                      | 3,071                               | 520                 | 475                 | 520                 | 520              | 105                | 95                 |
| Outcome mean                                  | 6.47                                | 6.57                | 6.56                | 6.57                | 6.57             | 6.49               | 6.45               |
| Outcome SD                                    | 0.66                                | 0.70                | 0.71                | 0.70                | 0.70             | 0.59               | 0.59               |

*Notes:* Observations are county-years. The outcome is the average interest rate paid by farmers on mortgage debt. Data are from the Agricultural Censuses in 1920 and 1940. In the triple-difference specifications, the treatment effect is interacted with the county share of mortgaged farms in 1930. In the double-difference specifications, the sample is restricted to counties in the top quartile of the 1930 mortgage-share distribution (Q4). Columns marked *Border* restrict the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment; for example, all counties along the Arizona–New Mexico border are assigned to one border-pair fixed effect. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table 12: Effects of Moratoria on Loan Quantity

|                                               | Outcome: Log Loans (\$/farm) |                   |                   |                   |                  |                   |                  |                  |
|-----------------------------------------------|------------------------------|-------------------|-------------------|-------------------|------------------|-------------------|------------------|------------------|
|                                               | (1)                          | (2)               | (3)               | (4)               | (5)              | (6)               | (7)              | (8)              |
| Post=1 $\times$ Moratoria                     | -0.022<br>(0.084)            | -0.106<br>(0.163) | -0.106<br>(0.163) | -0.044<br>(0.120) | 0.004<br>(0.155) | -0.102<br>(0.280) | 0.106<br>(0.126) | 0.075<br>(0.094) |
| Post=1 $\times$ Moratoria $\times$ % Mortgage | 0.022<br>(0.071)             | -0.081<br>(0.157) | -0.081<br>(0.157) | -0.033<br>(0.119) | 0.100<br>(0.147) | -0.218<br>(0.309) |                  |                  |
| County FE                                     | ✓                            | ✓                 | ✓                 | ✓                 | ✓                | ✓                 | ✓                | ✓                |
| Year FE                                       | ✓                            | ✓                 | ✓                 | ✓                 | ✓                | ✓                 | ✓                | ✓                |
| Border segment $\times$ Post FE               |                              |                   |                   | ✓                 |                  |                   |                  | ✓                |
| New Deal controls                             |                              |                   |                   |                   | ✓                |                   |                  |                  |
| Dust Bowl controls                            |                              |                   |                   |                   |                  | ✓                 |                  |                  |
| Border counties                               |                              | ✓                 | ✓                 | ✓                 | ✓                | ✓                 | ✓                | ✓                |
| High % mortgage counties                      |                              |                   |                   |                   |                  |                   | ✓                | ✓                |
| Period                                        | 1925-1940                    | 1925-1940         | 1925-1940         | 1925-1940         | 1925-1940        | 1925-1940         | 1925-1940        | 1925-1940        |
| Adjusted $R^2$                                | 0.90                         | 0.92              | 0.92              | 0.93              | 0.92             | 0.92              | 0.92             | 0.93             |
| Obs                                           | 30,169                       | 5,005             | 5,005             | 4,598             | 4,883            | 4,883             | 991              | 906              |
| Counties                                      | 2,219                        | 369               | 369               | 339               | 346              | 346               | 73               | 67               |
| Outcome mean                                  | 1,470                        | 1,836             | 1,836             | 1,833             | 1,836            | 1,836             | 1,014            | 1,062            |
| Outcome SD                                    | 30,267                       | 12,102            | 12,102            | 12,486            | 12,102           | 12,102            | 2,384            | 2,497            |

*Notes:* Observations are county-years. The outcome is the log of the dollar value of loans held at national banks in the county, normalized by the count of farms in 1930. Data are from [Correia et al. \(2026\)](#). In the triple-difference specifications, the treatment effect is interacted with the county share of mortgaged farms in 1930. In the double-difference specifications, the sample is restricted to counties in the top quartile of the 1930 mortgage-share distribution ( $Q_4$ ). Columns marked *Border* restrict the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment; for example, all counties along the Arizona–New Mexico border are assigned to one border-pair fixed effect. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table 13: Effects of Moratoria on Loan Quantity

|                                 | <i>Outcome: Log of Mortgage Debt per Farm (\$/farm)</i> |                     |                     |                     |                    |                   |                   |
|---------------------------------|---------------------------------------------------------|---------------------|---------------------|---------------------|--------------------|-------------------|-------------------|
|                                 | (1)                                                     | (2)                 | (3)                 | (4)                 | (5)                | (6)               | (7)               |
| Post=1 × Moratoria              | -0.003<br>(0.054)                                       | -0.184**<br>(0.093) | -0.230**<br>(0.097) | -0.238**<br>(0.104) | -0.273*<br>(0.162) | -0.049<br>(0.071) | -0.056<br>(0.079) |
| Post=1 × Moratoria × % Mortgage | -0.029<br>(0.064)                                       | -0.187*<br>(0.098)  | -0.221**<br>(0.095) | -0.258**<br>(0.111) | -0.248*<br>(0.141) |                   |                   |
| County FE                       | ✓                                                       | ✓                   | ✓                   | ✓                   | ✓                  | ✓                 | ✓                 |
| Year FE                         | ✓                                                       | ✓                   | ✓                   | ✓                   | ✓                  | ✓                 | ✓                 |
| Border segment × Post FE        |                                                         |                     | ✓                   |                     |                    |                   | ✓                 |
| New Deal controls               |                                                         |                     |                     | ✓                   |                    |                   |                   |
| Dust Bowl controls              |                                                         |                     |                     |                     | ✓                  |                   |                   |
| Border counties                 |                                                         | ✓                   | ✓                   | ✓                   | ✓                  | ✓                 | ✓                 |
| High % mortgage counties        |                                                         |                     |                     |                     |                    | ✓                 | ✓                 |
| Period                          | 1920-1940                                               | 1920-1940           | 1920-1940           | 1920-1940           | 1920-1940          | 1920-1940         | 1920-1940         |
| Adjusted $R^2$                  | 0.92                                                    | 0.89                | 0.90                | 0.89                | 0.89               | 0.84              | 0.80              |
| Obs                             | 9,183                                                   | 1,555               | 1,418               | 1,555               | 1,555              | 313               | 279               |
| Counties                        | 3,069                                                   | 520                 | 475                 | 520                 | 520                | 105               | 95                |
| Outcome mean                    | 4,320                                                   | 3,846               | 3,951               | 3,846               | 3,846              | 4,886             | 5,208             |
| Outcome SD                      | 3,491                                                   | 3,387               | 3,517               | 3,387               | 3,387              | 3,505             | 3,650             |

*Notes:* Observations are county-years. The outcome is the log of the dollar value of mortgage debt per farm mortgage. Data are from the Agricultural Censuses in 1920 and 1940. In the triple-difference specifications, the treatment effect is interacted with the county share of mortgaged farms in 1930. In the double-difference specifications, the sample is restricted to counties in the top quartile of the 1930 mortgage-share distribution (Q4). Columns marked *Border* restrict the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment; for example, all counties along the Arizona–New Mexico border are assigned to one border-pair fixed effect. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table 14: Decomposition of Risk Premium

|                                              | Annual crisis-with-moratorium probability, $\lambda$ |                  |                  |
|----------------------------------------------|------------------------------------------------------|------------------|------------------|
|                                              | $\lambda = 5\%$                                      | $\lambda = 10\%$ | $\lambda = 15\%$ |
| <b>Panel A. Stay-only states</b>             |                                                      |                  |                  |
| Observed spread (bps)                        | 27.9                                                 | 27.9             | 27.9             |
| Expected moratorium loss                     | -0.9                                                 | -1.8             | -2.7             |
| Priced observable selection                  | 1.7                                                  | 1.7              | 1.7              |
|                                              | (1.5)                                                | (1.5)            | (1.5)            |
| Residual                                     | 27.1                                                 | 28.0             | 28.9             |
| Residual share (%)                           | 97                                                   | 101              | 104              |
| <b>Panel B. Deficiency-regulation states</b> |                                                      |                  |                  |
| Observed spread (bps)                        | 35.4                                                 | 35.4             | 35.4             |
| Expected moratorium loss                     | -0.9                                                 | -1.8             | -2.7             |
| Priced observable selection                  | 3.8                                                  | 3.8              | 3.8              |
|                                              | (2.4)                                                | (2.4)            | (2.4)            |
| Residual                                     | 32.5                                                 | 33.4             | 34.3             |
| Residual share (%)                           | 92                                                   | 94               | 97               |

*Notes:* This table decomposes the moratorium interest-rate spread into three components, as developed in the text. The *observed spread* is the top-quartile double-difference estimate of moratorium adoption on the average farm mortgage interest rate (Post  $\times$  Moratoria; county, year, and border-pair  $\times$  post fixed effects). Panels A and B report separate estimates for stay-only states and for states that also regulated deficiency judgments, both identified against the same non-adopting control group. *Expected moratorium loss* is a risk-neutral, competitive-lender upper bound on the lender's expected recovery loss from a future stay, shown for annual crisis probabilities  $\lambda \in \{5\%, 10\%, 15\%\}$ . *Priced observable selection* reprices the estimated shift in observable borrower-risk characteristics (in logs: value per farm, machinery per cropland acre, and tractor share) at the 1930 pricing gradient; because it captures only observable shifts, it is a lower bound on total selection, and the *Residual* is correspondingly an upper bound on the combined risk-premium and unobserved-selection channels. Selection-component standard errors (in parentheses) are computed by the delta method. Regressions are weighted by 1930 farmland area, and standard errors are clustered at the county level within the border sample.

Table 15: Effects of Moratoria on Migration

|                               | Linked Migration (1930-1940) |                      |                     | Migration (1935-1940) |                      |                    |
|-------------------------------|------------------------------|----------------------|---------------------|-----------------------|----------------------|--------------------|
|                               | (1)                          | (2)                  | (3)                 | (4)                   | (5)                  | (6)                |
| Moratoria $\times$ % Mortgage | -0.020<br>(0.023)            | -0.014<br>(0.009)    | 0.072*<br>(0.041)   | -0.038*<br>(0.022)    | -0.041***<br>(0.013) | 0.045<br>(0.055)   |
| Moratoria                     | -0.020<br>(0.018)            | -0.032***<br>(0.010) | 0.077**<br>(0.037)  | -0.043**<br>(0.019)   | -0.062***<br>(0.012) | 0.031<br>(0.051)   |
| % Mortgage                    | 0.066***<br>(0.007)          | 0.061***<br>(0.004)  | 0.059***<br>(0.014) | 0.072***<br>(0.011)   | 0.063***<br>(0.007)  | 0.060**<br>(0.026) |
| Demographic Controls          | No                           | Yes                  | No                  | No                    | Yes                  | No                 |
| Economic Controls             | No                           | Yes                  | No                  | No                    | Yes                  | No                 |
| Outcome Mean                  | 0.23                         | 0.20                 | 0.23                | 0.28                  | 0.25                 | 0.29               |
| Outcome SD                    | 0.10                         | 0.08                 | 0.11                | 0.12                  | 0.11                 | 0.14               |
| Sample                        | All                          | All                  | Border              | All                   | All                  | Border             |
| Obs                           | 3,083                        | 2,331                | 474                 | 3,083                 | 2,331                | 474                |
| Adjusted $R^2$                | 0.09                         | 0.42                 | 0.20                | 0.05                  | 0.27                 | 0.08               |

Notes: Observations are counties. *Linked Migration (1930–1940)* measures migration using the linked sample between the 1930 and 1940 Censuses (Ruggles et al., 2025). *Migration (1935–1940)* uses the five-year migration question in the 1940 Census (Ruggles et al., 2024). Demographic controls are age, age squared, and an indicator for White. Economic controls are the logs of the number of farms and the number of manufacturing establishments in 1930. Standard errors are heteroskedasticity-robust. Regressions are weighted by 1930 farmland area.

Appendix for

**The Economic Consequences of  
Foreclosure Suspensions in the Great Depression**

Pranjal Drall  
Ewan Rawcliffe

For Online Publication

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## A Additional Figures and Tables

Figure A1: State Politics at the Passage of Moratoria

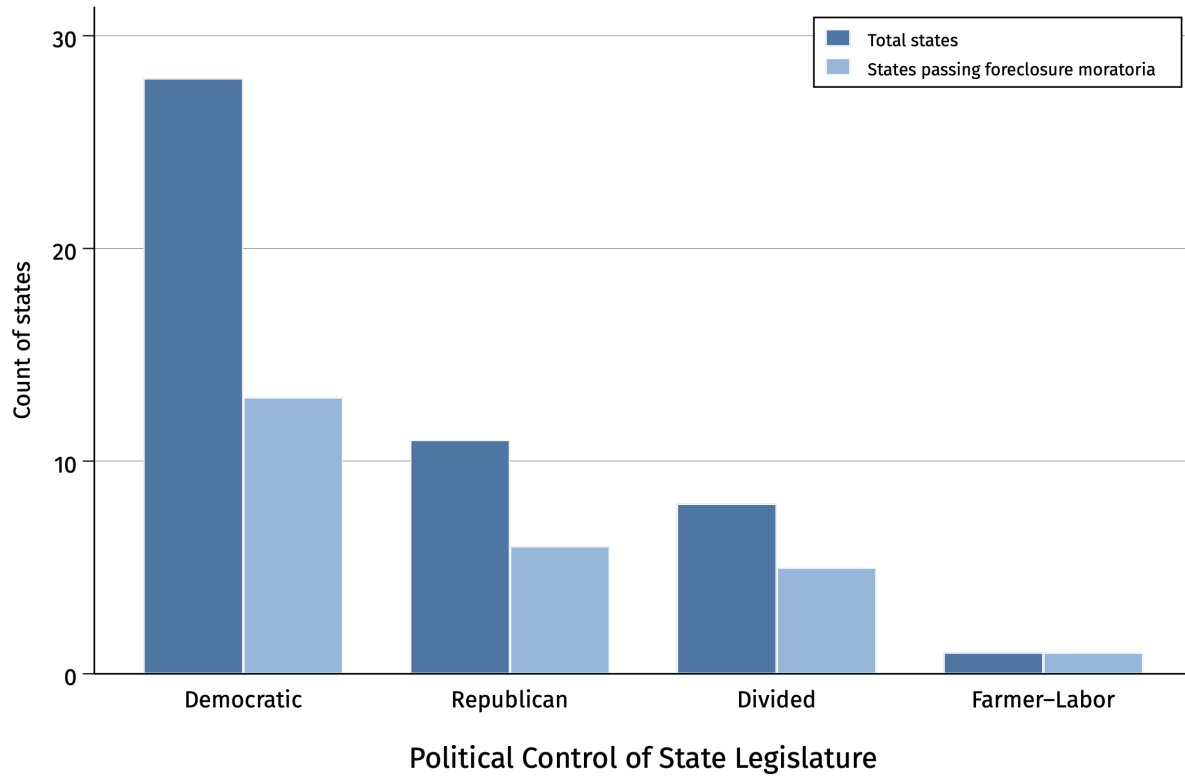


Figure A2: Predictors of Mortgage Exposure in 1930

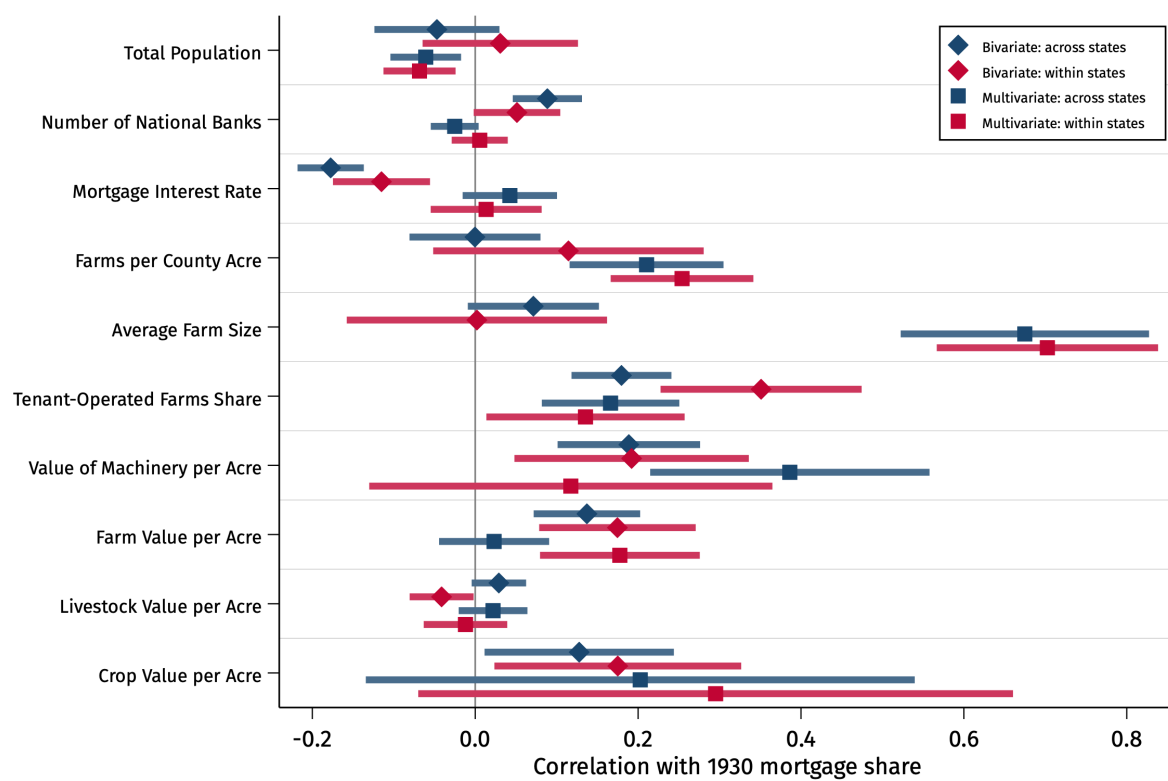


Figure A3: Balance of Border Counties Sample

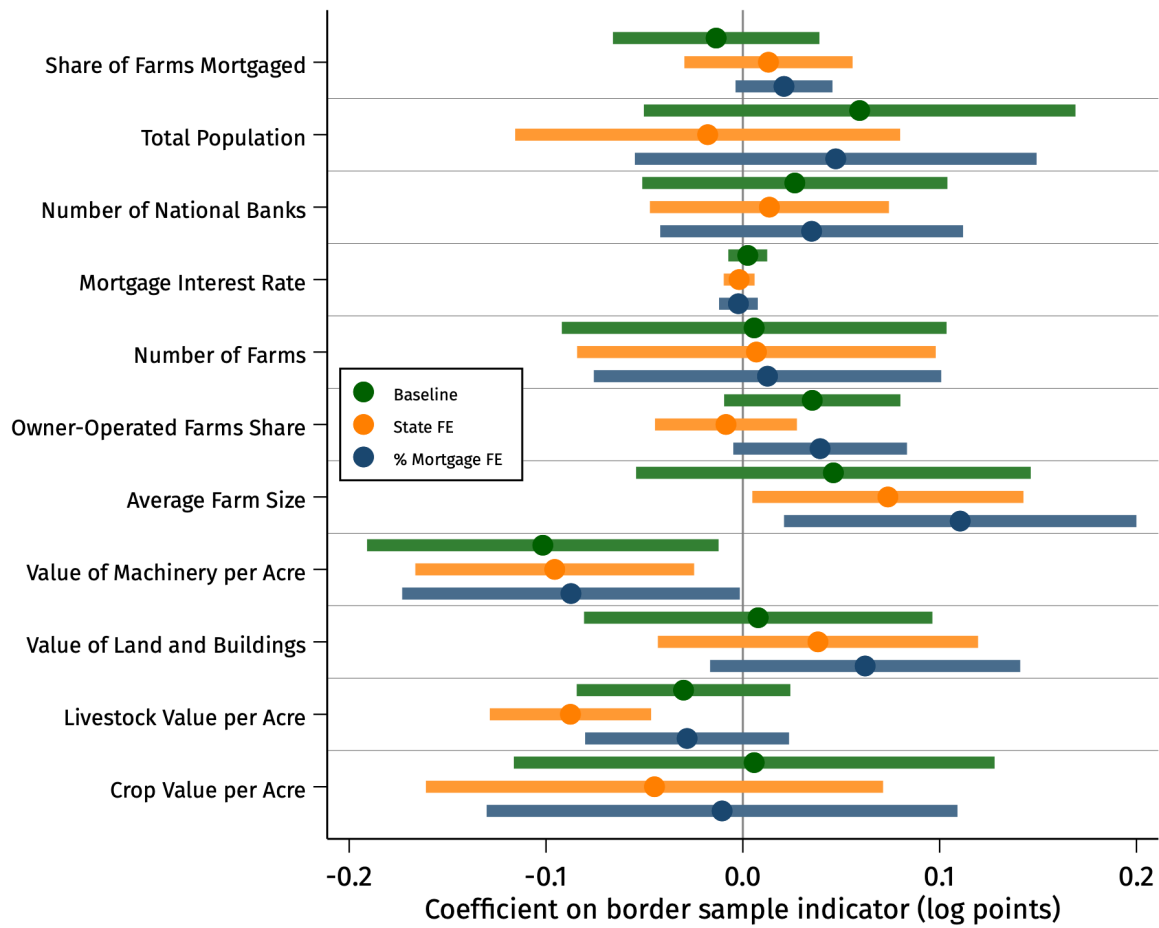
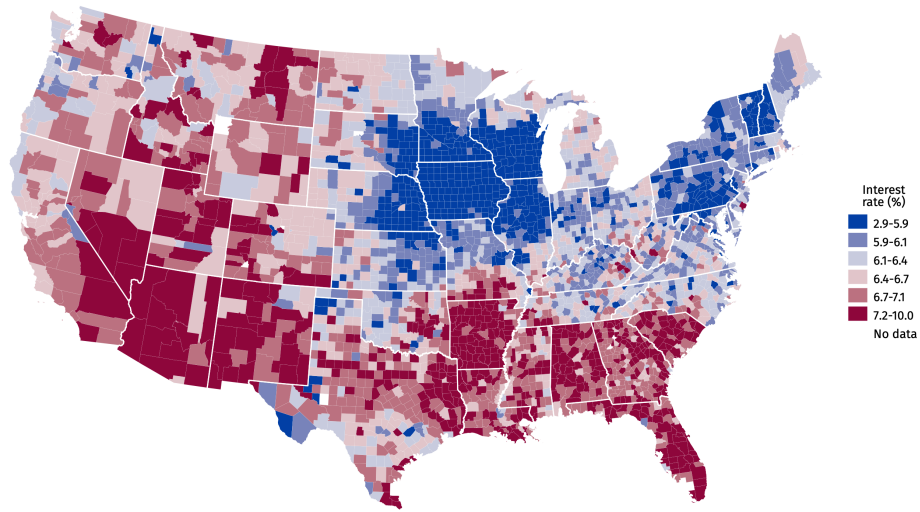
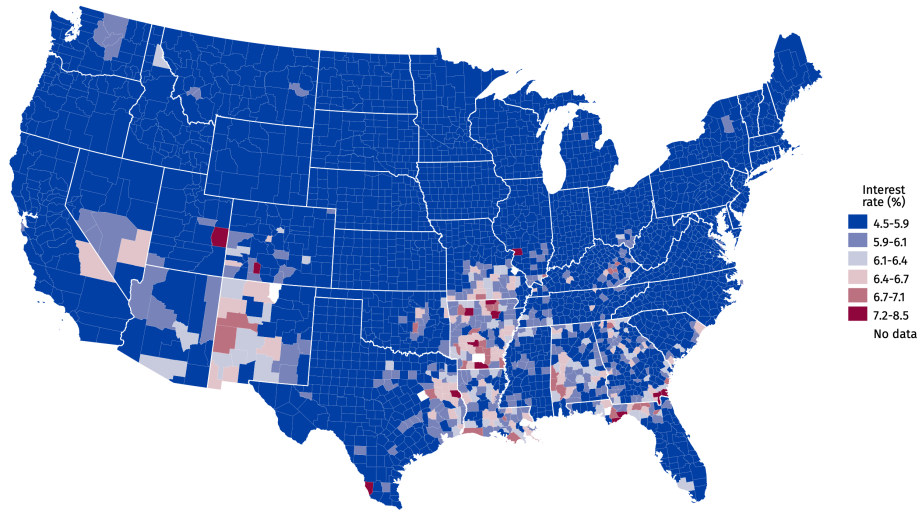


Figure A4: Variation in Farm Mortgage Interest Rates (1930–1940)



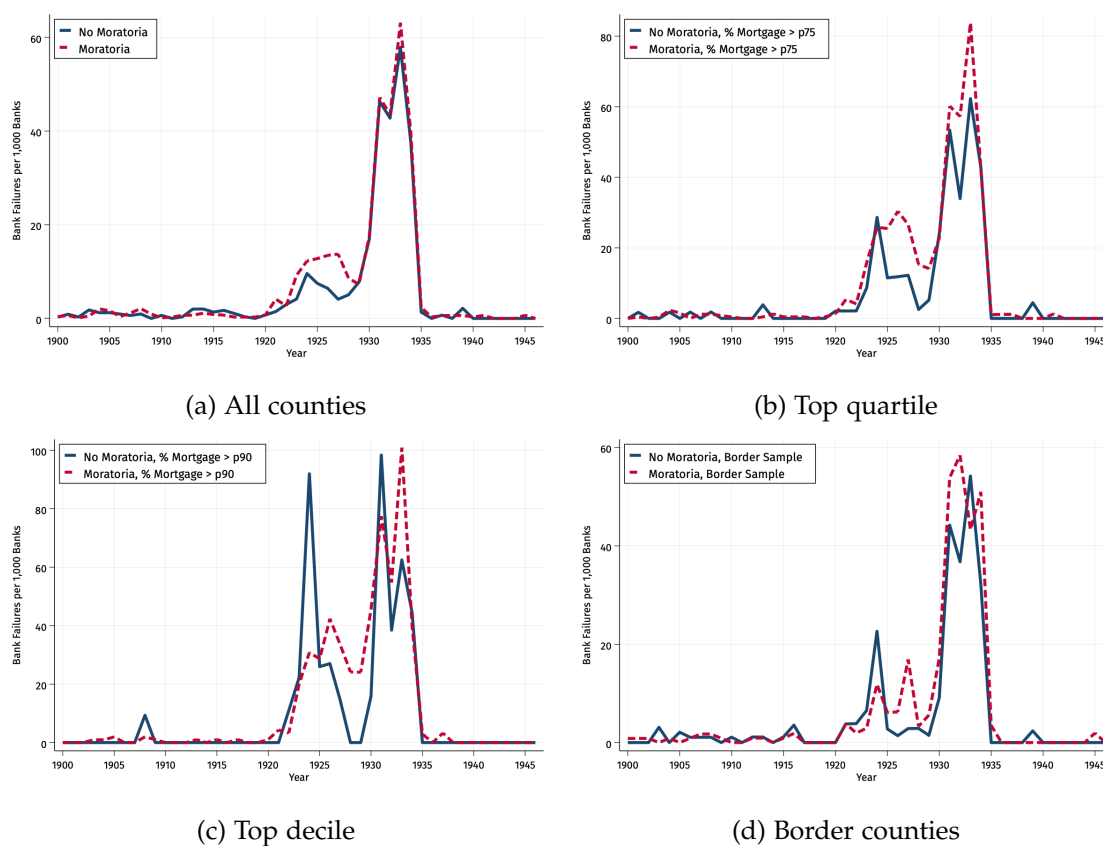
(a) Farm mortgage interest rates in 1930



(b) Farm mortgage interest rates in 1940

*Notes:* The maps show county-level farm mortgage interest rates in 1930 (upper panel) and 1940 (lower panel). A common color scale is used in both panels to highlight the substantial convergence in interest rates over the decade. Data are from [Haines et al. \(2018\)](#).

Figure A5: Bank Failure Rates



Notes: Figure plots bank failure rates over time in moratorium and non-moratorium states. Panels show data from all counties, counties in the top quartile and decile of the 1930 mortgage share distribution, and counties in the border sample. Data are from [Correia et al. \(2026\)](#).

Table A1: Effects of Moratoria on Occupational Transitions, Other Years

|                                      | Stay in Ag.         |                      | Stay Owner           |                     | Ag. to Manuf.        |                     |
|--------------------------------------|---------------------|----------------------|----------------------|---------------------|----------------------|---------------------|
|                                      | (1)                 | (2)                  | (3)                  | (4)                 | (5)                  | (6)                 |
| <i>Linked Censuses: 1930 to 1950</i> |                     |                      |                      |                     |                      |                     |
| Moratoria $\times$ % Mortgage        | 0.058***<br>(0.010) | 0.036*<br>(0.018)    | 0.014<br>(0.010)     | 0.015<br>(0.016)    | -0.018***<br>(0.005) | -0.013<br>(0.016)   |
| Moratoria                            | 0.061***<br>(0.010) | 0.034*<br>(0.019)    | 0.019*<br>(0.010)    | 0.028<br>(0.019)    | -0.022***<br>(0.005) | -0.011<br>(0.014)   |
| % Mortgage                           | 0.025***<br>(0.007) | -0.004<br>(0.009)    | 0.027***<br>(0.007)  | 0.004<br>(0.010)    | -0.006**<br>(0.003)  | -0.017*<br>(0.010)  |
| Outcome Mean                         | 0.56                | 0.54                 | 0.36                 | 0.35                | 0.10                 | 0.10                |
| Outcome SD                           | 0.09                | 0.09                 | 0.09                 | 0.09                | 0.05                 | 0.06                |
| Sample                               | All                 | Border               | All                  | Border              | All                  | Border              |
| Obs                                  | 2,333               | 476                  | 2,333                | 476                 | 2,333                | 476                 |
| Adjusted $R^2$                       | 0.29                | 0.01                 | 0.32                 | 0.01                | 0.45                 | 0.04                |
| <i>Linked Censuses: 1920 to 1930</i> |                     |                      |                      |                     |                      |                     |
| Moratoria $\times$ % Mortgage        | 0.003<br>(0.006)    | 0.016<br>(0.020)     | -0.007<br>(0.008)    | 0.014<br>(0.030)    | 0.002<br>(0.002)     | 0.004<br>(0.007)    |
| Moratoria                            | 0.025**<br>(0.010)  | 0.020<br>(0.028)     | 0.012<br>(0.013)     | 0.034<br>(0.043)    | -0.006*<br>(0.003)   | 0.004<br>(0.011)    |
| % Mortgage                           | -0.013**<br>(0.005) | -0.034***<br>(0.010) | -0.020***<br>(0.008) | -0.030**<br>(0.015) | -0.004**<br>(0.002)  | -0.009**<br>(0.004) |
| Demographic Controls                 | Yes                 | No                   | Yes                  | No                  | Yes                  | No                  |
| Economic Controls                    | Yes                 | No                   | Yes                  | No                  | Yes                  | No                  |
| Outcome Mean                         | 0.70                | 0.68                 | 0.50                 | 0.49                | 0.06                 | 0.07                |
| Outcome SD                           | 0.08                | 0.09                 | 0.10                 | 0.11                | 0.03                 | 0.03                |
| Sample                               | All                 | Border               | All                  | Border              | All                  | Border              |
| Obs                                  | 2,763               | 472                  | 2,763                | 472                 | 2,763                | 472                 |
| Adjusted $R^2$                       | 0.35                | 0.05                 | 0.25                 | 0.02                | 0.40                 | 0.02                |

Notes: Observations are counties. Data are from [Ruggles et al. \(2024\)](#), [Ruggles et al. \(2025\)](#), and [Haines et al. \(2010\)](#). County-level means are constructed from the linked IPUMS full-count sample. The sample is restricted to men active in the labor force in both the baseline and endline years who worked in agriculture in the baseline year. Demographic controls are age, age squared, and an indicator for White. Economic controls are the logs of the number of farms and the number of manufacturing establishments in the baseline year. Standard errors are heteroskedasticity-robust. Regressions are weighted by farmland area in the baseline year.

Table A2: Effects of Moratoria on Farm Ownership Structure

|                                               | Triple Difference    |                      |                  | Double Difference   |                  |                  |
|-----------------------------------------------|----------------------|----------------------|------------------|---------------------|------------------|------------------|
|                                               | (1)                  | (2)                  | (3)              | (4)                 | (5)              | (6)              |
| Post=1 $\times$ Moratoria $\times$ % Mortgage | -0.036<br>(0.030)    | 0.063<br>(0.051)     | 0.087<br>(0.064) |                     |                  |                  |
| Post=1 $\times$ % Mortgage                    | -0.044***<br>(0.013) | -0.074***<br>(0.027) | 0.014<br>(0.025) |                     |                  |                  |
| Moratoria                                     | -0.105***<br>(0.006) |                      |                  | -0.014**<br>(0.006) |                  |                  |
| Post=1 $\times$ Moratoria                     | -0.039<br>(0.025)    | 0.075<br>(0.060)     | 0.085<br>(0.070) | 0.013<br>(0.013)    | 0.076<br>(0.057) | 0.040<br>(0.042) |
| County FE                                     | Yes                  | Yes                  | Yes              | Yes                 | Yes              | Yes              |
| Year FE                                       | Yes                  | Yes                  | Yes              | Yes                 | Yes              | Yes              |
| Border Pair $\times$ Post FE                  | No                   | No                   | Yes              | No                  | No               | Yes              |
| Sample                                        | All                  | Border               | Border           | Q4                  | Q4+Border        | Q4+Border        |
| Period                                        | 1900-1959            | 1900-1959            | 1900-1959        | 1900-1959           | 1900-1959        | 1900-1959        |
| Adjusted $R^2$                                | 0.69                 | 0.61                 | 0.62             | 0.52                | 0.42             | 0.40             |
| Obs                                           | 33,646               | 5,675                | 5,187            | 8,432               | 1,144            | 1,036            |
| Counties                                      | 3,088                | 523                  | 478              | 769                 | 105              | 95               |

Notes: Observations are county-years. The outcome is the log share of farms operated by owners in a county. Data are from the Agricultural Censuses, 1900–1960. In the triple-difference specifications, the treatment effect is interacted with the county share of mortgaged farms in 1930. In the double-difference specifications, the sample is restricted to counties in the top quartile of the 1930 mortgage-share distribution (Q4). Columns marked *Border* restrict the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment; for example, all counties along the Arizona–New Mexico border are assigned to one border-pair fixed effect. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table A3: Effects of Moratoria on Agriculture by Leaveout Region

|                                               | (1)<br>Value         | (2)<br>Farms        | (3)<br>Farm Size     | (4)<br>Owner Share   | (5)<br>Tractors      | (6)<br>Capital/Labor | (7)<br>Crop Prod.  | (8)<br>Animal Prod. |
|-----------------------------------------------|----------------------|---------------------|----------------------|----------------------|----------------------|----------------------|--------------------|---------------------|
| <b>Baseline: All Regions</b>                  |                      |                     |                      |                      |                      |                      |                    |                     |
| Post=1 $\times$ Moratoria $\times$ % Mortgage | -0.167***<br>(0.038) | 0.230***<br>(0.060) | -0.323***<br>(0.081) | -0.036<br>(0.030)    | 0.000<br>(0.001)     | -0.209***<br>(0.048) | 0.057<br>(0.100)   | 0.044<br>(0.078)    |
| Adjusted R <sup>2</sup>                       | 0.93                 | 0.88                | 0.93                 | 0.69                 | 0.75                 | 0.93                 | 0.82               | 0.73                |
| Obs                                           | 33,648               | 33,692              | 33,650               | 33,646               | 18,402               | 8,686                | 27,598             | 27,496              |
| Counties                                      | 3,088                | 3,089               | 3,088                | 3,088                | 3,076                | 2,924                | 3,086              | 3,085               |
| <b>Excluded Region: Northeast</b>             |                      |                     |                      |                      |                      |                      |                    |                     |
| Post=1 $\times$ Moratoria $\times$ % Mortgage | -0.184***<br>(0.038) | 0.227***<br>(0.063) | -0.347***<br>(0.080) | -0.036<br>(0.031)    | 0.001<br>(0.001)     | -0.237***<br>(0.050) | 0.053<br>(0.105)   | 0.075<br>(0.079)    |
| Adjusted R <sup>2</sup>                       | 0.93                 | 0.88                | 0.93                 | 0.68                 | 0.75                 | 0.93                 | 0.82               | 0.72                |
| Obs                                           | 31,275               | 31,316              | 31,274               | 31,271               | 17,106               | 8,108                | 25,654             | 25,552              |
| Counties                                      | 2,872                | 2,873               | 2,872                | 2,872                | 2,860                | 2,709                | 2,870              | 2,869               |
| <b>Excluded Region: Midwest</b>               |                      |                     |                      |                      |                      |                      |                    |                     |
| Post=1 $\times$ Moratoria $\times$ % Mortgage | -0.165***<br>(0.053) | 0.250***<br>(0.069) | -0.424***<br>(0.082) | 0.018<br>(0.028)     | 0.001*<br>(0.001)    | -0.107*<br>(0.063)   | 0.231**<br>(0.104) | 0.099<br>(0.085)    |
| Adjusted R <sup>2</sup>                       | 0.94                 | 0.87                | 0.93                 | 0.67                 | 0.75                 | 0.92                 | 0.83               | 0.64                |
| Obs                                           | 22,050               | 22,094              | 22,052               | 22,048               | 12,070               | 5,560                | 18,102             | 18,011              |
| Counties                                      | 2,031                | 2,032               | 2,031                | 2,031                | 2,020                | 1,882                | 2,030              | 2,029               |
| <b>Excluded Region: South Atlantic</b>        |                      |                     |                      |                      |                      |                      |                    |                     |
| Post=1 $\times$ Moratoria $\times$ % Mortgage | -0.112***<br>(0.039) | 0.234***<br>(0.062) | -0.314***<br>(0.079) | -0.021<br>(0.033)    | 0.000<br>(0.000)     | -0.166***<br>(0.053) | 0.121<br>(0.105)   | 0.022<br>(0.084)    |
| Adjusted R <sup>2</sup>                       | 0.92                 | 0.88                | 0.93                 | 0.68                 | 0.77                 | 0.93                 | 0.82               | 0.74                |
| Obs                                           | 29,253               | 29,257              | 29,257               | 29,252               | 16,037               | 7,591                | 24,010             | 23,908              |
| Counties                                      | 2,675                | 2,675               | 2,675                | 2,675                | 2,674                | 2,559                | 2,674              | 2,674               |
| <b>Excluded Region: South</b>                 |                      |                     |                      |                      |                      |                      |                    |                     |
| Post=1 $\times$ Moratoria $\times$ % Mortgage | -0.139**<br>(0.059)  | 0.120***<br>(0.039) | -0.073<br>(0.101)    | -0.089***<br>(0.020) | -0.002***<br>(0.001) | -0.250***<br>(0.058) | -0.236<br>(0.150)  | -0.005<br>(0.148)   |
| Adjusted R <sup>2</sup>                       | 0.90                 | 0.88                | 0.93                 | 0.84                 | 0.77                 | 0.94                 | 0.81               | 0.71                |
| Obs                                           | 22,771               | 22,815              | 22,773               | 22,773               | 12,444               | 5,827                | 18,661             | 18,605              |
| Counties                                      | 2,095                | 2,096               | 2,095                | 2,095                | 2,083                | 1,971                | 2,093              | 2,092               |
| <b>Excluded Region: West</b>                  |                      |                     |                      |                      |                      |                      |                    |                     |
| Post=1 $\times$ Moratoria $\times$ % Mortgage | -0.115***<br>(0.035) | 0.284***<br>(0.057) | -0.295***<br>(0.084) | -0.098***<br>(0.035) | 0.000<br>(0.001)     | -0.335***<br>(0.039) | 0.014<br>(0.108)   | -0.098<br>(0.063)   |
| County FE                                     | Yes                  | Yes                 | Yes                  | Yes                  | Yes                  | Yes                  | Yes                | Yes                 |
| Year FE                                       | Yes                  | Yes                 | Yes                  | Yes                  | Yes                  | Yes                  | Yes                | Yes                 |
| Adjusted R <sup>2</sup>                       | 0.93                 | 0.89                | 0.94                 | 0.67                 | 0.75                 | 0.94                 | 0.81               | 0.82                |
| Obs                                           | 29,250               | 29,293              | 29,251               | 29,247               | 15,954               | 7,661                | 23,970             | 23,914              |
| Counties                                      | 2,680                | 2,681               | 2,680                | 2,680                | 2,668                | 2,576                | 2,678              | 2,677               |

Notes: Observations are county-years. Each panel re-estimates the baseline triple-difference specification while excluding the indicated U.S. Census Bureau region. Outcomes are in logs. Data are from the Agricultural Censuses in 1920 and 1940. Lower-order interactions are included in all specifications but not reported. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table A4: Effects of Moratoria on Farmland Value

|                                 | Outcome: Log of Value per Farmland Acre (\$/acre) |                    |                     |                  |                    |                     |                  |                    |                     |
|---------------------------------|---------------------------------------------------|--------------------|---------------------|------------------|--------------------|---------------------|------------------|--------------------|---------------------|
|                                 | (1)                                               | (2)                | (3)                 | (4)              | (5)                | (6)                 | (7)              | (8)                | (9)                 |
| Post=1 × Moratoria              | 0.143**<br>(0.056)                                |                    | 0.263***<br>(0.099) | 0.096<br>(0.067) | 0.149**<br>(0.058) | 0.077<br>(0.083)    | 0.058<br>(0.037) | 0.197**<br>(0.080) | 0.148***<br>(0.051) |
| Post=1 × Moratoria × % Mortgage | 0.156**<br>(0.063)                                | 0.137**<br>(0.057) | 0.264**<br>(0.104)  | 0.076<br>(0.070) | 0.159**<br>(0.066) | 0.241***<br>(0.086) |                  |                    |                     |
| County FE                       | ✓                                                 | ✓                  | ✓                   | ✓                | ✓                  | ✓                   | ✓                | ✓                  | ✓                   |
| Year FE                         | ✓                                                 | ✓                  | ✓                   | ✓                | ✓                  | ✓                   | ✓                | ✓                  | ✓                   |
| State × Year FE                 |                                                   | ✓                  |                     |                  |                    |                     |                  |                    |                     |
| Border segment × Post FE        |                                                   |                    |                     | ✓                |                    |                     |                  |                    | ✓                   |
| New Deal controls               |                                                   |                    |                     |                  | ✓                  |                     |                  |                    |                     |
| Dust Bowl controls              |                                                   |                    |                     |                  |                    | ✓                   |                  |                    |                     |
| Border counties                 |                                                   |                    | ✓                   | ✓                |                    |                     |                  | ✓                  | ✓                   |
| High % mortgage counties        |                                                   |                    |                     |                  |                    |                     | ✓                | ✓                  | ✓                   |
| Period                          | 1900-1959                                         | 1900-1959          | 1900-1959           | 1900-1959        | 1900-1959          | 1900-1959           | 1900-1959        | 1900-1959          | 1900-1959           |
| Adjusted $R^2$                  | 0.92                                              | 0.95               | 0.93                | 0.94             | 0.92               | 0.92                | 0.91             | 0.91               | 0.92                |
| Obs                             | 33,643                                            | 33,643             | 5,677               | 5,189            | 33,643             | 33,643              | 8,425            | 1,147              | 1,039               |
| Counties                        | 3,088                                             | 3,088              | 523                 | 478              | 3,088              | 3,088               | 768              | 105                | 95                  |
| Outcome mean                    | 49                                                | 49                 | 43                  | 44               | 49                 | 49                  | 50               | 39                 | 42                  |
| Outcome SD                      | 53                                                | 53                 | 52                  | 53               | 53                 | 53                  | 61               | 54                 | 57                  |

*Notes:* Observations are county-years. The outcome is log farmland value per acre. Data are from the Agricultural Censuses, 1900–1960. In the triple-difference specifications, the treatment effect is interacted with the county share of mortgaged farms in 1930. In the double-difference specifications, the sample is restricted to counties in the top quartile of the 1930 mortgage-share distribution ( $Q_4$ ). Columns marked *Border* restrict the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table A5: Effects of Moratoria on Capital/Labor Ratio

|                                 | Outcome: Log of Capital/Labor Ratio (\$/worker) |                    |                   |                   |                      |                      |                      |                   |                   |
|---------------------------------|-------------------------------------------------|--------------------|-------------------|-------------------|----------------------|----------------------|----------------------|-------------------|-------------------|
|                                 | (1)                                             | (2)                | (3)               | (4)               | (5)                  | (6)                  | (7)                  | (8)               | (9)               |
| Post=1 × Moratoria              | -0.243***<br>(0.046)                            |                    | -0.187<br>(0.142) | -0.133<br>(0.201) | -0.244***<br>(0.049) | -0.320***<br>(0.085) | -0.141***<br>(0.034) | -0.113<br>(0.150) | -0.141<br>(0.157) |
| Post=1 × Moratoria × % Mortgage | -0.209***<br>(0.048)                            | -0.129*<br>(0.069) | -0.193<br>(0.119) | -0.135<br>(0.186) | -0.202***<br>(0.052) | -0.214***<br>(0.075) |                      |                   |                   |
| County FE                       | ✓                                               | ✓                  | ✓                 | ✓                 | ✓                    | ✓                    | ✓                    | ✓                 | ✓                 |
| Year FE                         | ✓                                               | ✓                  | ✓                 | ✓                 | ✓                    | ✓                    | ✓                    | ✓                 | ✓                 |
| State × Year FE                 |                                                 | ✓                  |                   |                   |                      |                      |                      |                   |                   |
| Border segment × Post FE        |                                                 |                    |                   | ✓                 |                      |                      |                      |                   | ✓                 |
| New Deal controls               |                                                 |                    |                   |                   | ✓                    |                      |                      |                   |                   |
| Dust Bowl controls              |                                                 |                    |                   |                   |                      | ✓                    |                      |                   |                   |
| Border counties                 |                                                 |                    | ✓                 | ✓                 |                      |                      |                      | ✓                 | ✓                 |
| High % mortgage counties        |                                                 |                    |                   |                   |                      |                      | ✓                    | ✓                 | ✓                 |
| Period                          | 1930-1945                                       | 1930-1945          | 1930-1945         | 1930-1945         | 1930-1945            | 1930-1945            | 1930-1945            | 1930-1945         | 1930-1945         |
| Adjusted $R^2$                  | 0.93                                            | 0.95               | 0.88              | 0.87              | 0.93                 | 0.93                 | 0.80                 | 0.62              | 0.52              |
| Obs                             | 8,688                                           | 8,688              | 1,466             | 1,333             | 8,688                | 8,688                | 2,153                | 293               | 262               |
| Counties                        | 2,925                                           | 2,925              | 496               | 452               | 2,925                | 2,925                | 725                  | 99                | 90                |
| Outcome mean                    | 509                                             | 509                | 2,253             | 2,508             | 509                  | 509                  | 1,154                | 6,563             | 8,016             |
| Outcome SD                      | 18,151                                          | 18,151             | 44,924            | 47,585            | 18,151               | 18,151               | 29,423               | 78,034            | 86,570            |

Notes: Observations are county-years. The outcome is the log capital-labor ratio in agriculture. Data are from the Agricultural Censuses, 1900–1960. In the triple-difference specifications, the treatment effect is interacted with the county share of mortgaged farms in 1930. In the double-difference specifications, the sample is restricted to counties in the top quartile of the 1930 mortgage-share distribution (Q4). Columns marked *Border* restrict the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table A6: Effects of Moratoria on Capital per Cropland Acre

|                                 | Outcome: Log of Value of Machinery per Cropland Acre (\$/acre) |                      |                    |                      |                      |                      |                      |                    |                      |
|---------------------------------|----------------------------------------------------------------|----------------------|--------------------|----------------------|----------------------|----------------------|----------------------|--------------------|----------------------|
|                                 | (1)                                                            | (2)                  | (3)                | (4)                  | (5)                  | (6)                  | (7)                  | (8)                | (9)                  |
| Post=1 × Moratoria              | -0.360***<br>(0.048)                                           |                      | -0.143*<br>(0.078) | -0.270***<br>(0.078) | -0.363***<br>(0.049) | -0.439***<br>(0.063) | -0.172***<br>(0.030) | -0.143*<br>(0.072) | -0.165***<br>(0.058) |
| Post=1 × Moratoria × % Mortgage | -0.339***<br>(0.064)                                           | -0.339***<br>(0.088) | -0.134*<br>(0.074) | -0.278***<br>(0.081) | -0.347***<br>(0.065) | -0.392***<br>(0.055) |                      |                    |                      |
| County FE                       | ✓                                                              | ✓                    | ✓                  | ✓                    | ✓                    | ✓                    | ✓                    | ✓                  | ✓                    |
| Year FE                         | ✓                                                              | ✓                    | ✓                  | ✓                    | ✓                    | ✓                    | ✓                    | ✓                  | ✓                    |
| State × Year FE                 |                                                                | ✓                    |                    |                      |                      |                      |                      |                    |                      |
| Border segment × Post FE        |                                                                |                      |                    | ✓                    |                      |                      |                      |                    | ✓                    |
| New Deal controls               |                                                                |                      |                    |                      | ✓                    |                      |                      |                    |                      |
| Dust Bowl controls              |                                                                |                      |                    |                      |                      | ✓                    |                      |                    |                      |
| Border counties                 |                                                                |                      | ✓                  | ✓                    |                      |                      |                      | ✓                  | ✓                    |
| High % mortgage counties        |                                                                |                      |                    |                      |                      |                      | ✓                    | ✓                  | ✓                    |
| Period                          | 1925-1945                                                      | 1925-1945            | 1925-1945          | 1925-1945            | 1925-1945            | 1925-1945            | 1925-1945            | 1925-1945          | 1925-1945            |
| Adjusted $R^2$                  | 0.85                                                           | 0.87                 | 0.91               | 0.93                 | 0.85                 | 0.85                 | 0.88                 | 0.81               | 0.86                 |
| Obs                             | 12,272                                                         | 12,272               | 2,078              | 1,898                | 12,272               | 12,272               | 3,067                | 416                | 376                  |
| Counties                        | 3,073                                                          | 3,073                | 520                | 475                  | 3,073                | 3,073                | 767                  | 104                | 94                   |
| Outcome mean                    | 12.88                                                          | 12.88                | 17.70              | 18.74                | 12.88                | 12.88                | 14.39                | 7.01               | 7.37                 |
| Outcome SD                      | 53.36                                                          | 53.36                | 79.33              | 83.66                | 53.36                | 53.36                | 63.92                | 4.91               | 5.23                 |

Notes: Observations are county-years. The outcome is log capital per acre of cropland. Data are from the Agricultural Censuses, 1900–1960. In the triple-difference specifications, the treatment effect is interacted with the county share of mortgaged farms in 1930. In the double-difference specifications, the sample is restricted to counties in the top quartile of the 1930 mortgage-share distribution (Q4). Columns marked *Border* restrict the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table A7: Effects of Moratoria on Cropland Share of Farmland

|                                 | Outcome: % of Farm in Cropland |                     |                     |                   |                     |                     |                    |                    |                    |
|---------------------------------|--------------------------------|---------------------|---------------------|-------------------|---------------------|---------------------|--------------------|--------------------|--------------------|
|                                 | (1)                            | (2)                 | (3)                 | (4)               | (5)                 | (6)                 | (7)                | (8)                | (9)                |
| Post=1 × Moratoria              | 0.267***<br>(0.085)            |                     | 0.332***<br>(0.086) | 0.047<br>(0.100)  | 0.276***<br>(0.089) | 0.388**<br>(0.148)  | 0.111**<br>(0.051) | 0.192**<br>(0.085) | 0.093**<br>(0.044) |
| Post=1 × Moratoria × % Mortgage | 0.302***<br>(0.092)            | 0.252***<br>(0.059) | 0.277***<br>(0.085) | -0.044<br>(0.152) | 0.316***<br>(0.097) | 0.423***<br>(0.106) |                    |                    |                    |
| County FE                       | ✓                              | ✓                   | ✓                   | ✓                 | ✓                   | ✓                   | ✓                  | ✓                  | ✓                  |
| Year FE                         | ✓                              | ✓                   | ✓                   | ✓                 | ✓                   | ✓                   | ✓                  | ✓                  | ✓                  |
| State × Year FE                 |                                | ✓                   |                     |                   |                     |                     |                    |                    |                    |
| Border segment × Post FE        |                                |                     |                     | ✓                 |                     |                     |                    |                    | ✓                  |
| New Deal controls               |                                |                     |                     |                   | ✓                   |                     |                    |                    |                    |
| Dust Bowl controls              |                                |                     |                     |                   |                     | ✓                   |                    |                    |                    |
| Border counties                 |                                |                     | ✓                   | ✓                 |                     |                     |                    | ✓                  | ✓                  |
| High % mortgage counties        |                                |                     |                     |                   |                     |                     | ✓                  | ✓                  | ✓                  |
| Period                          | 1925-1959                      | 1925-1959           | 1925-1959           | 1925-1959         | 1925-1959           | 1925-1959           | 1925-1959          | 1925-1959          | 1925-1959          |
| Adjusted $R^2$                  | 0.94                           | 0.94                | 0.91                | 0.91              | 0.94                | 0.94                | 0.97               | 0.93               | 0.93               |
| Obs                             | 24,522                         | 24,522              | 4,153               | 3,793             | 24,522              | 24,522              | 6,139              | 836                | 756                |
| Counties                        | 48                             | 48                  | 523                 | 478               | 48                  | 48                  | 36                 | 105                | 95                 |
| Outcome mean                    | 0.42                           | 0.42                | 0.36                | 0.37              | 0.42                | 0.42                | 0.48               | 0.38               | 0.39               |
| Outcome SD                      | 0.22                           | 0.22                | 0.23                | 0.23              | 0.22                | 0.22                | 0.23               | 0.22               | 0.23               |

Notes: Observations are county-years. The outcome is the share of farmland in cropland. Data are from the Agricultural Censuses, 1900–1960. In the triple-difference specifications, the treatment effect is interacted with the county share of mortgaged farms in 1930. In the double-difference specifications, the sample is restricted to counties in the top quartile of the 1930 mortgage-share distribution (Q4). Columns marked *Border* restrict the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table A8: Effects of Moratoria on Crop Yields

|                                 | Outcome: Log of Crop Value per Cropland Acre (\$/acre) |                   |                   |                    |                     |                     |                      |                    |                     |
|---------------------------------|--------------------------------------------------------|-------------------|-------------------|--------------------|---------------------|---------------------|----------------------|--------------------|---------------------|
|                                 | (1)                                                    | (2)               | (3)               | (4)                | (5)                 | (6)                 | (7)                  | (8)                | (9)                 |
| Post=1 × Moratoria              | -0.238**<br>(0.096)                                    |                   | 0.218*<br>(0.128) | -0.077<br>(0.160)  | -0.224**<br>(0.100) | -0.424**<br>(0.174) | -0.306***<br>(0.071) | 0.243**<br>(0.099) | 0.254***<br>(0.093) |
| Post=1 × Moratoria × % Mortgage | -0.235**<br>(0.106)                                    | -0.122<br>(0.166) | -0.021<br>(0.128) | -0.357*<br>(0.186) | -0.204*<br>(0.113)  | -0.351**<br>(0.160) |                      |                    |                     |
| County FE                       | ✓                                                      | ✓                 | ✓                 | ✓                  | ✓                   | ✓                   | ✓                    | ✓                  | ✓                   |
| Year FE                         | ✓                                                      | ✓                 | ✓                 | ✓                  | ✓                   | ✓                   | ✓                    | ✓                  | ✓                   |
| State × Year FE                 |                                                        | ✓                 |                   |                    |                     |                     |                      |                    |                     |
| Border segment × Post FE        |                                                        |                   |                   | ✓                  |                     |                     |                      |                    | ✓                   |
| New Deal controls               |                                                        |                   |                   |                    | ✓                   |                     |                      |                    |                     |
| Dust Bowl controls              |                                                        |                   |                   |                    |                     | ✓                   |                      |                    |                     |
| Border counties                 |                                                        |                   | ✓                 | ✓                  |                     |                     |                      | ✓                  | ✓                   |
| High % mortgage counties        |                                                        |                   |                   |                    |                     |                     | ✓                    | ✓                  | ✓                   |
| Period                          | 1925-1959                                              | 1925-1959         | 1925-1959         | 1925-1959          | 1925-1959           | 1925-1959           | 1925-1959            | 1925-1959          | 1925-1959           |
| Adjusted R <sup>2</sup>         | 0.60                                                   | 0.71              | 0.68              | 0.72               | 0.60                | 0.61                | 0.65                 | 0.73               | 0.72                |
| Obs                             | 21,425                                                 | 21,425            | 3,627             | 3,312              | 21,425              | 21,425              | 5,370                | 732                | 662                 |
| Counties                        | 3,067                                                  | 3,067             | 521               | 476                | 3,067               | 3,067               | 768                  | 105                | 95                  |
| Outcome mean                    | 20.12                                                  | 20.12             | 20.89             | 21.46              | 20.12               | 20.12               | 17.11                | 17.82              | 19.25               |
| Outcome SD                      | 17.85                                                  | 17.85             | 16.60             | 17.10              | 17.85               | 17.85               | 21.39                | 21.65              | 23.26               |

Notes: Observations are county-years. The outcome is crop output per acre of cropland. Data are from the Agricultural Censuses, 1900–1960. In the triple-difference specifications, the treatment effect is interacted with the county share of mortgaged farms in 1930. In the double-difference specifications, the sample is restricted to counties in the top quartile of the 1930 mortgage-share distribution (Q4). Columns marked *Border* restrict the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table A9: Effects of Moratoria on Harvested Cropland

|                                 | Outcome: % Harvested Cropland |                     |                   |                     |                     |                      |                   |                  |                  |
|---------------------------------|-------------------------------|---------------------|-------------------|---------------------|---------------------|----------------------|-------------------|------------------|------------------|
|                                 | (1)                           | (2)                 | (3)               | (4)                 | (5)                 | (6)                  | (7)               | (8)              | (9)              |
| Post=1 × Moratoria              | -0.124*<br>(0.068)            |                     | -0.020<br>(0.065) | -0.121<br>(0.076)   | -0.124*<br>(0.068)  | -0.148*<br>(0.082)   | -0.042<br>(0.089) | 0.039<br>(0.051) | 0.061<br>(0.044) |
| Post=1 × Moratoria × % Mortgage | -0.138**<br>(0.052)           | -0.070**<br>(0.028) | -0.098<br>(0.065) | -0.227**<br>(0.098) | -0.140**<br>(0.053) | -0.166***<br>(0.062) |                   |                  |                  |
| County FE                       | ✓                             | ✓                   | ✓                 | ✓                   | ✓                   | ✓                    | ✓                 | ✓                | ✓                |
| Year FE                         | ✓                             | ✓                   | ✓                 | ✓                   | ✓                   | ✓                    | ✓                 | ✓                | ✓                |
| State × Year FE                 |                               | ✓                   |                   |                     |                     |                      |                   |                  |                  |
| Border segment × Post FE        |                               |                     |                   | ✓                   |                     |                      |                   |                  | ✓                |
| New Deal controls               |                               |                     |                   |                     | ✓                   |                      |                   |                  |                  |
| Dust Bowl controls              |                               |                     |                   |                     |                     | ✓                    |                   |                  |                  |
| Border counties                 |                               |                     | ✓                 | ✓                   |                     |                      |                   | ✓                | ✓                |
| High % mortgage counties        |                               |                     |                   |                     |                     |                      | ✓                 | ✓                | ✓                |
| Period                          | 1925-1959                     | 1925-1959           | 1925-1959         | 1925-1959           | 1925-1959           | 1925-1959            | 1925-1959         | 1925-1959        | 1925-1959        |
| Adjusted $R^2$                  | 0.34                          | 0.57                | 0.31              | 0.34                | 0.33                | 0.34                 | 0.33              | 0.37             | 0.38             |
| Obs                             | 24,517                        | 24,517              | 4,151             | 3,791               | 24,517              | 24,517               | 6,138             | 835              | 755              |
| Counties                        | 48                            | 48                  | 523               | 478                 | 48                  | 48                   | 36                | 105              | 95               |
| Outcome mean                    | 0.84                          | 0.84                | 0.81              | 0.81                | 0.84                | 0.84                 | 0.87              | 0.83             | 0.84             |
| Outcome SD                      | 0.12                          | 0.12                | 0.13              | 0.13                | 0.12                | 0.12                 | 0.13              | 0.10             | 0.10             |

Notes: Observations are county-years. The outcome is the share of cropland that is harvested. Data are from the Agricultural Censuses, 1900–1960. In the triple-difference specifications, the treatment effect is interacted with the county share of mortgaged farms in 1930. In the double-difference specifications, the sample is restricted to counties in the top quartile of the 1930 mortgage-share distribution (Q4). Columns marked *Border* restrict the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table A10: Effects of Moratoria on Share of Farms with Tractors

|                                 | Outcome: Share of Farms with Tractors |                      |                     |                    |                     |                      |                   |                      |                    |
|---------------------------------|---------------------------------------|----------------------|---------------------|--------------------|---------------------|----------------------|-------------------|----------------------|--------------------|
|                                 | (1)                                   | (2)                  | (3)                 | (4)                | (5)                 | (6)                  | (7)               | (8)                  | (9)                |
| Post=1 × Moratoria              | -0.013<br>(0.024)                     |                      | -0.073**<br>(0.032) | -0.054*<br>(0.029) | -0.018<br>(0.025)   | -0.064***<br>(0.024) | -0.006<br>(0.011) | -0.068***<br>(0.025) | -0.049*<br>(0.027) |
| Post=1 × Moratoria × % Mortgage | -0.066**<br>(0.029)                   | -0.065***<br>(0.019) | -0.071**<br>(0.033) | -0.053*<br>(0.027) | -0.068**<br>(0.030) | -0.110***<br>(0.027) |                   |                      |                    |
| County FE                       | ✓                                     | ✓                    | ✓                   | ✓                  | ✓                   | ✓                    | ✓                 | ✓                    | ✓                  |
| Year FE                         | ✓                                     | ✓                    | ✓                   | ✓                  | ✓                   | ✓                    | ✓                 | ✓                    | ✓                  |
| State × Year FE                 |                                       | ✓                    |                     |                    |                     |                      |                   |                      |                    |
| Border segment × Post FE        |                                       |                      |                     | ✓                  |                     |                      |                   |                      | ✓                  |
| New Deal controls               |                                       |                      |                     |                    | ✓                   |                      |                   |                      |                    |
| Dust Bowl controls              |                                       |                      |                     |                    |                     | ✓                    |                   |                      |                    |
| Border counties                 |                                       |                      | ✓                   | ✓                  |                     |                      |                   | ✓                    | ✓                  |
| High % mortgage counties        |                                       |                      |                     |                    |                     |                      | ✓                 | ✓                    | ✓                  |
| Period                          | 1925-1959                             | 1925-1959            | 1925-1959           | 1925-1959          | 1925-1959           | 1925-1959            | 1925-1959         | 1925-1959            | 1925-1959          |
| Adjusted $R^2$                  | 0.91                                  | 0.94                 | 0.90                | 0.92               | 0.91                | 0.91                 | 0.90              | 0.89                 | 0.89               |
| Obs                             | 21,459                                | 21,459               | 3,635               | 3,320              | 21,459              | 21,459               | 5,369             | 735                  | 665                |
| Counties                        | 3,075                                 | 3,075                | 521                 | 476                | 3,075               | 3,075                | 767               | 105                  | 95                 |
| Outcome mean                    | 0.20                                  | 0.20                 | 0.17                | 0.17               | 0.20                | 0.20                 | 0.31              | 0.28                 | 0.28               |
| Outcome SD                      | 0.17                                  | 0.17                 | 0.15                | 0.15               | 0.17                | 0.17                 | 0.18              | 0.18                 | 0.18               |

Notes: Observations are county-years. The outcome is the share of farms using tractors. Data are from the Agricultural Censuses, 1900–1960. In the triple-difference specifications, the treatment effect is interacted with the county share of mortgaged farms in 1930. In the double-difference specifications, the sample is restricted to counties in the top quartile of the 1930 mortgage-share distribution (Q4). Columns marked *Border* restrict the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table A11: Effects of Moratoria on Extensive Margin

|                                 | Outcome: Manufacturing Index Extensive Margin (z-score) |                  |                   |                  |                   |                   |                  |                   |                  |
|---------------------------------|---------------------------------------------------------|------------------|-------------------|------------------|-------------------|-------------------|------------------|-------------------|------------------|
|                                 | (1)                                                     | (2)              | (3)               | (4)              | (5)               | (6)               | (7)              | (8)               | (9)              |
| Post=1 × Moratoria              | 0.019<br>(0.071)                                        |                  | -0.014<br>(0.039) | 0.018<br>(0.065) |                   |                   | 0.048<br>(0.105) | -0.000<br>(0.033) | 0.045<br>(0.032) |
| Post=1 × Moratoria × % Mortgage | -0.041<br>(0.036)                                       | 0.028<br>(0.042) | -0.014<br>(0.040) | 0.048<br>(0.094) | 0.146*<br>(0.079) | -0.073<br>(0.048) |                  |                   |                  |
| County FE                       | ✓                                                       | ✓                | ✓                 | ✓                | ✓                 | ✓                 | ✓                | ✓                 | ✓                |
| Year FE                         | ✓                                                       |                  | ✓                 | ✓                |                   |                   | ✓                | ✓                 | ✓                |
| State × Year FE                 |                                                         | ✓                |                   |                  | ✓                 | ✓                 |                  |                   |                  |
| Border segment × Post FE        |                                                         |                  |                   | ✓                |                   |                   |                  |                   | ✓                |
| New Deal controls               |                                                         |                  |                   |                  | ✓                 |                   |                  |                   |                  |
| Dust Bowl controls              |                                                         |                  |                   |                  |                   | ✓                 |                  |                   |                  |
| Border counties                 |                                                         |                  | ✓                 | ✓                |                   |                   |                  | ✓                 | ✓                |
| High % mortgage counties        |                                                         |                  |                   |                  |                   |                   | ✓                | ✓                 | ✓                |
| Period                          | 1920-1958                                               | 1920-1958        | 1920-1958         | 1920-1958        | 1920-1958         | 1920-1958         | 1920-1958        | 1920-1958         | 1920-1958        |
| Adjusted $R^2$                  | 0.60                                                    | 0.61             | 0.90              | 0.90             | 0.69              | 0.61              | 0.49             | 0.93              | 0.93             |
| Obs                             | 24,597                                                  | 24,597           | 4,163             | 3,803            | 24,597            | 24,597            | 6,135            | 840               | 760              |
| Counties                        | 3,088                                                   | 3,088            | 523               | 478              | 3,088             | 3,088             | 767              | 105               | 95               |
| Outcome mean                    | 0                                                       | 0                | 0                 | 0                | 0                 | 0                 | -0               | -0                | -0               |
| Outcome SD                      | 1                                                       | 1                | 2                 | 2                | 1                 | 1                 | 1                | 1                 | 1                |

*Notes:* Observations are county-years. The outcome is the manufacturing extensive-margin principal-component index. Manufacturing data are from the Census of Manufactures and related historical county compilations. In the triple-difference specifications, the treatment effect is interacted with the county share of mortgaged farms in 1930. In the double-difference specifications, the sample is restricted to counties in the top quartile of the 1930 mortgage-share distribution ( $Q_4$ ). Columns marked *Border* restrict the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table A12: Effects of Moratoria on Intensive Margin

|                                 | Outcome: Manufacturing Index Intensive Margin (z-score) |                    |                   |                   |                    |                   |                   |                   |                   |
|---------------------------------|---------------------------------------------------------|--------------------|-------------------|-------------------|--------------------|-------------------|-------------------|-------------------|-------------------|
|                                 | (1)                                                     | (2)                | (3)               | (4)               | (5)                | (6)               | (7)               | (8)               | (9)               |
| Post=1 × Moratoria              | -0.238***<br>(0.076)                                    |                    | 0.010<br>(0.174)  | -0.185<br>(0.182) |                    |                   | -0.108<br>(0.078) | -0.074<br>(0.158) | -0.173<br>(0.169) |
| Post=1 × Moratoria × % Mortgage | -0.192***<br>(0.073)                                    | -0.203*<br>(0.110) | -0.043<br>(0.149) | -0.177<br>(0.161) | -0.196*<br>(0.113) | -0.171<br>(0.143) |                   |                   |                   |
| County FE                       | ✓                                                       | ✓                  | ✓                 | ✓                 | ✓                  | ✓                 | ✓                 | ✓                 | ✓                 |
| Year FE                         | ✓                                                       |                    | ✓                 | ✓                 |                    |                   | ✓                 | ✓                 | ✓                 |
| State × Year FE                 |                                                         | ✓                  |                   |                   | ✓                  | ✓                 |                   |                   |                   |
| Border segment × Post FE        |                                                         |                    |                   | ✓                 |                    |                   |                   |                   | ✓                 |
| New Deal controls               |                                                         |                    |                   |                   | ✓                  |                   |                   |                   |                   |
| Dust Bowl controls              |                                                         |                    |                   |                   |                    | ✓                 |                   |                   |                   |
| Border counties                 |                                                         |                    | ✓                 | ✓                 |                    |                   |                   | ✓                 | ✓                 |
| High % mortgage counties        |                                                         |                    |                   |                   |                    |                   | ✓                 | ✓                 | ✓                 |
| Period                          | 1920-1958                                               | 1920-1958          | 1920-1958         | 1920-1958         | 1920-1958          | 1920-1958         | 1920-1958         | 1920-1958         | 1920-1958         |
| Adjusted $R^2$                  | 0.82                                                    | 0.84               | 0.79              | 0.80              | 0.84               | 0.84              | 0.83              | 0.79              | 0.81              |
| Obs                             | 19,443                                                  | 19,443             | 3,268             | 2,996             | 19,443             | 19,443            | 4,261             | 542               | 492               |
| Counties                        | 2,901                                                   | 2,901              | 486               | 442               | 2,901              | 2,901             | 663               | 86                | 77                |
| Outcome mean                    | -0                                                      | -0                 | -0                | 0                 | -0                 | -0                | -0                | -0                | -0                |
| Outcome SD                      | 1                                                       | 1                  | 1                 | 1                 | 1                  | 1                 | 1                 | 1                 | 1                 |

Notes: Observations are county-years. The outcome is the manufacturing intensive-margin principal-component index. Manufacturing data are from the Census of Manufactures and related historical county compilations. In the triple-difference specifications, the treatment effect is interacted with the county share of mortgaged farms in 1930. In the double-difference specifications, the sample is restricted to counties in the top quartile of the 1930 mortgage-share distribution (Q4). Columns marked *Border* restrict the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

Table A13: Effects of Moratoria on Intensive Margin (1920–1947)

|                                 | Outcome: Manufacturing Index Intensive Margin (z-score) |                     |                   |                   |                     |                   |                     |                   |                   |
|---------------------------------|---------------------------------------------------------|---------------------|-------------------|-------------------|---------------------|-------------------|---------------------|-------------------|-------------------|
|                                 | (1)                                                     | (2)                 | (3)               | (4)               | (5)                 | (6)               | (7)                 | (8)               | (9)               |
| Post=1 × Moratoria              | -0.313***<br>(0.080)                                    |                     | -0.080<br>(0.187) | -0.287<br>(0.200) |                     |                   | -0.173**<br>(0.073) | -0.124<br>(0.149) | -0.270<br>(0.171) |
| Post=1 × Moratoria × % Mortgage | -0.265***<br>(0.082)                                    | -0.229**<br>(0.108) | -0.134<br>(0.167) | -0.265<br>(0.181) | -0.226**<br>(0.112) | -0.212<br>(0.149) |                     |                   |                   |
| County FE                       | ✓                                                       | ✓                   | ✓                 | ✓                 | ✓                   | ✓                 | ✓                   | ✓                 | ✓                 |
| Year FE                         | ✓                                                       |                     | ✓                 | ✓                 |                     |                   | ✓                   | ✓                 | ✓                 |
| State × Year FE                 |                                                         | ✓                   |                   |                   | ✓                   | ✓                 |                     |                   |                   |
| Border segment × Post FE        |                                                         |                     |                   | ✓                 |                     |                   |                     |                   | ✓                 |
| New Deal controls               |                                                         |                     |                   |                   | ✓                   |                   |                     |                   |                   |
| Dust Bowl controls              |                                                         |                     |                   |                   |                     | ✓                 |                     |                   |                   |
| Border counties                 |                                                         |                     | ✓                 | ✓                 |                     |                   |                     | ✓                 | ✓                 |
| High % mortgage counties        |                                                         |                     |                   |                   |                     |                   | ✓                   | ✓                 | ✓                 |
| Period                          | 1920-1945                                               | 1920-1945           | 1920-1945         | 1920-1945         | 1920-1945           | 1920-1945         | 1920-1945           | 1920-1945         | 1920-1945         |
| Adjusted $R^2$                  | 0.83                                                    | 0.84                | 0.79              | 0.81              | 0.84                | 0.84              | 0.81                | 0.80              | 0.79              |
| Obs                             | 14,081                                                  | 14,081              | 2,364             | 2,167             | 14,081              | 14,081            | 3,090               | 393               | 358               |
| Counties                        | 2,799                                                   | 2,799               | 471               | 428               | 2,799               | 2,799             | 630                 | 82                | 73                |
| Outcome mean                    | -0                                                      | -0                  | -0                | 0                 | -0                  | -0                | -0                  | -0                | -0                |
| Outcome SD                      | 1                                                       | 1                   | 1                 | 1                 | 1                   | 1                 | 1                   | 1                 | 1                 |

*Notes:* Observations are county-years. The outcome is the manufacturing intensive-margin principal-component index. Manufacturing data span 1920–1947 and are from the Census of Manufactures and related historical county compilations. In the triple-difference specifications, the treatment effect is interacted with the county share of mortgaged farms in 1930. In the double-difference specifications, the sample is restricted to counties in the top quartile of the 1930 mortgage-share distribution ( $Q_4$ ). Columns marked *Border* restrict the sample to counties adjacent to a state border across which moratorium status differs. *Border Pair FE* indicates specifications that include fixed effects for each border segment. Standard errors are clustered at the county level. Regressions are weighted by 1930 farmland area.

## B Data Appendix

Here, we describe our treatment of three borderline coding choices for our treatment variable.

Although we follow [Alston \(1984\)](#), three statutes sit at the boundary of what can be considered a moratorium law. Arkansas briefly postponed foreclosure filings under Act 21 of 1933, but its stronger 1935 measure, which abolished deficiency judgments, was struck down. Maryland's 1933 law restricted certain foreclosures unless a quarter of the holders' total bond value consented. This was initially invalidated but eventually restored. We reason that consent from a quarter of bondholders is not a significant barrier. Oregon's 1933 joint resolution only advised courts to delay sales, leaving its legal force uncertain. We code these as untreated in the baseline, but our results are qualitatively invariant to coding them as treated.

Illinois, by contrast, clearly enacted a moratorium, but it was not included in the report because the law had expired by the time the report was published. It is listed in the report as "Moratorium period provided by legislation in this State has now expired." House Bill 507 (1933), however, authorized courts to stay foreclosures on pre-June 1933 mortgages and allowed the governor or insurance commissioner to halt proceedings on insurance-held loans. The law expired in mid-1935, but was real and binding for two years. Therefore, we follow [Alston \(1984\)](#) and include it.<sup>1</sup>

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<sup>1</sup>Charles Bunn, "The Impairment of Contracts: Mortgage and Insurance Moratoria," 1 *U. Chi. L. Rev.* 495 (1933). J. Douglass Poteat, "State Legislative Relief for the Mortgage Debtor During the Depression," 5 *Law & Contemp. Probs.* 517 (1938).